Effects of Subsidies on Welfare and Market Structure in the U.S. Broadband Industry*

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Job Market Paper

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Abstract

How should governments divide broadband subsidies between consumer discounts and infrastructure grants? I study the Affordable Connectivity Program (ACP) and the Broadband Equity, Access, and Deployment (BEAD) program, combining state-year event studies with a tract-level structural model of demand, pricing, and entry estimated on 170,419 provider-tier products in 66,454 census tracts built from FCC, ACS, and NTIA data. In high-eligibility states, ACP was followed by lower prices—roughly 7–10 percent at peak response—a shift toward faster plans, and lower concentration; BEAD effects are smaller and delayed, consistent with construction lags. Counterfactual simulations imply a BEAD benefit-cost ratio of roughly 26–27 discounted over the infrastructure's lifetime and an ACP ratio of 1.01–1.03 under targeted eligibility. ACP's returns concentrate where broadband exists but remains unaffordable; BEAD's where infrastructure is missing. The optimal allocation depends on the binding constraint: affordability in served low-income markets, deployment in unserved ones.

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1 Introduction

In 2020, roughly 21 million American households lacked fixed broadband service.[1] The shortfall falls disproportionately on low-income and rural communities, and its consequences run through job search, schooling, and healthcare. Congress responded with the Infrastructure Investment and Jobs Act (IIJA) of 2021, which allocated approximately $65 billion to broadband. Two flagship programs account for most of that response. The Affordable Connectivity Program (ACP) was a $14.2 billion demand-side subsidy that gave eligible low-income households monthly discounts of up to $30 on broadband bills. The Broadband Equity, Access, and Deployment (BEAD) program is a $42.45 billion supply-side grant program funding deployment in unserved and underserved areas. The two programs share a goal—wider, more affordable broadband—but address different constraints. ACP lowers the consumer-facing price where service already exists. BEAD lowers the fixed cost of building service where private deployment has not materialized.

The two constraints reflect distinct market failures. On the demand side, affordability binds: even where service exists, low income keeps many households below the participation margin. By early 2021, 15 percent of U.S. home-broadband users reported trouble paying for service, and broadband subscription rates among households below 200 percent of the federal poverty line ran about 20 percentage points below the national average [Pew Research Center, 2021]. On the supply side, deployment binds: fixed costs are high enough to make broadband privately unprofitable in low-density areas even when the social return to connectivity is large—the classic wedge that justifies public subsidy [Bresnahan and Reiss, 1991, Kearns, 2024]. Market structure adds a third consideration. The average census tract in my data is served by 2.36 fixed-broadband providers, and the median within-tract Herfindahl–Hirschman index is 5,472, more than twice the threshold conventionally associated with concentrated markets. In markets this concentrated, a subsidy that enlarges the addressable market or lowers the cost of entry can move posted prices and product menus. These are equilibrium responses that subscriber counts alone cannot reveal.

I ask three questions. First, did ACP and BEAD change broadband prices, product offerings, and local market structure after IIJA? Second, how large are the welfare gains from each program once firms can enter or adjust local product positions? Third, are demand-side and supply-side subsidies substitutes, or do they address different parts of the digital divide? ACP exhausted its funding in June 2024, leaving policymakers to decide whether a successor affordability program is worth financing. BEAD is the largest federal broadband deployment

[1] Federal Communications Commission [2020] reported 21.3 million Americans lacking access to fixed broadband at 25/3 Mbps speeds based on FCC Form 477 data (see page 19).

program in U.S. history, but its returns depend on whether subsidies induce entry in markets where deployment is privately unprofitable. Standard evaluations that count new subscribers or dollars per connection do not answer these questions, especially in local broadband markets with few providers and substantial infra-marginal households.

To answer them, I combine reduced-form and structural evidence. The reduced-form analysis uses continuous-treatment event studies to compare states with different exposure to ACP eligibility and BEAD funding before and after IIJA. ACP exposure is the share of households below 200 percent of the federal poverty level, measured from the 2020 American Community Survey before the program launched. BEAD intensity is the log of allocation per unserved location, fixed by a federal formula over pre-grant counts. Both intensities are therefore predetermined with respect to post-IIJA outcomes. Because all states are exposed at the same time, identification comes from cross-state differences in intensity rather than in treatment timing, so the negative-weighting and contamination problems of staggered-adoption designs do not arise. The identifying assumption is that, conditional on state and year fixed effects and state-specific linear trends, states with different exposure would have trended similarly absent the programs; pre-period coefficients close to zero across outcomes support that assumption. One feature of the data works against finding BEAD effects: the Urban Rate Survey samples urban tracts and under-represents the rural markets BEAD is designed to serve, so I read the BEAD event-study estimates as conservative lower bounds.

The event studies establish that the programs moved prices, product menus, and concentration. They cannot, by themselves, answer the welfare questions. A falling price is consistent with pass-through, strategic repricing, or a shifting plan mix, and the reduced form cannot distinguish among these mechanisms. State averages wash out the tract-level variation in market structure and cost that underlies the response. And no event study can measure the gains from a policy design that was never implemented, or decompose welfare into consumer and producer components. The structural model takes up exactly these questions. It is a two-stage game played in each census tract: providers first choose which speed tiers to offer, paying a fixed cost for each activated product position, and then set prices in Bertrand-Nash competition. I estimate a nested-logit demand system with the differentiation instruments of Gandhi and Houde [2019], recover markups and marginal costs from the pricing first-order conditions, and bound fixed costs of entry using the scalable discrete-game method of Fan and

Yang [2024], which requires only that observed product configurations be undominated.[2] This framework fits the policy setting naturally: broadband subsidies change both the demand margin and product availability, so a model that treats entry and pricing jointly is the right tool. The discrete-game approach also avoids solving the entry game inside the estimation loop, which matters with tens of thousands of markets and many potential provider-tier positions. The two parts of the analysis are linked by construction. The tract-level share of households below 200 percent of the federal poverty level that defines ACP eligibility in the counterfactuals is the same variable that measures treatment intensity in the event study, and the model's 2020 baseline predates both programs, so the counterfactuals start from the undisturbed market whose response the event studies trace.

The data combine several official federal sources. For the reduced-form analysis, I build a balanced panel of 52 U.S. states from 2014 to 2025. The panel combines FCC Urban Rate Survey prices and product characteristics, ACP eligibility exposure from the American Community Survey, and BEAD allocations from federal program records. For the structural analysis, I construct a pre-IIJA product-level dataset. The demand sample contains 170,419 observed broadband products in 66,454 census tracts, defined at the tract-provider-speed-tier level. The entry sample augments these observed products with county-footprint feasible positions, yielding 727,601 potential provider-tier-tract positions across 70,854 tracts. Product characteristics, prices, adoption shares, and demographics all come from publicly available federal sources.

The results show that the programs operate through different margins. In the event study, ACP is associated with lower prices, a shift toward faster plans, and lower concentration in high-eligibility states. Moving from the 25th to the 75th percentile of ACP eligibility exposure is associated with average broadband prices roughly 7–10 percent lower at peak response. Effects on high-speed plans reach 15–18 percent and persist through the second and third post-launch years. The product mix shifts in tandem: the low-speed plan share contracts and the high-speed share expands, the pattern expected when subsidized households raise demand for faster service and providers upgrade their menus to meet it. BEAD effects are smaller and appear with a two-to-three year lag, consistent with a program whose institutional clock—allocations announced in June 2023, state proposals due that December, construction only

[2] The key estimates line up with external benchmarks, which validates the model's primitives. The mean own-price elasticity of -4.99 falls within the range of product-level broadband and pay-TV estimates: -4.2 to -4.7 in Kearns [2024], -5.9 in Goetz [2019], and -4.1 to -5.4 in Crawford et al. [2019]. The median Lerner index of 20.0 % is comparable to the 16–45 % range Nevo [2001] reports for ready-to-eat cereals and to the magnitudes Crawford et al. [2019] find in cable television. The recovered fixed-cost bounds are of the same order of magnitude as the annual deployment costs Kearns [2024] recovers from revealed deployment decisions (roughly \$404,000–\$710,000 per census block for de novo entry) and consistent with industry estimates of the annualized cost of keeping fixed-broadband plant in service. Section 5 reports the full comparisons.

afterward—delays any market response. The timing is itself informative: effects appear where the construction timeline predicts, and not before, which points to entry as the operative channel rather than a near-null result. These estimates should be read as the leading edge of the program, not its full effect. BEAD construction extends through 2026 and beyond, so the event window closes before most funded deployment is in service. This is a further reason the structural model is necessary: the counterfactuals simulate the completed program against the pre-IIJA baseline and recover its steady-state effects, which no event study conducted during rollout can measure.

The counterfactuals translate these patterns into welfare. A 75 percent BEAD fixed-cost subsidy induces 6,665–8,648 new firm-market entries and brings service to 6,128–8,054 previously unserved tracts—most of the roughly nine thousand tracts that begin with no provider at all. Consumer surplus rises from \$13.64–\$14.07 billion per month at baseline to \$15.41–\$15.54 billion. A \$30 ACP discount raises consumer surplus to \$14.45–\$14.89 billion per month, operating almost entirely through demand expansion among eligible households: the consumer-facing price falls by roughly nine dollars on average while the gross posted price barely moves, and overall take-up rises by about 1.5 percentage points. By design, the counterfactual holds gross provider prices fixed, so this estimate isolates the demand-expansion channel and excludes the equilibrium price responses the event study documents.

The distributional results show that the two programs' gains are nearly non-overlapping. BEAD's gains are determined by baseline market structure: monopoly tracts gain about \$31 per household per month, five times the gain in tracts with three or more providers, while gains are nearly flat across income quartiles. ACP's gains follow the opposite pattern: the lowest-income quartile gains more than twice the highest, while gains are largely invariant to provider count. This answers the third question: the programs are complements rather than substitutes. Each reaches a part of the market the other cannot—BEAD the deployment margin, ACP the affordability margin—with limited overlap in served monopoly and duopoly markets.

The cost-benefit analysis sets these gains against fiscal outlays. BEAD's one-time grants generate recurring surplus over the infrastructure lifetime: at a 3% discount rate and a 25-year horizon, the standard assumption for long-lived broadband infrastructure assets,[3] the 75 percent subsidy has a benefit-cost ratio of 25.85–27.44. The ratio declines with the subsidy rate—35.58–43.82 at 25 percent—reflecting diminishing returns as higher rates reach harder-to-serve markets, and the lower bound remains 12.51 even at the most conservative discount-rate and horizon combination. ACP's recurring outlays nearly break even: the benefit-cost ratio is 1.01–1.03 at the \$30 discount under targeted eligibility, with welfare gains

[3] See Grunvalds et al. [2017] for broadband infrastructure lifetime.

offsetting outlays, a small positive residual from marginal subscribers, and a pure transfer to infra-marginal households. The comparison does not imply that one instrument dominates the other. The two ratios price different instruments—a capital investment in unserved markets against a transfer in served ones—and together they support a sequencing principle: build where private deployment is uneconomic, then target affordability support where price remains the binding constraint.

The paper contributes to three related literatures. First, it provides the first tract-level structural evaluation of ACP and BEAD, the two central broadband programs created by IIJA. Existing work shows that broadband adoption depends on prices, income, local market structure, and provider choice [Rosston et al., 2010, Nevo et al., 2016, Flamm and Varas, 2022, Espín and Rojas, 2024, Kearns, 2024]. I build on that evidence in two ways: by using policy exposure as the source of identifying variation, and by tracing the market-structure responses—entry, product repositioning, and concentration—that aggregate adoption studies cannot capture.[4]

Second, the paper brings Fan–Yang-style scalable entry bounds to broadband infrastructure. The approach is closest to Eizenberg [2014], Ciliberto et al. [2021], and Fan and Yang [2024], but the application is new: I use these bounds to recover fixed costs and evaluate entry responses across tens of thousands of local broadband markets. The setting also stresses the method in a useful direction. With 727,601 potential provider-tier positions, an estimator that requires solving the entry game inside the estimation loop would be infeasible; the moment-inequality approach requires only that observed configurations be undominated, which is what makes a national tract-level entry analysis tractable.

Third, the paper provides a unified evaluation of demand-side and supply-side broadband subsidies. By placing a consumer discount and a fixed-cost deployment subsidy inside the same structural welfare framework, I compare fiscal efficiency across instruments that are usually evaluated separately. The comparison matters for policy design because the two instruments were funded by the same legislative appropriation yet target different market failures; evaluating them in a common framework makes the allocation question—which dollar goes where—answerable rather than rhetorical.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting, data, and local market structure. Section 3 presents the reduced-form evidence. Section 4 develops the structural model, including timing assumptions, identification, and a discussion of multiple equilibria. Section 5 reports estimation results and structural parameter diagnostics.

[4] Concentration, high prices, and limited provider choice are well documented in U.S. broadband [Flamm and Varas, 2022]. Downstream consequences for labor markets, education, and health are documented in Forman et al. [2005], Goldfarb and Prince [2008], and Goolsbee and Klenow [2002].

Section 6 presents counterfactual policy simulations and distributional effects across market types. Section 7 reports the cost-benefit analysis, including sensitivity to discount rate and infrastructure horizon. Section 8 concludes.

2 Industry Background and Data

2.1 Industry Background

2.1.1 Current Landscape

The technologies available to a household depend primarily on location, which determines both local competitive pressure and the geography of the access gap. Broadband service is delivered through four main platforms. Digital subscriber line (DSL) transmits data over legacy copper telephone lines. Cable modem service uses coaxial television networks. Fiber-optic broadband carries data through optical fibers using light signals. Fixed wireless access connects households through radio links from nearby transmission towers. These technologies differ substantially in speed, reliability, and infrastructure requirements. Fiber generally provides the highest speeds and lowest latency but requires significant upfront investment in network deployment. By contrast, fixed wireless offers lower throughput but can be deployed more cost-effectively in rural and sparsely populated areas because it does not require extensive trenching or new wired infrastructure. Providers compete on speed, data allowances, and price. Telephone and cable incumbents still dominate, but national fiber entrants, fixed-wireless operators, and a handful of municipal systems now contest local markets.[5]

Two facts define the market and explain why Congress intervened. Service is expensive relative to household income, and most households cannot choose among more than two or three providers [Flamm and Varas, 2022]. Both facts hold sharply in my data. The average census tract is served by just 2.36 fixed-broadband providers. The median within-tract provider Herfindahl–Hirschman Index (HHI) is 5,472—more than double the 2,500 threshold conventionally associated with concentrated markets. The shortfall is not spread evenly. Fiber, the platform with the best performance, concentrates in cities and suburbs. Rural households still depend on aging copper DSL whose speeds and reliability fall well short of the federal broadband standard.

These conditions produce the digital divide: a gap in reliable internet access that tracks income and geography closely. Its consequences run through job search, healthcare, and

[5] Wilson [2025] studies the effects of municipal broadband entry on private ISP investment. See Appendix A for an overview of broadband technology types.

schooling [Forman et al., 2005, Goldfarb and Prince, 2008, Goolsbee and Klenow, 2002]. Federal policy has responded along the same two margins that define the underlying failures: lowering the price of service that already exists and paying to build service where private providers have not.

2.1.2 Federal Broadband Subsidies

The 2021 Infrastructure Investment and Jobs Act (IIJA) committed roughly $65 billion to broadband through two programs that address opposite sides of the market. The Affordable Connectivity Program (ACP), funded at $14.2 billion, subsidized household bills directly. The Broadband Equity, Access, and Deployment (BEAD) program, funded at $42.45 billion, subsidizes deployment in underserved areas.

ACP succeeded the Emergency Broadband Benefit (EBB), which reached nearly nine million low-income households at up to $50 per month between February and December 2021. ACP scaled far larger. At its peak, it enrolled 23 million households with a monthly discount of up to $30 ($75 on qualifying Tribal lands) and a one-time device discount of up to $100.[6] Its funding lapsed in June 2024 when Congress declined to reappropriate. This full cycle—launch in 2022 through funding exhaustion in 2024—falls within the event-study window. The program's termination also motivates the central policy question of whether a successor affordability program is worth financing. The welfare and fiscal-efficiency analysis in Sections 6–7 addresses this question directly.

BEAD is the largest federal program ever to subsidize broadband providers directly. Funds are allocated to states by formula on the count of underserved locations. These are broadband-serviceable locations lacking service entirely or lacking reliable service at 25/3 Mbps with latency below 100 milliseconds.[7] Because allocations are formula-based and tied to pre-grant counts of unserved locations, treatment intensity is predetermined with respect to post-IIJA broadband outcomes. The program's slow institutional clock explains the lagged effects documented below. IIJA passed in November 2021, but allocations were not announced until June 2023. Initial state proposals were due in December 2023. Subgrantee selection, permitting, and construction follow only afterward. Deployment is expected to extend through 2026 and beyond. The counterfactuals are anchored to a pre-IIJA baseline. They are therefore best read as the program's steady-state effects once deployment is complete, not as forecasts for any single intermediate year.

[6] https://www.fcc.gov/acp

[7] https://broadbandusa.ntia.doc.gov

2.1.3 Economic Rationale

Each program targets a distinct market failure; that distinction structures the empirical strategy and the counterfactual analysis. On the demand side, affordability binds: even where service exists, low income and limited digital literacy keep many households below the participation margin. By early 2021, 15 percent of U.S. home-broadband users reported trouble paying for high-speed internet service, and 2020 ACS data show broadband subscription rates among households below 200 percent of the federal poverty line about 20 percentage points below the national average [Pew Research Center, 2021]. A demand-side subsidy addresses this gap directly, lowering the effective price until price-constrained households subscribe.

On the supply side, deployment binds: fixed costs are high enough to make broadband privately unprofitable in low-density or low-income areas even when the social return to connectivity is large. This wedge between private and social returns to infrastructure is the classic case for public subsidy [Bresnahan and Reiss, 1991, Kearns, 2024]. BEAD lowers the provider's private cost of entry, but its net effect depends on how much private deployment it displaces.

The two failures are largely non-overlapping, which makes the programs complements rather than substitutes. A demand subsidy does nothing where no infrastructure exists. In wholly unserved areas, no discount can induce adoption of a product that is not offered. A deployment subsidy has limited reach on affordability where competitive infrastructure already exists and price, not access, is the binding constraint. The boundary is less sharp in served monopoly or duopoly markets. In those markets, a second provider may face high entry costs, and BEAD-induced entry can discipline prices through Bertrand competition. Each program therefore addresses the failure the other cannot in most markets, with some overlap at the frontier.

Market structure opens a third channel. With most households facing one to three providers [Flamm and Varas, 2022], broadband is an oligopoly, and a demand subsidy that enlarges the addressable market can move posted prices and product menus—equilibrium responses that subscriber counts miss. The reduced-form event study tests for these responses; the counterfactuals then hold gross provider prices fixed, isolating the demand-expansion channel from provider price adjustment.

2.2 Data Sources and Estimation Samples

A central input of this paper is a new tract-level dataset of the U.S. broadband market on the eve of IIJA. No single federal source contains the information needed to estimate broadband supply, demand, and entry jointly: plan-level prices, product-level deployment, household demographics, and adoption are collected by different agencies at different units of observation. I assemble five publicly available federal sources into a nationally representative pre-IIJA dataset linking prices, product characteristics, market structure, and adoption at census-tract resolution. Three construction steps make the merged data usable at this scale: a brand-first price-matching hierarchy that assigns a price to every one of the 727,601 potential product positions, a manual normalization of provider names into 1,732 canonical corporate identifiers, and a market-share construction validated against an independent national benchmark. Each step is described with its source below.

2.2.1 Data Sources

FCC Urban Rate Survey (URS). The FCC conducts the Urban Rate Survey annually. The survey draws a stratified random sample of ISPs operating in urban census tracts and collects each plan's advertised download speed, upload speed, technology type, and monthly price.[8] It is one of the few nationally representative sources of plan-level broadband prices and is now a standard input to structural work on broadband competition and infrastructure policy [Espín and Rojas, 2024, Kearns, 2024]. I use the URS as the primary source of product prices in both the state-year panel and the structural sample. Assigning a survey price to every tract-level product is the first construction step. I match prices through a six-level brand-first hierarchy that proceeds from a state-provider-tier match down to a national tier average, selecting within each level the URS plan closest in download speed, and deflate prices to constant 2025 dollars using the CPI-U. The hierarchy achieves complete coverage: all 727,601 potential product positions receive a price, so no position drops out of the entry analysis for lack of price data. Coverage does not come at the cost of match quality: 63.9 percent of positions match on the provider's own plans, and only 1.4 percent fall to the coarsest national-tier level. Appendix D documents the full hierarchy and the match-quality distribution.

One important limitation of the URS is its frame: because it samples urban tracts, it under-represents the rural markets that BEAD is designed to serve. As a result, the state-year event-study estimates for BEAD tend toward zero—they capture urban and suburban response while rural deployment effects are outside the sample frame. I treat the URS-based BEAD estimates as conservative lower bounds on the program's full impact.

FCC Form 477 Deployment Data. Form 477, collected semi-annually, requires all ISPs to report the maximum advertised download and upload speeds of every fixed residential

[8] https://www.fcc.gov/economics-analytics/industry-analysis-division/urban-rate-survey-data-resources

broadband service they offer at the census-block level.[9] I restrict to residential fixed broadband and aggregate to the census-tract × provider × speed-tier level. The December 2020 release defines the structural baseline. It identifies which provider-tier products are observed in each tract ($Y_{fgt} = 1$) and which are feasible but not offered ($Y_{fgt} = 0$), the two inputs to the Fan–Yang moment inequalities. Form 477 deployment records also construct the county-footprint potential entry set by identifying all provider-tier pairs active anywhere in the county. A known limitation of Form 477 is overstatement of coverage. A block is marked as served if a provider *could* serve a household in the block at that speed, which overstates actual availability in multi-dwelling-unit buildings and rural blocks. Since I use December 2020 data before the post-IIJA transition to the Broadband Data Collection system, this overstatement is uniform across the cross-section. It therefore does not differentially affect the treatment variation used for identification. Provider names in Form 477 vary substantially across subsidiaries and legacy brands, and treating each filing name as a distinct firm would badly overstate competition. The second construction step consolidates these names into 1,732 canonical corporate identifiers using a manually constructed crosswalk, so that markups, market structure, and entry decisions are attributed to the economic decision-maker rather than to its filing labels; details are in Appendix C.

American Community Survey (ACS). The Census Bureau's ACS provides annual small-area estimates of demographic and socioeconomic conditions at the census-tract level.[10] I use the ACS 2019 five-year release, the pre-IIJA vintage closest to the 2020 structural baseline. At the tract level, I extract the number of housing units, median household income, poverty rate (Table C17002, households below 200 % FPL), broadband subscription rate (Table B28002), renter share, college-educated share, age-65+ share, and share with no internet. These variables enter the model as market-size measures, eligibility proxies for ACP counterfactuals, and demand-side controls in both the CSS market-share construction and the structural demand estimation. The poverty-rate variable is the share of the tract population with income below 200 % of the federal poverty level. It is also the ACP treatment intensity FPL200_i in the reduced-form event study, providing a direct bridge between the two parts of the empirical strategy.

FCC Broadband Adoption Data. The FCC publishes tract-level fixed broadband adoption statistics derived from Form 477 subscription filings. The data report the share of households with fixed broadband connections by speed threshold in five binned ranges (10,

[9] Data available at https://broadband477map.fcc.gov/#/data-download; 2014–2023.
[10] Data available at https://www.census.gov/programs-surveys/acs.

30, 50, 70, and 90 percent of households).[11] I use the ≥10/1 Mbps adoption bin to anchor the outside-option share $s_{0t} = 1 - S_t^{\text{fixed}}$ in each tract. This outside-option share is the key object the CSS market-share construction—the third construction step—requires. The 10/1 Mbps threshold aligns with the speed-tier definition used throughout the structural model (Tier H: ≥10/1 Mbps; Tier L: below). The construction passes an external check it was not built to satisfy: the implied national subscriber count is 93.7 million households at ≥10/1 Mbps, within about 1 % of the FCC national benchmark of 94.5 million (Table 3).

BEAD Program Allocation Records. State-level BEAD allocations, announced by the NTIA in June 2023, are determined by a federal formula based on the certified count of broadband-serviceable locations lacking reliable service at 25/3 Mbps or entirely unserved.[12] The formula-based allocation of BEAD funds is the key property exploited in the event study: funding intensity is determined by certified unserved-location counts rather than by post-IIJA broadband outcomes. I measure treatment intensity as $\log \text{BEAD}_i$, the log of total BEAD allocation per certified unserved location, which is fixed by the formula and predetermined relative to the 2022–2025 outcome window.

2.2.2 Sample construction and the two estimation samples

These five sources feed two estimation samples, each matched to a distinct empirical question.

The *state-year policy panel* covers 52 states (including the District of Columbia and Puerto Rico) from 2014 to 2025 ($N = 624$ state-year cells). It pairs Urban Rate Survey plan-level outcomes—log prices by speed tier, plan-composition shares, and provider concentration—with ACP eligibility exposure (FPL200_i from the 2020 ACS) and BEAD funding intensity ($\log \text{BEAD}_i$ from NTIA). This panel supports the continuous-treatment event studies in Section 3.

The *tract-level structural sample* is a single 2020 cross-section built around the Form 477 deployment file. I merge URS prices, ACS demographics, and FCC adoption shares, then restrict to tracts with at least one product with a positive imputed price. The resulting *demand estimation sample* contains 170,419 products (provider-tier-tract triples with $Y_{fgt} = 1$) across 66,454 census tracts and 1,732 canonical providers. The *entry sample* augments the observed products with county-footprint feasible positions, yielding 727,601 potential provider-tier-tract

[11] December 2020 release available at https://docs.fcc.gov/public/attachments/DOC-395959A1.pdf.

[12] State allocation totals available at https://www.ntia.gov/funding-programs/internet-all/broadband-equity-access-and-deployment-bead-program/program-documentation/state-allocation-totals.

positions across 70,854 tracts.[13] The county-footprint rule captures nearby but not-yet-offered configurations without positing entry from providers with no local infrastructure. The two samples have different units of observation and answer different questions. The demand sample identifies preference and price-sensitivity parameters from observed market shares. The entry sample identifies fixed-cost parameters from which positions are offered and which are not.

Data quality, not only identification, motivates this split between a state-year panel and a single-year structural cross-section. A consistent tract-year panel is not constructible over 2014–2025: the FCC's post-2021 transition from Form 477 to the Broadband Data Collection system redefined geographic units and coverage measurement in ways that would inject spurious dynamics into a long tract panel. The state-year aggregation absorbs that structural break; the 2020 cross-section predates both ACP enrollment and any BEAD-induced deployment, so the structural baseline—and every counterfactual built on it—reflects a market untouched by either program. Appendix B reports file-level details, merge keys, and sample restrictions; the full replication package is available upon request.

2.3 Descriptive Statistics

Table 1 summarizes the demand estimation sample. The typical plan costs \$74.06 per month. High-speed plans dominate the 2020 product menu: 90.8 % of products are classified as Tier H, reflecting the retirement of legacy sub-10 Mbps DSL by most carriers. Despite the preponderance of high-speed products, consumer choice remains thin. The average tract is served by just 2.36 providers offering 2.56 products, and over half of tracts have at most two providers (Table 4). Median household income is \$65,422. The poverty share below 200 % FPL—the eligible population for ACP—is 32.6 %.

[13] The demand estimation sample (66,454 tracts) is smaller than the full entry sample (70,854 tracts) because 4,400 tracts with observed products did not match when merging demand and tract-level covariates. The resulting sample remains large enough to reliably recover the demand and supply estimates.

Table 1: Summary Statistics — Estimation Sample

Variable	Mean	Std. Dev.	P25	P75
Panel A. Product and Market Characteristics				
Price ($/month)	74.06	19.64	63.66	85.99
Download speed (Gbps)	0.555	0.767	0.026	1.000
Upload speed (Gbps)	0.313	0.748	0.006	0.500
Tier H indicator	0.908	—	—	—
Products per tract	2.56	1.39	2	3
Providers per tract	2.32	1.16	2	3
Panel B. Tract Demographics				
Population (tract)	4,533	2,416	2,957	5,615
Median household income ($)	65,422	32,562	44,083	78,750
Share renters (%)	35.24	22.36	17.79	49.02
Share college educated (%)	29.91	19.12	15.54	40.47
Share age 65+ (%)	16.64	7.63	11.76	20.31
Share below 200% FPL (%)	32.55	18.18	18.56	43.71

Notes: Price is reported in dollars per month; estimation uses price divided by 100. Tier H denotes service with download speed $\geq$ 10 Mbps and upload speed $\geq$ 1 Mbps. Demographic variables are measured at the census-tract level using the 2019 ACS 5-year estimates and merged to each product observation. The poverty measure is derived from ACS Table C17002 (income-to-poverty ratio below 2.0). Statistics are computed over 170,419 product-level observations across 66,454 tract markets.

2.4 Market Share Construction

Estimating broadband demand confronts one binding data limitation: product-level market shares are not readily observed. The FCC discloses where providers operate, their technologies, and their advertised speeds, but never the subscriber count for a given provider-speed product in a given tract—the exact object the demand system requires.

I recover these shares by combining observed adoption totals with product availability, following the logic of Crawford et al. [2019]. Two steps suffice. FCC tract-level adoption data, reported in binned ranges for any fixed broadband and for service at or above 10/1 Mbps, anchor total adoption and the outside option. Logit weights built from product characteristics and demographics then allocate that total across the available provider-speed-tier products. The resulting shares are inferred, not observed subscriber counts, and should be read as such. *Implementation details, the allocation equation, and validation diagnostics are in Appendix E.*

Table 2 reports summary statistics for the constructed market shares.

Table 2: Market Share Summary Statistics

Variable	Mean	Std. Dev.	P25	P75
Product market share $\hat{s}_{jt}$	0.322	0.208	0.182	0.417
Within-tier share $\hat{s}_{j\|g,t}$	0.456	0.245	0.278	0.527
Outside-option share s_{0t}	0.191	0.149	0.100	0.300
Provider HHI	5,472	2,576	3,463	5,415

Notes: Shares validated such that $\sum_j s_{jt} + s_{0t} = 1$ in every tract. Product shares are inferred using FCC adoption information and product-level data (see Appendix E.2). Within-tier share is computed as $\hat{s}_{j|g,t} = \hat{s}_{jt}/\sum_{k\in g}\hat{s}_{kt}$, where $g \in \{H, L\}$ denotes the speed tier. The Herfindahl–Hirschman Index is computed within tracts using provider shares among fixed-broadband subscribers and is scaled from 0 to 10,000. The demand estimation sample contains 170,419 observed provider-tier products in 66,454 census tracts.

The constructed sample reproduces national aggregates closely. Table 3 benchmarks the entry sample against FCC totals: the 70,854 tracts cover 118.6 million households (95.9 % of the national benchmark), and the inferred adoption shares imply 93.7 million residential fixed connections at ≥10/1 Mbps, within about 1 % of the FCC national figure. Appendix E reports the model-implied market shares by provider as a further internal consistency check.

Table 3: Aggregate Validation of the Structural Sample

Moment	Structural sample	FCC national benchmark	Coverage
Households	118.59 million	123.67 million	95.90%
Residential fixed connections, ≥10/1 Mbps	93.70 million	94.45 million	99.21%

Notes: The entry sample contains 70,854 census tracts; the demand estimation sample contains 66,454 tracts. Residential fixed connections are computed as tract households multiplied by the FCC-based 10/1 adoption-share bin. Product speed tiers in the structural model remain defined using advertised 10/1 Mbps availability. FCC national benchmarks are from Federal Communications Commission [2022]. Coverage is the structural-sample value divided by the FCC national benchmark. These moments validate the scale of the structural sample against national residential broadband aggregates; they are not provider-level subscriber counts.

Broadband market structure. How many providers serve a tract governs both the competitive pressure on price and the room for subsidy-induced entry, so the distribution of provider counts is a primitive of the analysis. Table 4 reports it.

Most tracts are monopolies or duopolies. Nearly a fifth have exactly one fixed-broadband

Table 4: Provider Competition in the Sample

Number of providers in tract	Share of tracts (%)	Share of households (%)
One provider	21.63	20.77
Two providers	42.07	40.94
Three or more providers	36.30	38.30

Notes: Provider counts are based on canonical providers with positive constructed product shares in the tract. Sample: 66,454 census tracts in the demand estimation sample, December 2020.

provider, and nearly two-thirds have at most two. The 20.77 percent of households in monopoly tracts face no intra-market price competition at all, and only 36.30 percent of tracts reach three providers or more. This thin competition is itself part of what the federal programs are designed to address—and it is precisely what gives subsidy-induced entry room to move consumer welfare.

3 Reduced-Form Evidence

3.1 State-Year Panel and Treatment Variables

The event study traces how broadband prices, product menus, and concentration moved in states more exposed to each program. The panel is balanced over 52 states (including Puerto Rico and the District of Columbia) from 2014 to 2025, yielding $N = 624$ state-year cells. Outcomes are built from FCC Urban Rate Survey plan records. I aggregate them to state-year plan-count-weighted average prices and plan-composition shares, then deflate prices to 2025 USD by the CPI-U. Six outcomes span three economic margins: prices (log average price and log prices by speed tier), product mix (plan shares below and above the 25/3 Mbps threshold), and concentration (provider HHI). Table 5 reports descriptive statistics.

Panel structure and outcome construction. The study window spans both program rollouts while providing sufficient pre-treatment observations to assess parallel trends. For both ACP and BEAD, 2021 serves as the enactment year. The sample therefore includes seven pre-enactment years (2014–2020), the enactment year (2021), and four post-enactment years (2022–2025). For ACP specifically, the post-enactment period includes program implementation (2022–2024) and one follow-up year (2025). This structure yields five pre-

Table 5: Descriptive Statistics – Main Outcomes

Outcome	N	Mean	SD	Min	Median	Max
Panel A. Prices						
Log avg. real price	597	4.583	0.190	3.827	4.556	5.254
Log price (<25/3 Mbps)	563	4.200	0.210	3.314	4.223	5.073
Log price (≥25/3 Mbps)	587	4.703	0.223	4.169	4.673	5.483
Panel B. Plan Composition						
Plan share (<25/3 Mbps)	597	0.261	0.260	0.000	0.171	1.000
Plan share (≥25/3 Mbps)	597	0.739	0.260	0.000	0.829	1.000
Panel C. Market Structure						
Provider HHI	597	0.524	0.238	0.146	0.477	1.000

Notes: Unit of observation is state × year, 2014–2025 ($N = 624$ state-year cells before missing). These six outcomes are the main outcomes in the event studies and difference-in-differences regressions. Prices are weighted-average plan prices deflated to 2025 USD using the Consumer Price Index for Urban Consumers. Plan shares and the HHI are bounded on the interval $[0, 1]$.

treatment leads, which helps detect potential violations of the parallel-trends assumption. Four to five post-treatment years then allow dynamic treatment effects to be traced over time.

One feature of the data biases the BEAD estimates toward zero. The Urban Rate Survey samples urban tracts, so the panel captures urban and suburban conditions and under-represents the rural markets BEAD is built to serve. If BEAD's price and concentration effects concentrate in rural areas—as the structural results below imply—the URS-based estimates understate its true impact, making them a conservative lower bound. ACP, which works through demand expansion in already-served and largely urban markets, is better aligned with the URS frame and is measured more directly.

Both treatment intensities are predetermined and time-invariant, which limits the endogeneity concerns that plague contemporaneous treatment measures. The ACP eligibility share (FPL200_i) is taken from the 2020 ACS, before launch. The BEAD intensity ($\log \text{BEAD}_i$) is fixed by a federal formula over pre-grant unserved-location counts. Neither responds to ISP behavior or state outcomes over the window. This supports the assumption that exposure is uncorrelated with differential trends.

ACP exposure. I measure state-level ACP eligibility exposure as the share of households with incomes below 200% of the federal poverty level ($\mathrm{FPL200}_i$), drawn from the 2020 American Community Survey. This is a time-invariant, pre-determined cross-sectional measure that varies across states and proxies for the potential demand stimulus ACP can generate. The program became active in 2022; I set $\mathrm{Post}_t^{\mathrm{ACP}} = \mathbf{1}[t \geq 2022]$.

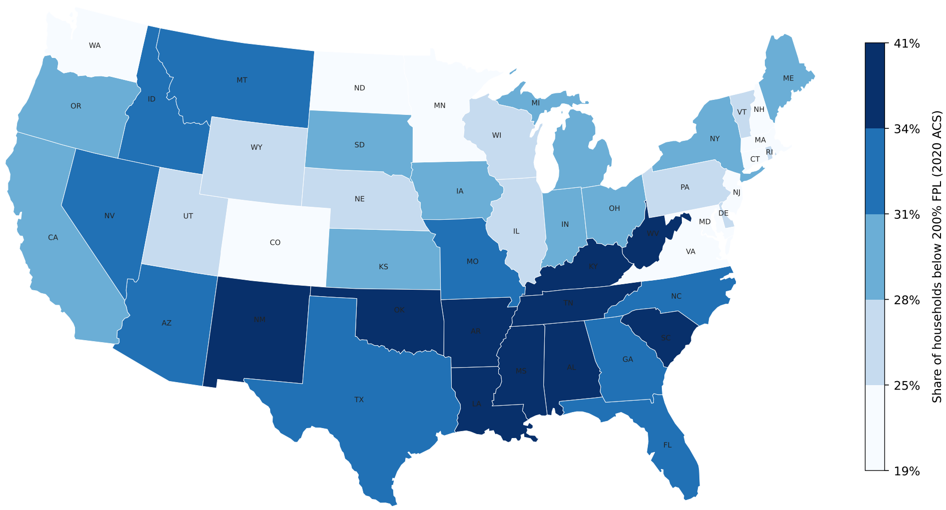

Figure 1: State-level ACP exposure: share of households with incomes below 200% of the federal poverty level ($\mathrm{FPL200}_i$), measured from the 2020 American Community Survey. Darker shading indicates higher eligibility exposure. Colour breaks at sample quintiles.

BEAD exposure. I measure state-level BEAD intensity as the log of BEAD allocation (USD) per unserved broadband location ($\log \mathrm{BEAD}_i$). A higher value indicates that a state receives more funding per targeted location, implying a more intensive supply-side intervention. This measure is also time-invariant and predetermined by the federal allocation formula. BEAD was announced in 2023; I set $\mathrm{Post}_t^{\mathrm{BEAD}} = \mathbf{1}[t \geq 2023]$.

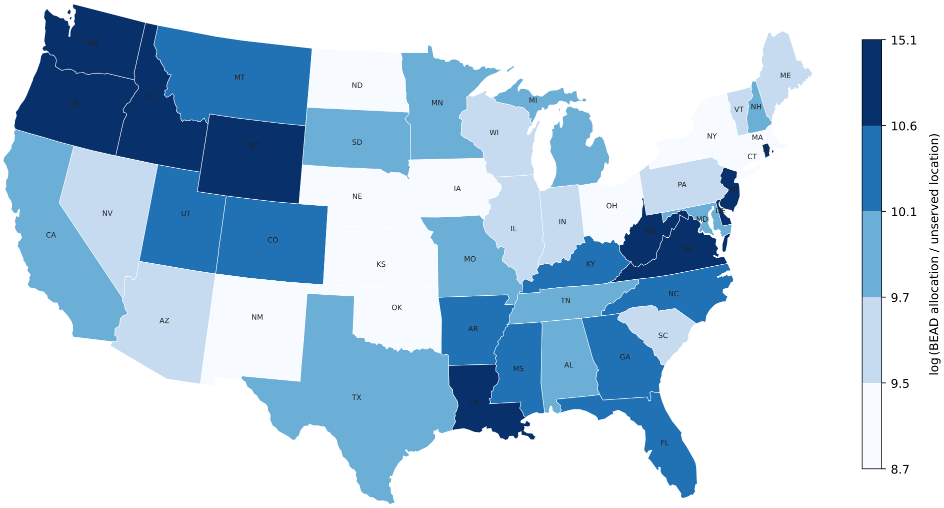

Figure 2: State-level BEAD exposure: log of BEAD allocation (USD) per unserved broadband location ($\log \mathrm{BEAD}_i$), based on NTIA allocation announcements (June 2023). Darker shading indicates higher funding intensity per targeted location. Colour breaks at sample quintiles.

3.2 Identification Strategy

For program $P \in \{\mathrm{ACP}, \mathrm{BEAD}\}$, let D_i^P denote the time-invariant treatment intensity for state i and let τ index event time relative to the relevant policy year. The dynamic specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^{\bar{\tau}^P} \beta_\tau^P \left(D_i^P \times \mathbf{1}[t = \bar{t}^P + \tau] \right) + \boldsymbol{\gamma}_i' t + \varepsilon_{it}, \tag{3.1}$$

where α_i are state fixed effects, λ_t are year fixed effects, and $\bar{t}^P$ is the event-year normalization for program P. For ACP, $\bar{t}^{\mathrm{ACP}} = 2022$, the first full implementation year. For BEAD, the dynamic event study is normalized to $\bar{t}^{\mathrm{BEAD}} = 2021$, the IIJA enactment year. The static post indicator instead begins in 2023, when allocations were announced. The omitted period is $\tau = -1$, corresponding to the year immediately before the relevant event year, so all estimates are relative to that baseline. Regressions are weighted by the number of plans n_{it}, and standard errors are clustered at the state level.

The role of fixed effects. State fixed effects α_i absorb all time-invariant determinants of broadband outcomes at the state level, including permanent differences in geographic coverage, regulatory environment, and demographic composition. Identification therefore exploits *within-state* variation in outcomes over time, purged of level differences across states.

Year fixed effects λ_t absorb common national shocks to broadband markets. These include economy-wide shifts in input costs, federal regulatory changes, and aggregate demand trends. The interaction of treatment intensity with year indicators then captures whether states more intensively exposed to each program moved differentially relative to the national trend.

State-specific trends. For both ACP and BEAD, I include state-specific linear time trends $\boldsymbol{\gamma}_i' t$ to account for the possibility that treatment intensity correlates with pre-existing state-level broadband trajectories. For ACP, states with higher poverty rates mechanically have higher eligibility exposure. They may also have followed different broadband price and adoption paths before the program launched. For BEAD, states with more unserved locations tend to be lower-density and lower-income. They may therefore have experienced systematically different infrastructure investment trends before IIJA enactment. The digital divide literature has documented that broadband adoption and pricing trends differ systematically across income and demographic groups [Goldfarb and Prince, 2008, Espín and Rojas, 2024]. Without state-specific trends, such pre-existing divergence would contaminate both sets of estimates. State-specific trends absorb smooth state-level deviations from the national year effects. The treatment coefficients $\hat{\beta}_\tau^{\text{ACP}}$ and $\hat{\beta}_\tau^{\text{BEAD}}$ are then identified from non-linear departures from each state's pre-existing trend that coincide with program rollout.

Unlike standard staggered-adoption event studies, all states are exposed to ACP and BEAD at the same time. Identification comes from cross-state differences in treatment intensity—ACP eligibility shares and BEAD funding intensity—rather than from differences in treatment timing. Consequently, the negative-weighting and contamination problems that arise when already-treated units serve as controls for later-treated units [Sun and Abraham, 2021, Goldsmith-Pinkham et al., 2020] do not apply in this setting. The event-study coefficients therefore measure how outcomes evolve over time in states with higher exposure relative to states with lower exposure.

Assumption 1 (Conditional parallel trends)**.** *After controlling for state fixed effects, year fixed effects, and state-specific linear time trends, states with different treatment intensities* D_i^P *would have experienced the same outcome trends in the absence of the policy. Formally,* $\mathbb{E}[\varepsilon_{it} \mid D_i^P, i, t] = 0$ *for all* $\tau < 0$.

The identifying assumption is that states with higher treatment intensity D_i^P were not on systematically different trajectories for reasons unrelated to the policy. Both FPL200_i and $\log \text{BEAD}_i$ are predetermined by pre-program demographic composition and federal allocation formulas. The main threat is therefore differential broadband trends before launch among states with higher exposure. The pre-period coefficients $\hat{\beta}_\tau^P$ for $\tau < 0$ serve as direct falsification tests of Assumption 1.

A secondary concern is anticipation of program benefits: ISPs or households might alter behavior before formal program launch in expectation of the policy. Under the IIJA timeline, BEAD was announced in November 2021 but allocations were not finalized until 2023. Anticipatory responses would therefore appear as effects at $\tau = 1$–2 (2022–2023), which is largely consistent with what I find: early effects are near zero. For ACP, the predecessor Emergency Broadband Benefit program ran from February to December 2021. Some anticipatory ISP behavior may therefore be absorbed in the $\tau = -1$ base period, which would lead me to understate ACP's total effect. Neither concern appears to drive the main qualitative results.

3.3 ACP: Event Study Results

Figure 3 presents event study estimates from (3.1) for six key outcomes. Event time is indexed relative to 2022 ($\tau = 0$); the dashed vertical line marks November 2021 (IIJA enactment). The omitted period is $\tau = -1$ (year 2021).

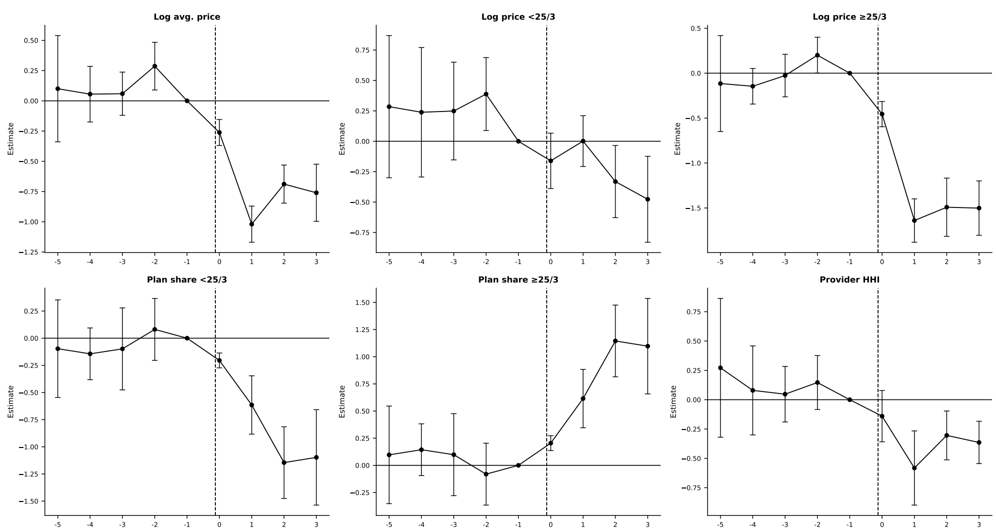

Figure 3: ACP event study estimates. Treatment intensity: share of households below 200% FPL (FPL200_i). TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2022 (ACP launch); the dashed vertical line marks $\tau = 0$. The omitted period is $\tau = -1$ (2021). Bars indicate 95% confidence intervals.

Pre-trends. Pre-period coefficients are statistically indistinguishable from zero across most outcomes, as required by the parallel trends assumption. Conditional on state and year fixed effects and state-specific trends, the pre-launch estimates for high-speed prices, plan shares, and provider HHI are all close to zero. The one exception is the log price of plans below 25/3 Mbps, which runs somewhat high at $\tau = -5$ to $\tau = -3$—though with confidence intervals wide enough to make little of it. That wrinkle is the reason the specification carries state-specific linear trends, which soak up any differential price trajectory across states at different poverty rates before the program begins.

Post-ACP effects. I find that ACP lowered prices and shifted providers toward faster-speed plans. After the 2022 launch, average broadband prices fall in higher-eligibility states. The log-average-price coefficient $\hat{\beta}_1^{\text{ACP}}$ is roughly -1.0 per unit of FPL200 share. With an interquartile range in state eligibility of about 0.10 (0.27 at P25, 0.37 at P75), a high-eligibility state saw prices fall about 10 percent more than a low-eligibility state at the peak ($\tau = 1$). The gap settles near 7 percent through $\tau = 2$–3. The effect is largest for high-speed plans. There, the response of -0.15 to -0.18 implies a 15–18 percent interquartile gap that persists through $\tau = 2$–3.

The product mix shifts in tandem: the low-speed plan share contracts and the high-speed share expands through $\tau = 2$–3, the pattern expected when subsidized households raise demand for faster service and ISPs upgrade their menus to meet it. Provider HHI also declines in more exposed states, though imprecisely.

These estimates are consistent with equilibrium responses to demand expansion. Prices fall and menus tilt toward faster plans in states where ACP enlarges the eligible population, which is inconsistent with a simple mechanical increase in subscriptions at unchanged prices and menus. The reduced form cannot disentangle whether the price movement reflects pass-through, strategic repricing, or composition; the structural model below isolates the demand-expansion channel that drives it.

3.4 BEAD: Event Study Results

Figure 4 presents event study estimates for BEAD. Event time is indexed relative to 2021 (IIJA enactment, $\tau = 0$); the omitted period is $\tau = -1$ (year 2020). The specification follows (3.1) with state and year fixed effects, state-specific linear time trends, plan-count weights, and state-clustered standard errors, as described in Section 3.

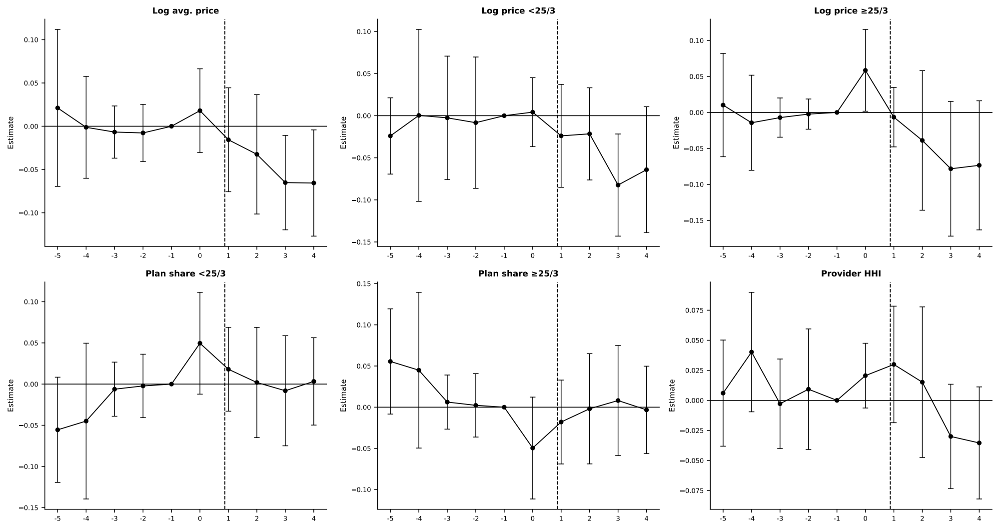

Figure 4: BEAD event study estimates. Treatment intensity: log BEAD allocation per unserved location ($\log \text{BEAD}_i$). TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2021 (IIJA enactment); the dashed vertical line marks $\tau = 0$. The omitted period is $\tau = -1$ (2020). Bars indicate 95% confidence intervals.

Pre-trends. Pre-period coefficients are small and statistically indistinguishable from zero across all outcomes. States that ultimately received more BEAD funding per unserved location were on similar trajectories in prices, technology, and market structure before the program was announced.

Post-BEAD effects. BEAD's estimated effects are smaller and emerge only at the edge of the event window, consistent with the program's construction timeline. The window runs through $\tau = 4$ (2025), and meaningful effects require networks to be built and activated before entrants can affect incumbent prices. Through the first two post-enactment years ($\tau = 1$–2, 2022–2023), every outcome is statistically indistinguishable from zero. Prices begin to move only at $\tau = 3$–4 (2024–2025), falling roughly 0.05–0.07 log points per log unit of BEAD intensity. Provider HHI turns negative over the same window. Plan composition does not respond throughout the window. The timing is informative. Effects appear where the construction timeline predicts, and not before, which points to entry as the operative channel rather than a near-null result. Equally important, the event window closes before the program does. With construction extending through 2026 and beyond, these estimates capture only the earliest funded deployments; the program's full effect is not observable in any event window available today. Quantifying it requires the structural counterfactuals of Section 6, which simulate the completed program against the pre-IIJA baseline and are best read as steady-state effects.

3.5 Summary Difference-in-Differences Estimates

Pooling the post-period into static two-way fixed-effects difference-in-differences estimates (Table F.1, Appendix F) summarizes the average effect and checks the event-study dynamics: genuine dynamic patterns should reappear as significant pooled coefficients of the same sign.

ACP shows significantly lower prices in higher-eligibility states (log average price -0.561, log high-speed price -0.922), a higher high-speed plan share ($+0.390$), and lower HHI (-0.306)—the same price and product-mix adjustments the event study traces. BEAD's six estimates are all small and insignificant, consistent with effects that emerge only in the later years and are muted by the urban URS frame. The flat plan-composition estimates are consistent with entry as the operative channel for BEAD rather than repositioning of incumbent product menus.

3.6 What the Reduced-Form Evidence Establishes

The event studies show that ACP is associated with lower prices and faster plan mixes, while BEAD effects emerge only after a two- to three-year lag. Several questions remain for the structural model. A falling price is consistent with direct pass-through, strategic repricing, or a shifting plan mix. The reduced form cannot distinguish among these mechanisms. State averages also wash out the tract-level variation in market structure, provider presence, and cost that underlies the response. No event study can measure the welfare gains from a policy design that was never implemented. Nor can it decompose welfare into consumer and producer components or recover the entry and product-choice margins behind the observed price and concentration patterns. Standard evaluations that focus on new subscribers or cost per connection miss these equilibrium responses [Goolsbee and Guryan, 2006]. The structural model quantifies them. The two analyses are complementary: the reduced form documents that the programs moved observable outcomes, and the structural model traces the channels and measures counterfactual welfare.

4 Structural Model

4.1 Model Setup and Timing

Markets and products. Markets are indexed by t and firms (or providers) by f. A *market* is a U.S. census tract in 2020, before the implementation of the major IIJA broadband programs.[14] A *product position* is a provider-speed-tier pair (f, g, t), where $g \in \{H, L\}$ denotes the IIJA high-speed tier and low-speed tier.

I distinguish three product sets. The observed set $\mathcal{J}_t^{obs}$ contains provider-tier products actually offered in tract t; these products enter the demand estimation and markup recovery. The potential set $\mathcal{J}_t^0$ contains provider-tier positions that are feasible in tract t. Feasible but unobserved positions are elements of $\mathcal{J}_t^0 \setminus \mathcal{J}_t^{obs}$. They do not enter the demand estimation as active products, but they are needed for entry and fixed-cost identification. The indicator Y_{fgt} equals one if provider f offers tier g in tract t and zero otherwise. The vector $Y_t = \{Y_{fgt}\}_{(f,g)\in\mathcal{J}_t^0}$ summarizes the product configuration. Provider entry in this model is *product positioning*. A provider that already serves some tracts in a county expands its product line by activating a tier in an additional tract or offering a second speed tier in an existing tract. The model thus captures both geographic expansion and product-line decisions within the local market.

The *outside option* is no fixed-broadband subscription. Market size M_t is the number of households in the tract, and the market share of the outside option is $s_{0t} = 1 - \sum_{j\in\mathcal{J}_t^{obs}} s_{jt}$.

Timing and information. The following two assumptions formalize the timing and the information structure.[15]

Assumption 2 (Timing)**.** *The game proceeds in two stages:*

1. *Product-positioning stage. Firms observe demand shifters X_t, cost shifters W_t, rivals' observed footprints, and the fixed-cost shocks ζ_{fgt} for all firms. Based on this information, each firm simultaneously chooses its portfolio $Y_{ft} \in \mathcal{Y}_{ft}$.*

2. *Pricing stage. Conditional on the realized product configuration Y_t, firms observe demand shocks ξ_{jt} and marginal-cost shocks ω_{jt} and simultaneously set monthly plan prices in a complete-information Bertrand-Nash game.*

[14] I use census-tract data and pre-2021 FCC Form 477 deployment data to ensure the baseline predates IIJA programs. This timing is important for the counterfactuals: simulated price discounts and fixed-cost reductions are applied to markets that do not already reflect ACP or BEAD effects.

[15] The timing assumption has been used in several studies. Examples include Fan and Yang [2024] for the brewery industry, Eizenberg [2014] in the personal computer industry, and Hidalgo [2024] for the broadband industry.

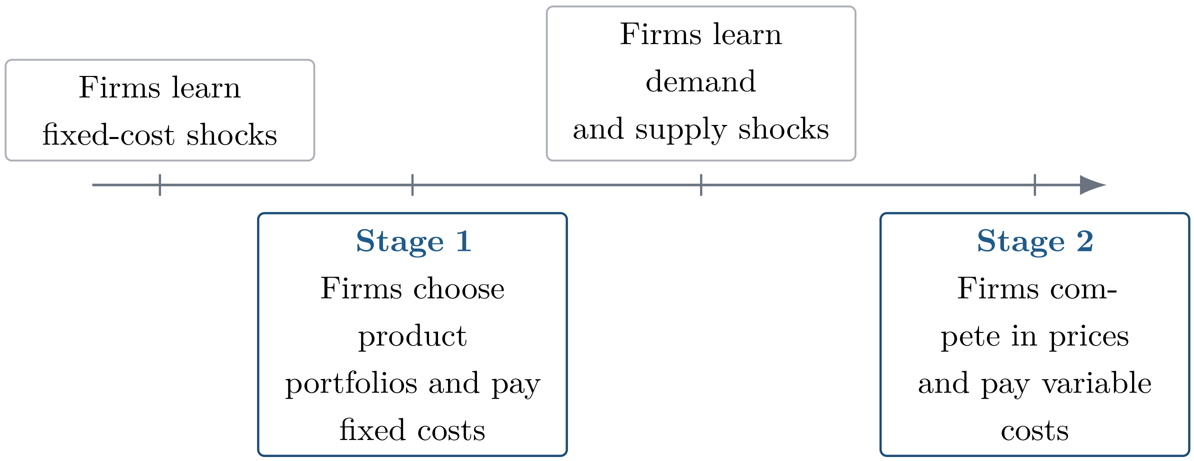

Figure 5: Timing of the structural entry and pricing game.

Assumption 3 (Information)**.** *Firms. At the product-positioning stage, fixed-cost shocks ζ_{fgt}, market characteristics (X_t, W_t), and rivals' potential market footprints are all common knowledge. The demand and cost shocks (ξ, ω) that will be realized in the second-stage pricing game are not yet known; only their joint distribution is common knowledge. Firms therefore compute expected variable profits by integrating over (ξ, ω).*

Econometrician. The shocks (ξ, ω, ζ) are unobserved. Market characteristics (X_t, W_t) and the set of observed product choices $\{Y_{fgt}\}$ are observed.

Splitting portfolio choice from pricing is standard in this literature [Mazzeo, 2002, Seim, 2006, Fan, 2013, Eizenberg, 2014] and reflects a difference in horizons. An ISP commits to building infrastructure months or years before it posts a price list. Complete information on fixed costs is a simplifying assumption, but a plausible one in this setting. Deployment costs depend on local permitting, right-of-way negotiations, and contractor availability that are largely observable within local infrastructure markets. Prices are publicly advertised and monitored by rivals, so a complete-information pricing stage is also a natural approximation.

Firm f's first-stage objective is

$$\max_{Y_{ft} \in \mathcal{Y}_{ft}} \mathbb{E}_{\xi,\omega} \left[r_t^f \left(Y_{ft}, Y_{-ft} \right) \mid I_{ft} \right] - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}, \tag{4.1}$$

where $Y_{ft} = (Y_{fHt}, Y_{fLt})$, Y_{-ft} denotes rivals' portfolios, and $r_t^f(\cdot)$ is variable profit from the pricing game. Because a provider can offer at most two tiers in a tract, $\mathcal{Y}_{ft}$ contains four portfolios: no product, high-speed only, low-speed only, and both tiers.

The obstacle to estimation is dimensionality. With about ten feasible provider-tier positions per tract, the configuration space Y_t contains hundreds to thousands of portfolios. Solving the entry game inside an estimation loop across 70,854 markets is infeasible. The scalable discrete-game method of Fan and Yang [2024] bypasses this problem. Rather than solving the game, it bounds the incremental profit from each position. These bounds imply moment inequalities for the fixed-cost parameters. The game is solved only afterward, when counterfactual policies are evaluated by best response under the new environment.

4.2 Consumer Demand

Household h's indirect utility from product j in market t is

$$U_{hjt} = \delta_{jt} + \nu_{hg} + (1-\rho)\varepsilon_{hjt}, \tag{4.2}$$

where δ_{jt} is mean utility, ν_{hg} is a group-level taste shock, and ε_{hjt} is Type-I extreme value.[16] Mean utility is

$$\delta_{jt} = X'_{jt}\beta + \alpha p_{jt} + \xi_{jt}. \tag{4.3}$$

The vector X_{jt} includes download speed, upload speed, their squares, and a high-speed tier indicator. The parameter α is the marginal utility of price and ξ_{jt} is unobserved quality.

All fixed-broadband products sit in a single inside nest and compete against the outside option of no subscription. The single nest is a deliberate tractability choice, defensible on two grounds. First, it applies uniformly across all 70,854 tracts. A speed-tier nest would be degenerate wherever only one tier is offered: the within-nest share equals one and the structure collapses to logit. Many rural tracts carry only a low-speed tier, so a tier nest would require dropping or special-casing those markets. Second, the single nest yields a unique closed-form Bertrand-Nash price equilibrium. The entry counterfactuals require this tractability because equilibrium prices must be recomputed at every iteration of the positioning game across tens of thousands of markets and fixed-cost draws. Richer structures, such as random coefficients or a second nest, can admit multiple pricing equilibria in multiproduct markets [Nocke and Schutz, 2018]. The Berry inversion [Berry, 1994] gives

$$\log(s_{jt}) - \log(s_{0t}) = X'_{jt}\beta + \alpha p_{jt} + \rho \log(s_{j|g,t}) + \xi_{jt}, \tag{4.4}$$

where $s_{j|g,t} = s_{jt}/(1-s_{0t})$. In the tract-level data, the left-hand side is the CSS mean utility recovered from product-level implied shares.

[16] This paper departs from Fan and Yang [2024], who employ a random-coefficients demand model. Although such models allow for richer substitution patterns, they may generate non-unique pricing equilibria in multiproduct settings [Nocke and Schutz, 2018], thus complicating welfare and entry counterfactuals. Therefore, I use the nested logit model, which delivers a unique closed-form pricing solution and ensures a stable mapping from demand primitives to prices (e.g., Berry [1994]). Tractability is essential for evaluating firms' endogenous product offerings and entry decisions. I nonetheless rely on Fan and Yang [2024]'s framework to recover fixed-cost parameters in the entry stage.

Price and the within-inside share are endogenous. I instrument for both using the quadratic differentiation instruments of Gandhi and Houde [2019]: for characteristic $k \in$ {download speed, upload speed},

$$z^{\text{within}}_{jt,k} = \sum_{l \neq j,\ \text{same tier}} (x_{lt,k} - x_{jt,k})^2, \qquad z^{\text{cross}}_{jt,k} = \sum_{l \neq j,\, \text{other tier}} (x_{lt,k} - x_{jt,k})^2, \tag{4.5}$$

yielding four excluded instruments that measure how differentiated product j is from its rivals in speed-characteristic space. The preferred specification absorbs county and provider fixed effects; identification of $\hat{\alpha}$ and $\hat{\rho}$ comes from within-county, within-provider variation in relative prices and shares across tracts.

4.3 Oligopoly Price Competition

In the pricing stage, all product configurations Y_t are realized and common knowledge. Firm f's variable profit is

$$r^f_t = \sum_{j \in \mathcal{J}^f_t} (p_{jt} - mc_{jt})\, M_t\, s_{jt}(p_t). \tag{4.6}$$

Differentiating with respect to p_{jt} and setting to zero gives the Bertrand-Nash first-order condition:

$$s_{jt} + \sum_{k \in \mathcal{J}^f_t} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0. \tag{4.7}$$

Stacking over all products yields the linear system:

$$s_t + (H_t \odot \Omega_t)(p_t - mc_t) = 0, \tag{4.8}$$

where H_t is the $J_t \times J_t$ matrix of demand derivatives with $(H_t)_{jk} = \partial s_{kt}/\partial p_{jt}$, and Ω_t is the ownership matrix with $(\Omega_t)_{jk} = 1$ if j and k are offered by the same firm. The system is solved for the markup vector:

$$\hat{\mu}^{BN}_t = -(H_t \odot \Omega_t)^{-1} s_t, \tag{4.9}$$

and marginal costs are recovered as $\hat{mc}_{jt} = p_{jt} - \hat{\mu}^{BN}_{jt}$. I then estimate the cost regression

$$\log(\hat{mc}_{jt}) = X'_{jt}\gamma + \omega_{jt}, \tag{4.10}$$

4.4 Product Positioning and Fixed Costs

The supply-side object is a product-positioning decision. A product position is a provider $\times$ speed-tier pair in a census tract. Let f index canonical providers, $g \in \{H, L\}$ index speed tiers, and t index census tracts. The primitive decision is

$$Y_{fgt} = \mathbf{1}\{\text{provider } f \text{ offers speed tier } g \text{ in tract } t\}. \tag{4.11}$$

This definition treats entry as product positioning rather than entry into the national broadband industry. It captures both geographic expansion into a tract and product-line expansion within a tract. For example, a provider that already offers the low-speed tier in tract t enters the high-speed position if it also begins offering the high-speed tier.

Potential product positions. The observed data are organized at the tract $\times$ canonical-provider $\times$ speed-tier level, so each observed row corresponds to $Y_{fgt} = 1$. To estimate fixed costs, I also need feasible positions that are not chosen. I define the potential set using local provider footprints rather than assuming every national provider can enter every tract. Let $c(t)$ denote the county containing tract t. The baseline potential set is

$$\mathcal{J}_t^0 = \{(f, g) : \exists\, t' \in c(t) \text{ such that } Y_{fgt'} = 1\}\,. \tag{4.12}$$

Thus, provider-tier position (f, g) is feasible in tract t if that same provider-tier pair is observed somewhere else in the county. This restriction is economically natural: broadband expansion depends on nearby network presence, local permitting, and rights of way. It also keeps the potential choice set local enough to avoid treating providers with no plausible footprint as potential entrants.

For every feasible but unobserved position in $\mathcal{J}_t^0$, I impute product characteristics using predetermined information: provider-tier-state means, then provider-tier national means, then tier means. Observed positions retain their observed characteristics.

Table 6 reports the implied dimensions. The average tract contains 3.20 observed products but 10.24 potential provider-tier positions. A full market-level portfolio model would require evaluating $2^{|\mathcal{J}_t^0|}$ configurations per market (up to thousands at the 75th percentile), motivating the Fan and Yang [2024] approach below.

The potential product set is used exclusively for the entry model and fixed-cost identification. Demand estimation and markup recovery use only the observed product set.

Table 6: Dimension of Observed and Potential Product Sets

Product-set definition	Mean	P25	Median	P75	P90
Observed products in tract, $\mathcal{J}_t^{obs}$	3.20	2	3	4	5
Observed providers in tract	2.32	1	2	3	4
County-footprint positions, $\lvert\mathcal{J}_t^0\rvert$	10.24	6	9	13	19
County providers	8.05	4	7	10	16

Notes: The unit of observation is a census tract. Observed products are provider-tier products present in the tract. The county-footprint potential set includes provider-tier positions observed somewhere in the same county as the tract.

4.5 Multiple Equilibria and Equilibrium Selection

Product positioning generically has multiple Nash equilibria: a single fixed-cost draw can support several mutually best-response configurations. This multiplicity has been recognized since Bresnahan and Reiss [1991] and Tamer [2003] and is central to Ciliberto et al. [2021] and Fan and Yang [2024]. It normally obstructs estimation because it severs the map from parameters to a unique prediction.

It does not obstruct estimation here. The moment-inequality approach asks only that the observed configuration be *not dominated*, a condition every Nash equilibrium satisfies, so the fixed-cost bounds remain valid whatever selection rule operates in the data. This is the decisive advantage of the Fan and Yang [2024] method over likelihood approaches that must assume a unique equilibrium.

The counterfactual stage does require solving for post-policy equilibria. I handle this as follows. For each market and each draw of fixed costs, I run iterated best-response from three starting points: the observed pre-policy configuration, the empty configuration, and the full feasible configuration.[17] The algorithm is declared converged when no firm can profitably deviate from the current portfolio, checked by a firm-by-firm Nash verification. If multiple fixed points are found from different starting configurations, I retain all of them. The reported national-level bounds combine equilibrium multiplicity and fixed-cost parameter uncertainty. They do so by stacking market-level lower and upper outcomes across equilibria and fixed-cost draws and reporting the 2.5th and 97.5th percentile bounds.

Markets where no fixed point is found within the iteration limit (fewer than one percent of markets) are assigned the pre-policy configuration. This conservative fallback ensures the bounds remain valid. It is likely to understate the response of hard-to-solve markets.

[17] Fan and Yang [2024] consider two starting points: the empty and the full feasible configuration.

Extensive convergence diagnostics are reported in Appendix H (Figure H.2). They show that the algorithm converges in the vast majority of markets and that convergence failures are not systematically correlated with market size or policy intensity.

The main measurement assumptions underlying the structural analysis—inferred product-level market shares, imputed URS prices, and a static 2020 baseline—are discussed in Appendix G, along with supporting robustness checks.

4.6 Identification

Identification proceeds in three sequential blocks. Demand parameters come from the Berry inversion. Marginal costs are a deterministic function of estimated demand through the Bertrand–Nash first-order conditions in (4.7). Fixed costs come from the discrete product-positioning game. Marginal costs require no separate identifying argument. The substantive content lies at the two ends of the model.

Demand. The linearized Berry inversion in (4.4) contains two endogenous regressors. Price p_{jt} correlates with unobserved quality ξ_{jt}. The within-nest log share $\ln s_{j|\text{in},t}$ correlates with ξ_{jt} mechanically and biases $\hat{\rho}$ toward one [Berry, 1994]. I instrument both using the quadratic differentiation instruments of Gandhi and Houde [2019]: for $k \in \{\text{download speed, upload speed}\}$,

$$z_{jt,k}^{\text{within}} = \sum_{l \neq j,\ \text{same tier}} (x_{lt,k} - x_{jt,k})^2, \qquad z_{jt,k}^{\text{cross}} = \sum_{l \neq j,\, \text{other tier}} (x_{lt,k} - x_{jt,k})^2, \tag{4.13}$$

yielding four excluded instruments that measure how differentiated product j is from its rivals in speed-characteristic space. County and provider fixed effects are absorbed prior to estimation; identification therefore comes from within-county, within-provider variation in relative prices and shares across tracts, instrumented by the local competitive configuration of rival speed characteristics. The exclusion restriction is that rivals' speed configurations are determined by pre-existing infrastructure footprints uncorrelated with the product-level demand residual after these absorptions.

Fixed costs: the Fan–Yang moment inequalities. Fixed costs are latent and identified from revealed product-positioning decisions. For each potential position $(f, g, t) \in \mathcal{J}_t^0$, define the lower and upper expected variable-profit bounds:

$$\Delta_{fgt}^{\text{lower}} = \mathbb{E}\Big[\Big(r_t^f(Y_{fgt} = 1,\, Y_{-fgt} = 1) - r_t^f(Y_{fgt} = 0,\, Y_{-fgt} = 1)\Big)\Big], \tag{4.14}$$

$$\Delta_{fgt}^{\text{upper}} = \mathbb{E}\Big[\,\pi_{fgt}^{\text{mono}}\Big], \tag{4.15}$$

where $r_t^f(\cdot)$ denotes variable profit in the pricing game and π_{fgt}^{mono} is monopoly variable profit. Both expectations are approximated by $B = 100$ paired draws of demand and cost shocks $(\hat{\xi}, \hat{\omega})$ from the estimated residual distributions. The bound Δ^{lower} is the incremental variable profit under maximum competition (all feasible rivals active); Δ^{upper} is variable profit under monopoly.

Nash optimality of the observed configuration implies one-sided restrictions on fixed costs. Under the parameterization $FC_{fgt} := W_{fgt}\theta + \sigma_{b(t)}\zeta_{fgt} = \mu_{fgt}^C(\vartheta) + \sigma_{b(t)}\zeta_{fgt}$ with $\zeta_{fgt} \sim N(0,1)$ and $\mu_{fgt}^C(\vartheta) = \theta_{b(t)} + \theta_H\, \mathbf{1}\{g = H\}$:

$$\Phi\left(\frac{\Delta_{fgt}^{\text{lower}} - \mu_{fgt}^C(\vartheta)}{\sigma_{b(t)}}\right) \;\le\; \Pr(Y_{fgt} = 1) \;\le\; \Phi\left(\frac{\Delta_{fgt}^{\text{upper}} - \mu_{fgt}^C(\vartheta)}{\sigma_{b(t)}}\right), \tag{4.16}$$

where $b\,(t) \in \{Small, Medium, Large\}$ based on tertile of market size M_t. This holds for every Nash equilibrium, so no equilibrium-selection assumption is required for estimation [Fan and Yang, 2024, Tamer, 2003, Ciliberto and Tamer, 2009]. Taking expectations of (4.16) against 33 nonnegative weight functions $h^{(k)}(X_{fgt}, W_{fgt})$, $k = 1, \ldots, 33$, yields $K = 66$ unconditional moment inequalities. The weight functions are built hierarchically, from coarse to fine conditioning cells. The first is the constant $h^{(1)} = 1$, which aggregates over all potential positions. Two speed-tier indicators $\mathbf{1}\{g = H\}$ and $\mathbf{1}\{g = L\}$ and three market-size indicators $\mathbf{1}\{b(t) = s\}$, $s \in \{\text{Small}, \text{Medium}, \text{Large}\}$, then condition on each dimension separately. The remaining 27 functions interact market size with the location of the expected-profit bounds. Let $q_1 < q_2 < q_3$ denote empirical quantiles of the pooled distribution of the profit bounds. Nine functions of the form $\mathbf{1}\{b(t) = s\} \cdot \mathbf{1}\{\Delta_{fgt}^{\text{lower}} > q_\ell\}$ select positions with high competitive lower bounds, where the lower inequality in (4.16) is most restrictive. Nine functions of the form $\mathbf{1}\{b(t) = s\} \cdot \mathbf{1}\{\Delta_{fgt}^{\text{upper}} < q_\ell\}$ select positions with low monopoly upper bounds, where the upper inequality is most restrictive. Nine bracketing functions of the form $\mathbf{1}\{b(t) = s\} \cdot \mathbf{1}\{q_\ell < \Delta_{fgt}^{\text{lower}} \le \Delta_{fgt}^{\text{upper}} < q_{\ell'}\}$, $\ell < \ell'$, select positions whose bounds fall inside a common interval, where the sharpness of observed entry behavior is most informative about the dispersion parameters σ_b. Conditioning on progressively finer cells in this way converts the single conditional restriction in (4.16) into a vector of unconditional moments that retains identifying power against parameters binding only in subpopulations of positions. Moments are averaged first within tracts, to respect strategic dependence within a market, and then across tracts, which are independent markets. The seven structural parameters $(\theta_H, \theta_{\text{Small}}, \theta_{\text{Medium}}, \theta_{\text{Large}}, \sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}})$ are disciplined jointly. Cross-market-size variation in offering rates identifies θ_b. Within-size-bin comparisons of high- versus low-speed positions identify θ_H. The sharpness with which entry responds to changes in the profit-change bounds identifies σ_b.

Heuristic identification of fixed-cost shocks. Write the fixed cost for product speed $s \in \{L, H\}$ in market-size bin $b \in \{S, M, Lg\}$ as

$$F_{b,s} = \theta_b + \theta_H \mathbf{1}\{s = H\} + \sigma_b \, \varepsilon,$$

where ε is a market-level shock common to all firms in a given market. The mean fixed costs therefore take the form

	Low speed	High speed
Small	θ_S	$\theta_S + \theta_H$
Medium	θ_M	$\theta_M + \theta_H$
Large	θ_{Lg}	$\theta_{Lg} + \theta_H$

The parameters are identified as follows. The market-size intercepts θ_b are identified by the central tendency of product-offering rates across markets within each bin: bins where entry is systematically more prevalent imply lower mean fixed costs. The high-speed premium θ_H is identified by comparing entry into high- versus low-speed products within the same bin, where both speed tiers share the common intercept θ_b, so that any systematic difference in offering rates is attributable solely to θ_H. The dispersion parameters σ_b are identified by cross-market variation in offering rates within each bin: bins where entry rates are nearly uniform across markets imply low σ_b, while bins where some markets see broad entry and others see little entry imply high σ_b. Because ε is a market-level shock, within-market variation across firms contributes no information about σ_b; identification relies entirely on the across-market distribution of entry outcomes within each bin. All parameters are estimated jointly: a candidate vector $(\theta_S, \theta_M, \theta_{Lg}, \theta_H, \sigma_S, \sigma_M, \sigma_{Lg})$ is retained only if it satisfies all moment inequalities simultaneously.

Andrews–Soares test inversion and confidence set. The identified set is estimated by inverting the modified method-of-moments test of Andrews and Soares [2010]. Let

$$\bar{Z}_M(\vartheta) = M^{-1} \sum_{m=1}^{M} Z_m(\vartheta)$$

be the sample moment vector at candidate ϑ, where $M = 70{,}854$ tract markets. The test statistic penalizes only violated inequalities:

$$T_M(\vartheta) = \sum_{k=1}^{K} \left[\frac{\sqrt{M}\, \bar{Z}_{M,k}(\vartheta)}{\hat{\sigma}_k(\vartheta)} \right]_+^2, \qquad [x]_+ \equiv \max\{x, 0\}, \tag{4.17}$$

where $\hat{\sigma}_k(\vartheta)$ is the empirical standard deviation of the k-th moment across markets. The 95 % confidence set is the collection of parameter vectors that pass the test:

$$\hat{\mathcal{C}} = \Big\{ \vartheta : T_M(\vartheta) \leq \hat{c}_M(\vartheta, 0.95) \Big\}, \tag{4.18}$$

where $\hat{c}_M(\vartheta, 0.95)$ is the generalized moment selection (GMS) critical value of Andrews and Soares [2010]. The GMS critical value adapts to the local geometry of the inequalities at each ϑ. Moment conditions that are comfortably satisfied receive conservative slack. This inflates $\hat{c}_M$ and makes the test easy to pass in the interior of the identified set. Binding or nearly binding inequalities instead receive tight critical values.

Formally, let

$$\bar{Z}^*_M(\vartheta) = \frac{1}{M} \sum_{m=1}^{M} Z^*_m(\vartheta)$$

denote the bootstrap sample mean of the moment vector, where $\{Z^*_m(\vartheta)\}_{m=1}^M$ is obtained by resampling the M tract markets with replacement. The critical value $\hat{c}_M(\vartheta, 0.95)$ is estimated from 999 simulated draws of the bootstrap empirical process

$$\sqrt{M}\Big(\bar{Z}^*_M(\vartheta) - \bar{Z}_M(\vartheta) \Big).$$

For each draw, the Andrews–Soares generalized moment selection (GMS) procedure is applied to determine which inequalities are binding or nearly binding, and the corresponding bootstrap test statistic is computed. The 95th percentile of the resulting distribution of bootstrap test statistics is taken as $\hat{c}_M(\vartheta, 0.95)$. This guarantees asymptotic validity—$\Pr(\vartheta_0 \in \hat{\mathcal{C}}) \geq 1 - \alpha$ for the true parameter value ϑ_0—while remaining sharper than a conservative approach that applies a common chi-squared critical value to all 66 inequalities simultaneously. The procedure below mimics the computational simplification used in Fan and Yang [2024].

Computational implementation. Direct evaluation of the confidence set $\hat{\mathcal{C}}$ on a dense grid over the seven-dimensional parameter space is computationally infeasible. I therefore approximate $\hat{\mathcal{C}}$ using a stochastic search algorithm. First, a reference point $\hat{\vartheta}$ is obtained by minimizing the criterion $Q_M(\vartheta) = T_M(\vartheta)$. I use the L-BFGS-B algorithm from seven starting values spanning the admissible parameter space, subject to the constraint $\sigma_b \geq 0$. Starting from $\hat{\vartheta}$, I run 50 independent local random-walk chains of 50 iterations each.[18] At each iteration, a candidate parameter vector is generated by adding a random perturbation to the current point. The proposal is retained whenever it satisfies the confidence-set membership condition $T_M(\vartheta) \leq \hat{c}_M(\vartheta, 0.95)$. This local exploration identifies a connected region of accepted parameter vectors around $\hat{\vartheta}$. To explore more distant portions of the confidence set, I next initiate 100 additional chains of 50 iterations from accepted points near the boundary of the locally explored region. These chains use perturbation scales five times larger than those employed in the local search. The larger steps allow the algorithm to trace the outer boundary of $\hat{\mathcal{C}}$ and discover disconnected or weakly connected regions. Across both stages, the search yields 656 accepted parameter vectors. For counterfactual simulations, I select 50 representative parameter vectors from the accepted set. Specifically, I project the 656 accepted vectors onto their first principal component and choose 50 approximately evenly spaced points along this direction. This procedure ensures that the simulation draws span the full extent of the estimated confidence region rather than clustering near its center. Reported ($[Q_{2.5}, Q_{97.5}]$) intervals across these 50 draws therefore incorporate both fixed-cost parameter uncertainty and equilibrium multiplicity uncertainty. Figure H.1 in Appendix H visualizes the procedure.

Universal dominance and its implications for identification. All 727,371 positions with valid profit bounds (of the 727,601 potential positions in the entry sample) satisfy $\Delta^{\text{lower}}_{fgt} > 0$. Every potential product earns positive incremental variable profit even under maximum competition; no position is dominated ($\Delta^{\text{upper}} < 0$). Economically, this reflects the high-margin structure of broadband in variable terms. For identification, universal dominance has a precise consequence. Since dominated-action logic never fires, the inequalities in (4.16) identify fixed costs entirely from the upper-bound side. Observed products reveal $FC_{fgt} \leq \Delta^{\text{upper}}_{fgt}$; non-offered products reveal $FC_{fgt} > \Delta^{\text{lower}}_{fgt} > 0$.

From demand to fixed costs. The two identification blocks are sequential. The profit-change bounds Δ^{lower} and Δ^{upper} are functions of the estimated demand system. Errors in $(\hat{\alpha}, \hat{\rho})$ therefore propagate into the moment inequalities. Following Eizenberg [2014], Fan and Yang [2024], and Aguirregabiria et al. [2024], demand estimates are treated as given when constructing the bounds. The strong first stage and tight demand estimates limit this propagation. The width of $\hat{\mathcal{C}}$ primarily reflects the partial-identification structure of the entry game rather than demand imprecision.

[18]Each proposal is drawn from a multivariate normal distribution centered at the current point with diagonal covariance. All components are perturbed independently.

5 Estimation Results

Estimation proceeds in the order the model is identified. Demand comes first, from the Berry inversion in (4.4), instrumenting price and the within-inside share with Gandhi and Houde [2019] quadratic differentiation instruments (4.13) and absorbing county and provider fixed effects. The demand estimates and the Bertrand–Nash conditions in (4.7) then recover tract-level markups and marginal costs through (4.9). Fixed costs come last, as bounds from the Fan and Yang [2024] positioning inequalities, requiring only that observed choices be undominated. The identification argument underlying each step is in Section 4.6; Appendix H details the fixed-cost moments and weight matrix, and Appendix H.2 gives the formal construction of the Andrews–Soares confidence set.

5.1 Consumer Demand

Table 7 reports four specifications. Columns (1) and (2) are OLS; columns (3) and (4) instrument for price and the within-inside share with Gandhi and Houde [2019] quadratic differentiation instruments. Even-numbered columns absorb county and provider fixed effects. Column (4) is the preferred specification.

The four columns illustrate that neither correction alone is sufficient. Column (1) yields $\hat{\rho} = 1.183$, above the admissible upper bound, because unobserved quality ξ_{jt} is positively correlated with the within-nest share. Adding county and provider fixed effects in column (2) reduces $\hat{\rho}$ only marginally, to 1.029. Absorbing fixed effects alone therefore does not resolve the quality-price correlation. Column (3) instruments for price and the within-share without absorbing fixed effects. The price coefficient reverses sign to +3.261 because the Gandhi and Houde [2019]instruments generate variation correlated with uncontrolled unobserved quality, producing an upward-biased estimate. Column (4), which instruments and absorbs fixed effects simultaneously, satisfies standard regularity conditions: $\hat{\alpha} = -1.491^{***}$, $\hat{\rho} = 0.863^{***}$. Each correction in isolation addresses one source of bias while leaving the other unresolved; both are required.

The nesting parameter $\hat{\rho} = 0.863$ indicates strong within-broadband substitution. Households treat fixed-broadband products as far closer substitutes for one another than for the outside good of no subscription. This is economically sensible: consumers who want internet service tend to compare plans rather than choose between internet and no internet. It is also essential for the counterfactual welfare calculations, which require demand responses to entry and price changes to be concentrated within the broadband market. Download speed enters positively and concavely, consistent with diminishing marginal utility from very high speeds. Upload speed is small and imprecise once county and provider effects are absorbed.

Table 7: Nested Logit Demand Estimates

	OLS		2SLS (GH instruments)	
	(1)	(2)	(3)	(4)
Price / \$100 [$\hat{\alpha}$]	-0.344^{***}	-0.127^{***}	$+3.261^{***}$	-1.491^{***}
	(0.016)	(0.023)	(0.182)	(0.471)
Within-share [$\hat{\rho}$]	1.183^{***}	1.029^{***}	1.242^{***}	0.863^{***}
	(0.004)	(0.005)	(0.043)	(0.033)
Download speed (Gbps)	0.366^{***}	0.015	-0.243^{***}	0.223^{***}
	(0.009)	(0.017)	(0.053)	(0.043)
Download speed^2	-0.052^{***}	-0.017^{***}	0.013	-0.041^{***}
	(0.005)	(0.006)	(0.019)	(0.008)
Upload speed (Gbps)	-0.012	0.087^{***}	0.679^{***}	-0.003
	(0.008)	(0.015)	(0.043)	(0.024)
Upload speed^2	0.018^{***}	0.009	-0.053^{***}	0.024^{***}
	(0.005)	(0.006)	(0.019)	(0.008)
Tier H indicator	0.027^{***}	n.s.	-0.622^{***}	n.s.
	(0.009)		(0.084)	
N	170,419			
R^2	0.452	0.256	—	—
County FEs	No	Yes	No	Yes
Provider FEs	No	Yes	No	Yes
Condition satisfied ($\hat{\rho} < 1$)	No	No	No	Yes
Mean own-price elasticity	+0.83	+1.96	−5.91	−4.99

Notes: Dependent variable: $\delta_{jt} = \ln(s_{jt}/s_{0t})$. HC1 robust standard errors in parentheses for OLS; heteroskedasticity-robust for 2SLS. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. n.s. = absorbed by fixed effects. GH instruments: four Gandhi and Houde [2019]) quadratic differentiation instruments on download and upload speed, computed within- and across-tier for each product (equation (4.13)). County and provider fixed effects absorbed by iterative two-way demeaning before estimation. Column (4) is the preferred specification and the only one satisfying the standard regularity condition $\hat{\rho} \in [0, 1)$ [Berry, 1994]. Positive mean elasticities in columns (1)–(2) reflect standard condition violations ($\hat{\rho} > 1$); positive price coefficient in column (3) reflects quality bias without fixed effects.

The mean own-price elasticity in the preferred specification is -4.99:

$$\varepsilon_{jj} = \frac{\hat{\alpha}}{100}\, p_{jt} \left[\frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}}\, s_{j|\text{in},t} - s_{jt} \right]. \tag{5.1}$$

Comparison with related estimates. The mean own-price elasticity of -4.99 falls within the range of recent product-level broadband and pay-TV estimates. Kearns [2024] reports -4.2 to -4.7, Goetz [2019] finds -5.9, and Crawford et al. [2019] finds -4.1 to -5.4 in cable television. The magnitude reflects the product-substitution margin within a local market rather than the extensive-margin adoption decision. Participation-margin estimates in the broadband literature (-1 to -4, Rosston et al. 2010, Nevo et al. 2016) are naturally smaller. An alternative state-level aggregation yields a mean elasticity of -5.87, confirming that the tract-level share construction does not mechanically drive the result.

Elasticity heterogeneity across market types. Own-price elasticities vary with market shares via (5.1), and this heterogeneity matters for the welfare comparison between programs. At a product share of $s_{jt} = 0.10$ with $\hat{\rho} = 0.863$, the formula yields an elasticity of approximately -2.1. At $s_{jt} = 0.45$, it yields approximately -8.6. Low-share markets, such as rural tracts with few products and high outside-option shares, are relatively inelastic. High-share markets, such as dense urban tracts, are highly elastic. ACP delivers the largest consumer surplus gains where elasticity is highest, because more households are at the margin of subscribing and will respond to a price discount. BEAD acts through fixed-cost relief. Its gains are therefore less sensitive to demand elasticity and depend instead on the gap between variable profits and fixed costs in currently unserved tracts. This difference reinforces the geographic complementarity between the two programs documented in Section 6.3.

Instrument strength. Table 8 reports the Gandhi and Houde [2019] excluded-instrument coefficients and weak-identification diagnostics for column (4). The signs are economically transparent. Greater within-tier characteristic dispersion raises a product's own price because slack competitive pressure permits a premium. Greater cross-tier dispersion lowers the within-nest share because more distinct products capture fewer within-broadband switchers. The conditional Sanderson–Windmeijer F-statistics are 58.3 for price and 544.2 for the within-share, both well above the Stock–Yogo 10% critical value of 7.56 (2 endog., 4 instruments). The Cragg–Donald minimum eigenvalue of 986.2 confirms system-level identification.

Table 8: First-Stage Diagnostics: Preferred 2SLS Specification (Column 4)

	Price / \$100	$\ln s_{j\|\text{in},t}$
Gandhi-Houde excluded instruments		
Within-tier $\sum(x_d - x_{d'})^2$ (download)	$+0.0027^{***}$	-0.0010
	(0.0003)	(0.0022)
Cross-tier $\sum(x_d - x_{d'})^2$ (download)	-0.0073^{***}	-0.0532^{***}
	(0.0015)	(0.0128)
Within-tier $\sum(x_u - x_{u'})^2$ (upload)	-0.0024^{***}	-0.0045^{**}
	(0.0003)	(0.0020)
Cross-tier $\sum(x_u - x_{u'})^2$ (upload)	$+0.0070^{***}$	$+0.0508^{***}$
	(0.0015)	(0.0128)
Exogenous controls (speed, speed^2)	Yes	Yes
County fixed effects	Yes	Yes
Provider fixed effects	Yes	Yes
First-stage R^2	0.031	0.107
Angrist–Pischke Wald stat. ($\chi^2(4)$)	166.8	2,174.2
Sanderson–Windmeijer conditional F	58.3	544.2
Shea partial R^2	0.0054	0.0233
Cragg–Donald minimum eigenvalue stat.	986.2	

Notes: x_d and x_u denote download and upload speed (Gbps). Within-tier sums are over same-market same-tier rivals; cross-tier sums are over same-market other-tier rivals. Exogenous controls include download speed, download^2, upload speed, upload^2; county and provider FEs absorbed by iterative two-way demeaning. The Sanderson–Windmeijer conditional F tests one endogenous variable conditional on the other. Stock–Yogo (2002) 10% critical value ≈ 7.56 (2 endog., 4 instruments). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. HC1 robust standard errors.

5.2 Marginal Costs and Markups

The preferred supply specification (column 1 of Table 9) excludes fixed effects because the identifying variation in broadband costs runs across providers, technologies, and speed tiers. Rich fixed effects absorb precisely those dimensions. Column (2), with county and provider fixed effects, reverses signs on upload speed and drives RMSE sharply lower. This suggests that the fixed effects over-absorb the cross-sectional cost variation. The no-FE specification preserves economically interpretable estimates.

Table 9: Marginal-Cost Regression: $\ln(\hat{mc}_{jt}) = W'_{jt}\gamma + \omega_{jt}$

	(1) No FE *Preferred*	(2) Cty+Prov FE
Constant	-1.115^{***}	—
	(0.006)	
Download speed (Gbps)	0.242^{***}	0.141^{***}
	(0.010)	(0.010)
Download speed^2 (Gbps^2)	-0.026^{***}	-0.026^{***}
	(0.007)	(0.007)
Upload speed (Gbps)	-0.279^{***}	-0.052^{***}
	(0.009)	(0.009)
Upload speed^2 (Gbps^2)	0.029^{***}	0.017^{**}
	(0.007)	(0.007)
Tier H indicator	0.529^{***}	n.s.
	(0.006)	
N (products, $\hat{mc} > 0$)	154,032	154,032
R^2	0.204	0.012
RMSE	0.369	0.203
County + Provider FEs	No	Yes

Notes: Dependent variable $\ln(\hat{mc}_{jt})$ where $\hat{mc}_{jt} = p_{jt} - \hat{\mu}^{BN}_{jt}$; restricted to products with positive implied marginal costs (90.4% of the estimation sample). HC1 robust standard errors. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Tier H and the constant are collinear with county and provider FEs in column (2) and are absorbed. Column (1) is used to impute marginal costs for all 727,601 potential product positions in the entry sample.

The preferred specification implies that download speed raises marginal cost at a diminishing rate. The Tier H premium of $\exp(0.529) - 1 \approx 70$ percent reflects the substantially higher infrastructure investment required for high-speed broadband relative to legacy low-speed service. Upload speed is negatively related to marginal cost, consistent with fiber-grade networks combining high upload capacity with newer, more cost-efficient physical plant. The constant $\exp(-1.115) \approx \$33$ per month represents the baseline variable cost at zero measured speeds. High-speed products average approximately \$61.5 per month in marginal costs once the Tier H premium is applied. Of the 170,419 products in the estimation sample, 16,387 (9.62 %) yield negative implied marginal costs. These are concentrated among multiproduct firms, a well-documented feature of Bertrand–Nash markup recovery in differentiated-products markets [Nevo, 2001].

Conditional on positive marginal costs, the median Lerner index is 20.0 % and the mean 22.2 %. These estimates provide a useful internal check. For single-product firms, the first-order condition collapses to the inverse-elasticity rule. The mean Lerner of 20.4 % is close to $1/|\bar{\varepsilon}| = 1/4.99 = 20.0\,\%$ at the sample-mean elasticity.[19] Multiproduct firms mark up substantially more (mean 32.2 %, median 27.8 %). This is consistent with a firm that sells both tiers in a tract internalizing cannibalization between them. The distinction matters for the counterfactuals. Entry by a new provider disciplines prices in a way that adding a second tier by an incumbent does not, which is the margin on which BEAD operates.

Comparison with related estimates. The median Lerner index of 20.0 % is plausible relative to estimates in related markets. Nevo [2001] finds Lerner indices of 16–45 % for ready-to-eat cereals. Crawford et al. [2019] find similar magnitudes in cable television. Mean marginal costs of approximately \$59.5 per month are economically plausible given the capital intensity of fixed broadband infrastructure, and the Tier H–Tier L cost gap (\$61.5 versus \$38.4) reflects the heavier investment that faster connections require.

Tables 10 and 11 report the full distribution of markups, marginal costs, and Lerner indices by speed tier and firm type.

Several patterns merit discussion. High-speed (Tier H) products have lower Lerner indices than low-speed (Tier L) products (median 0.195 vs. 0.278). This is consistent with high-speed products facing greater effective competition: fiber and modern cable face more competing alternatives than legacy DSL. Tier L products, largely legacy DSL, exhibit higher markups because they operate as de facto monopolists in low-density areas where no competing technology exists. The substantially elevated Lerner indices for multiproduct firms (median 0.278 vs. 0.191 for single-product) reflect the internalization of cross-product cannibalization. A firm offering both tiers prices both higher than independent firms would [Nevo, 2001]. This has a direct counterfactual implication: BEAD-induced entry by a new provider generates competitive pressure that an incumbent adding a second tier does not.

Figure 6 provides key internal validity checks. Demand shocks are centered near zero and symmetric, confirming that the Gandhi and Houde [2019] instruments are uncorrelated with the demand residual after county and provider absorptions. Marginal-cost shocks are likewise

[19]The comparison is approximate: the inverse-elasticity benchmark uses the all-product mean elasticity; a tighter check would use the single-product-firm mean elasticity directly.

Table 10: Bertrand–Nash Markup Distribution by Speed Tier ($\hat{mc} > 0$ sample)

	All	Tier H	Tier L
Observations ($\hat{mc} > 0$)	154,032	140,303	13,729
% negative $\hat{mc}$ (full)	9.62%	—	—
Panel A: Markup $\hat{\mu}$ (\$/month)			
Mean	15.3	15.2	16.7
Median	14.3	14.2	15.3
P10	11.2	11.2	10.8
P90	20.1	19.8	23.8
Panel B: Marginal Cost $\hat{mc}$ (\$/month)			
Mean	59.5	61.5	38.4
Median	57.5	58.9	43.0
P10	35.2	37.5	15.4
P90	82.0	82.8	66.5
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$			
Mean	0.222	0.211	0.336
Median	0.200	0.195	0.278
P10	0.131	0.127	0.109
P90	0.326	0.305	0.584

Notes: Tier H: download ≥ 10 Mbps and upload ≥ 1 Mbps; Tier L: below. Markups and marginal costs in \$/month (estimation in \$/100, multiplied by 100). Lerner index $\hat{L} = \hat{\mu}/p$; computed on $\hat{mc} > 0$ subsample (90.4% of the 170,419 product estimation sample). Full sample has 9.62% negative implied marginal costs, concentrated in multiproduct firms.

symmetric. The markup distribution is right-skewed, consistent with the higher markups of multiproduct firms documented in Table 11. The Lerner median of 0.200 matches the inverse-elasticity benchmark $1/4.99 = 0.200$ evaluated at the mean own-price elasticity.

Table 11: Markup and Lerner Index by Firm Type ($\hat{mc} > 0$ sample)

	Single-Product	Multi-Product
Observations	143,333	27,086
% negative $\hat{mc}$	8.87%	13.54%
Panel A: Markup $\hat{\mu}$ (\$/month)		
Median	14.49	17.49
P90	23.99	223.61
Panel B: Marginal Cost $\hat{mc}$ (\$/month)		
Mean	62.41	42.96
Median	60.01	47.29
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$		
Mean	0.204	0.322
Median	0.191	0.278
P25	0.153	0.214
P75	0.240	0.381
P90	0.296	0.534

Notes: Single-product firms offer one speed tier in the tract; multiproduct firms offer both tiers. Markup mean is dominated by the right tail of products with near-zero marginal costs; the median is the more informative central tendency. Lerner index restricted to $\hat{mc} > 0$ subsample.

5.3 Fixed-Cost Estimates

I parameterize fixed costs with a market-size intercept, a high-speed premium, and market-size-specific shock dispersion:

$$FC_{fgt} = \underbrace{\theta_{b(t)}}_{\text{size intercept}} + \underbrace{\theta_H \mathbf{1}\{g = H\}}_{\text{speed premium}} + \underbrace{\sigma_{b(t)}\, \zeta_{fgt}}_{\text{position-level heterogeneity}}, \qquad \zeta_{fgt} \sim N(0,1), \tag{5.2}$$

where $b(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$ is the market-size tertile of tract t and $g \in \{H, L\}$ is the speed tier. There are seven structural parameters. The parameters $\theta_{\text{Small}}, \theta_{\text{Medium}}, \theta_{\text{Large}}$ capture size-specific mean fixed costs. The parameter θ_H captures the high-speed premium and is unconstrained in sign. The parameters $\sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}}$ capture size-specific shock dispersions. The identification logic follows from Section 4.6. Cross-tertile variation in offering

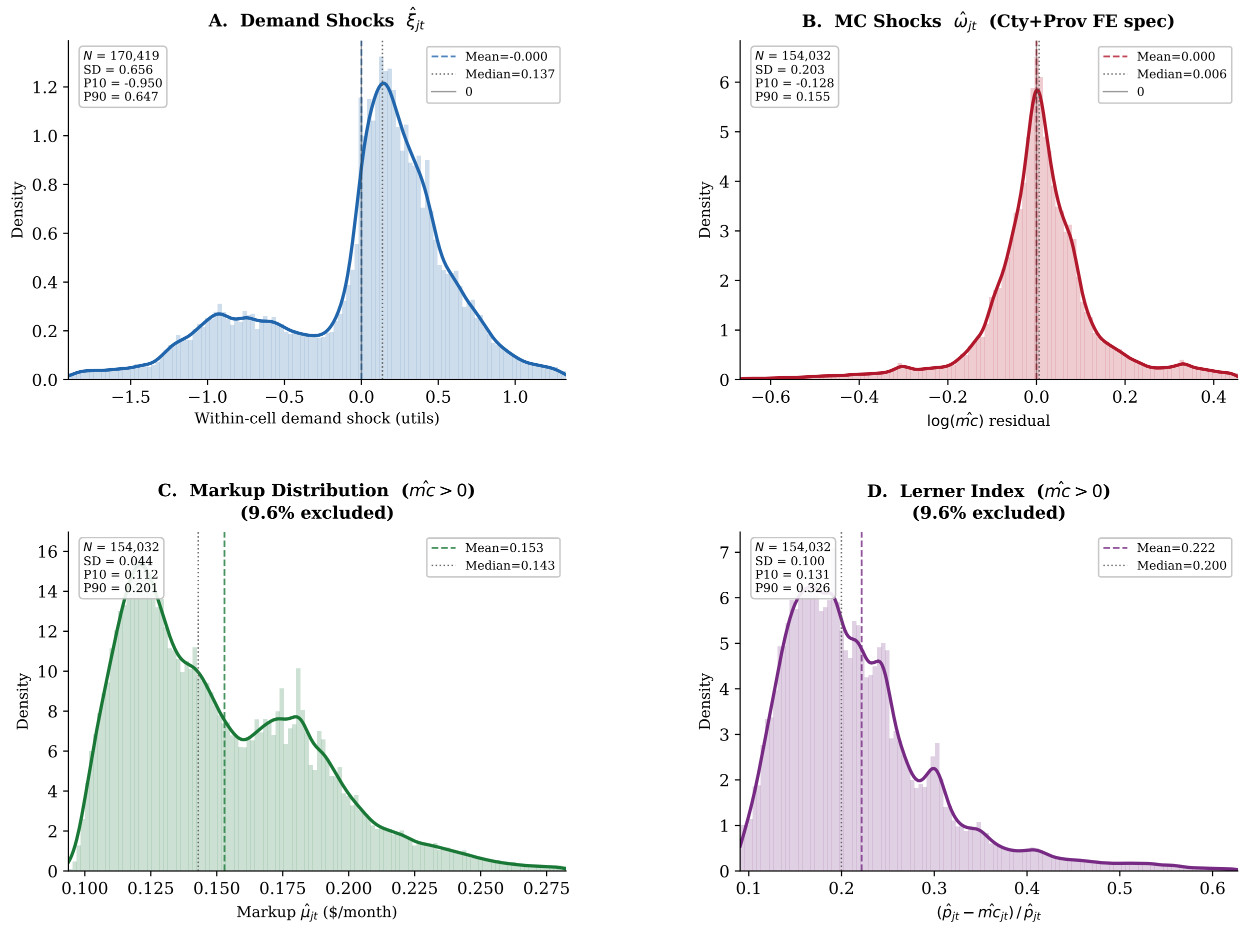

Figure 6: Structural diagnostics from the preferred demand and supply specification (2SLS, County and Provider FEs, Gandhi and Houde [2019] instruments). *Panel A*: kernel density of demand shocks $\hat{\xi}_{jt}$; approximately symmetric and centered near zero, consistent with the Gandhi and Houde [2019] orthogonality conditions. *Panel B*: kernel density of marginal-cost shocks $\hat{\omega}_{jt}$; symmetric and centered near zero. *Panel C*: Bertrand–Nash markup distribution; right-skewed with median near \$14.3, the elevated right tail driven by multiproduct firms. *Panel D*: Lerner index distribution; median 0.200, consistent with the inverse-elasticity benchmark $1/|\bar{\varepsilon}| = 1/4.99 = 0.200$.

rates disciplines θ_b. Within-tertile H vs. L comparisons discipline θ_H. The sharpness of the entry threshold disciplines σ_b.

Figure 7 reports profit-bound diagnostics. The median tightness ratio $\Delta^{\text{lower}}/\Delta^{\text{upper}}$ is only 0.017, indicating that competitive-profit lower bounds are small relative to monopoly-profit upper bounds. Observed positions have systematically higher ratios than unobserved positions, consistent with firms selecting into more profitable opportunities. Both bounds increase with market size, but upper bounds rise much more rapidly, reflecting the greater monopoly-profit potential of large markets. Finally, nearly all positions satisfy $\Delta^{\text{lower}} > 0$

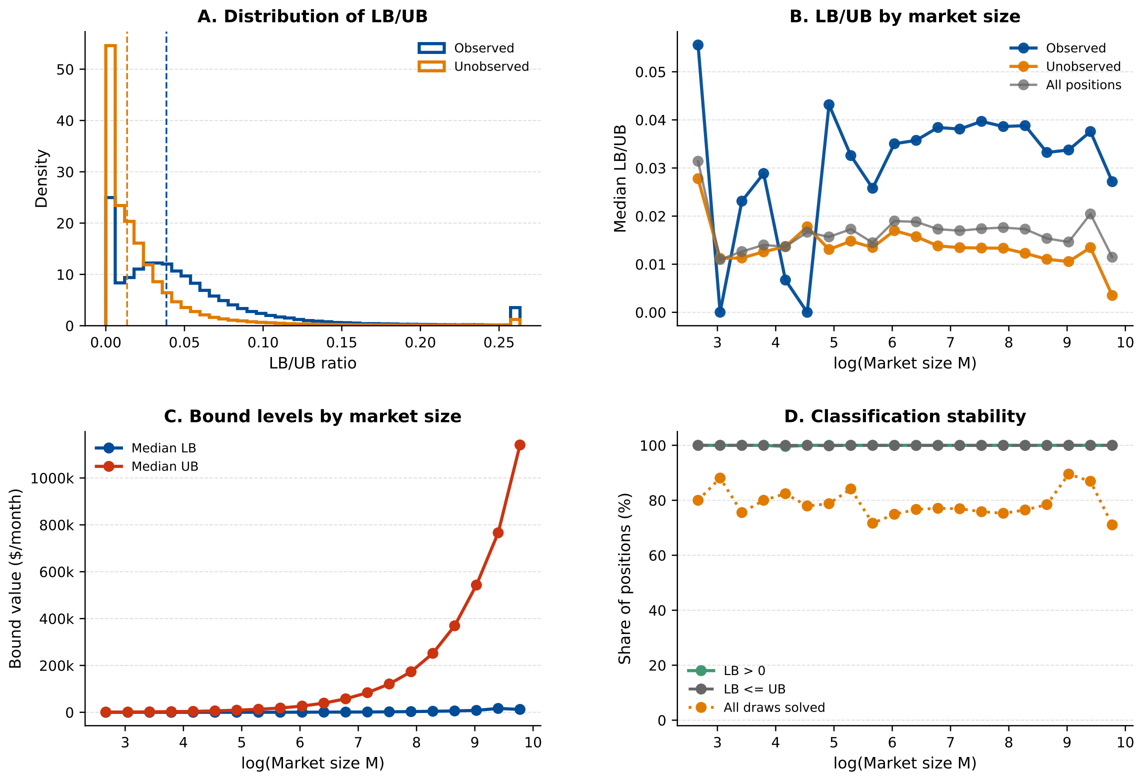

Figure 7: Fan–Yang expected variable-profit bound diagnostics. Observed positions exhibit systematically higher bound-tightness ratios than unobserved positions. Bound levels increase strongly with market size, especially on the upper-bound side. All valid positions satisfy $\Delta^{\text{lower}} > 0$ and $\Delta^{\text{lower}} \leq \Delta^{\text{upper}}$, implying no dominated positions and identification driven primarily by upper-bound restrictions.

and $\Delta^{\text{lower}} \leq \Delta^{\text{upper}}$, implying that no position is dominated. Identification therefore comes primarily from the upper-bound restriction on fixed costs. Appendix I reports additional diagnostics.

Table 12 reports the 95 % Andrews–Soares confidence set projected onto each of the seven structural parameters.

Table 12: Fixed-Cost Parameter Bounds: 95 % Andrews–Soares Confidence Set

	95 % projected bound ($/month)
θ_H (high-speed premium)	[−53,843; 25,081]
Market-size intercepts (θ_b)	
θ_{Small}	[7,357; 74,287]
θ_{Medium}	[5,155; 84,040]
θ_{Large}	[5,832; 82,783]
Market-size dispersion (σ_b)	
σ_{Small}	[0; 41,542]
σ_{Medium}	[0; 45,221]
σ_{Large}	[0; 37,891]

Note: Bounds are coordinate-wise projections of the 95 % confidence set $\hat{\mathcal{C}}$ (equation (4.18)) onto each structural parameter, constructed following Andrews and Soares [2010]. Search: 50 local Markov chains + 100 expansion chains; 656 accepted vectors; 999 bootstrap draws for GMS critical values. Lower bounds on σ_b touch zero, consistent with a nearly deterministic fixed-cost structure not being ruled out by the data. 50 representative parameter vectors from $\hat{\mathcal{C}}$ are used for all counterfactual simulations. Estimates in 2025 US dollars.

The monthly bounds correspond to the flow equivalent of annualized capital and operating expenditure per product position. The mean intercepts run from roughly $5,000 to $84,000 per month across size tertiles ($62,000–$1,008,000 a year). These magnitudes are consistent with structural estimates of broadband deployment costs. From revealed deployment decisions in Seattle, Kearns [2024] recovers annual fixed costs of roughly $404,000–$710,000 per census block for de novo entry depending on technology—the same order of magnitude as the annualized intercepts here—and notes that revealed-preference cost estimates exceed engineering-based industry figures because they embed opportunity costs and regulatory barriers, a point that applies equally to these bounds. They are also consistent with industry estimates of the cost of keeping fixed-broadband plant in service.[20] The intervals reflect

[20] https://businessplan-templates.com/blogs/running-costs/internet-service-provider

partial identification rather than sampling imprecision. The moment inequalities constrain fixed costs from above and below without forcing equality.

The 95 % projection for θ_H, [−\$53,843; \$25,081] per month, straddles zero and reaches into negative values. The data cannot reject that high-speed service has lower ongoing fixed costs than low-speed service once market size is held fixed. Modern fiber is expensive to build but inexpensive to maintain; aging copper DSL is the reverse. With construction costs sunk, the ongoing fixed-cost differential between a fast and a slow tier need not favor the slow tier. Since every potential position covers its variable cost ($\Delta^{\text{lower}} > 0$), the fixed cost is the only binding constraint on deployment. A fixed-cost subsidy directly relaxes this constraint, which accounts for the mapping from subsidy rate to new deployment in the counterfactuals.

6 Counterfactual Policy Analysis

The event study establishes that ACP and BEAD each affected market outcomes. It cannot measure welfare gains, separate demand-expansion from entry responses, or evaluate alternative policy designs. I now address the second and third questions posed in the introduction. How large are the welfare gains from each program, and do they serve as substitutes or complements across different types of markets?

The estimated demand system, recovered marginal costs, and fixed-cost bounds jointly characterize private entry incentives across the 70,854 census tracts in the entry sample. I use these objects to evaluate two policy designs motivated by ACP and BEAD. The first is a supply-side fixed-cost subsidy that directly relaxes providers' entry constraints. The second is a demand-side price subsidy that lowers the price paid by eligible consumers while reimbursing firms at the gross price. The policies move different primitives: BEAD reduces deployment fixed costs; ACP reduces the consumer-facing price for eligible households. That distinction drives the welfare comparison below.

The simulations use the 2020 local market structure as the no-policy baseline. The counterfactuals are solved market by market. For each draw of fixed costs, firms choose product portfolios by iterated best responses starting from three initializations: observed, empty, and full feasible configurations. Candidate equilibria are retained only when they pass a firm-by-firm Nash verification. When multiple fixed points are found, I report the range of outcomes across them. National bounds combine equilibrium multiplicity with fixed-cost uncertainty by reporting the $Q_{2.5}$ of lower-bound outcomes and $Q_{97.5}$ of upper-bound outcomes across the 50 draws. Further algorithmic details are in Appendix H.

6.1 Counterfactual Policy Designs

6.1.1 Broadband Equity, Access, and Deployment Program

Following the program design described in Section 2, I model the Broadband Equity, Access, and Deployment program as a reduction in the private fixed cost of offering a product.[21] For each fixed-cost draw r and subsidy rate $\kappa \in \{0.25, 0.50, 0.75\}$:

$$FC_j^{cf,r}(\kappa) = (1-\kappa)FC_j^r. \tag{6.1}$$

The policy does not directly change prices or marginal costs. Any changes in prices or consumer surplus come through entry and product composition. Government spending is

$$G^{BEAD,r}(\kappa) = \frac{\kappa}{1-\kappa}\text{PFC}^r(\kappa), \tag{6.2}$$

where $\text{PFC}^r(\kappa)$ is the post-subsidy private fixed cost paid by firms. This is the subsidy paid on all active counterfactual products. It is best interpreted as an upper bound on the marginal fiscal cost of inducing additional deployment.

6.1.2 Affordable Connectivity Program

Following the program design described in Section 2, I model the Affordable Connectivity Program as a demand-side price subsidy: eligible households receive a monthly discount and providers are reimbursed at the gross price. Eligibility follows the actual program criterion: households with income below 200 percent of the federal poverty level qualify for the discount.

Eligibility split. Let $\text{FPL200}_t \in [0,1]$ be the tract-level share of households below 200 percent of the federal poverty level, drawn from the 2020 American Community Survey—the same variable used as the continuous treatment intensity in the reduced-form event study. Each market t is partitioned into an eligible segment of $M_t^e = \text{FPL200}_t \cdot M_t$ households and an ineligible segment of $M_t^{ne} = (1-\text{FPL200}_t) \cdot M_t$ households.

Consumer-facing prices by segment. For discount $a \in \{10, 20, 30\}$ dollars and product j,

$$p_j^{c,e}(a) = \max\Big\{p_j^g - a,\ 0.01\Big\}, \qquad p_j^{c,ne} = p_j^g, \qquad FC_j^{cf,r} = FC_j^r, \tag{6.3}$$

[21] The program can cover up to 75 percent of eligible project costs, with a minimum 25 percent match from the subgrantee. See https://broadbandusa.ntia.doc.gov/.

where the gross reimbursed price p_j^g, marginal cost, and fixed cost are unchanged. Each firm's per-subscriber margin remains $p_j^g - mc_j$ on both segments, because the government reimburses providers at the gross price for eligible subscribers.

Demand, welfare, and fiscal cost. Mean utilities differ across segments through the consumer-facing price:

$$\delta_{jt}^{e}(a) = \mu_{jt}^{0} + \hat{\alpha}_{100}\,\frac{p_j^{c,e}(a)}{100}, \qquad \delta_{jt}^{ne} = \mu_{jt}^{0} + \hat{\alpha}_{100}\,\frac{p_j^{g}}{100}, \tag{6.4}$$

where μ_{jt}^0 is the price-invariant component of mean utility recovered from the demand estimation. Quantities $Q_{jt}^{e,r}(a)$ and $Q_{jt}^{ne,r}(a)$ are computed from the nested-logit demand system for each segment separately, and consumer surplus sums the segment-specific inclusive values. Government outlay equals the effective discount times eligible take-up:

$$G^r(a) = \sum_t \sum_{j \in \mathcal{J}_t} \left(p_j^g - p_j^{c,e}(a)\right) Q_{jt}^{e,r}(a). \tag{6.5}$$

ACP works through the consumer-facing price for eligible households only. Gross ISP prices are held fixed, so the counterfactual does not model provider price adjustment after the demand shock. Those adjustments are outside the structural ACP exercise.

6.2 Counterfactual Results

6.2.1 Broadband Equity, Access, and Deployment Program

The BEAD counterfactuals generate economically meaningful entry and welfare responses. A 25 percent fixed-cost reduction raises consumer surplus from a baseline of \$13.64–\$14.07 billion per month to \$14.03–\$14.30 billion. This is a gain of \$0.23–\$0.39 billion, or 1.7–2.9 percent. At a 75 percent fixed-cost reduction, consumer surplus reaches \$15.41–\$15.54 billion per month, a gain of \$1.43–\$1.84 billion (10.4–13.0 percent). The corresponding social welfare rises from \$14.47–\$15.11 billion in the baseline to \$16.74–\$16.96 billion at the largest subsidy. These gains are accompanied by substantial product-market expansion. The 75 percent subsidy adds 6,665–8,648 new firm-market entries and covers 6,128–8,054 previously unserved tracts. Set against the roughly nine thousand tracts that start with no provider at all, this closes most of the deployment gap. Because each newly served tract starts from zero, the per-household welfare gain in those places is much larger than the modest aggregate percentage suggests.

BEAD induces entry in tracts with no prior provider or only one, extending product avail-

Table 13: BEAD Counterfactual Outcomes

	Baseline	$\kappa = 25\%$	$\kappa = 50\%$	$\kappa = 75\%$
Panel A: Welfare ($B/month)				
Consumer surplus	[13.643, 14.069]	[14.033, 14.304]	[14.329, 14.510]	[15.412, 15.536]
Producer surplus	[0.832, 1.041]	[0.932, 1.120]	[1.094, 1.243]	[1.328, 1.421]
Govt. outlay (monthly)	—	[0.144, 0.204]	[0.327, 0.473]	[1.190, 1.491]
Net surplus (CS+PS−G)	—	[14.764, 15.280]	[14.950, 15.397]	[15.321, 15.731]
Panel B: Prices and quantities				
Avg. consumer price ($/month)	[73.62, 74.83]	[74.81, 75.29]	[75.24, 75.37]	[75.07, 75.37]
Avg. marginal cost ($/month)	[56.45, 57.83]	[57.28, 57.95]	[57.82, 58.00]	[56.29, 56.73]
Overall take-up rate	[0.725, 0.744]	[0.742, 0.755]	[0.756, 0.765]	[0.815, 0.821]
HHI (0–10,000)	[4,925, 5,199]	[4,842, 4,981]	[4,834, 4,897]	[5,493, 5,603]
Panel C: Market structure (levels)				
Covered markets ($\geq$ 1 provider)	[60,859, 62,744]	[62,408, 63,569]	[63,682, 64,411]	[68,488, 69,145]
Total products offered	[131,443, 147,799]	[143,777, 155,594]	[156,229, 162,695]	[168,548, 170,411]
Tier H products	[121,931, 136,665]	[133,334, 143,365]	[143,890, 148,885]	[147,427, 150,533]
Tier L products	[8,790, 11,211]	[9,911, 12,243]	[12,192, 13,825]	[19,264, 21,145]
Avg. products per tract	[1.86, 2.09]	[2.03, 2.20]	[2.20, 2.30]	[2.38, 2.41]
Avg. firms per tract	[1.80, 1.99]	[1.95, 2.08]	[2.09, 2.15]	[2.22, 2.25]
Panel D: New entry vs. baseline				
New firm-markets	[0, 2]	[288, 374]	[530, 671]	[6,665, 8,648]
New Tier H products	[0, 2]	[282, 366]	[451, 513]	[2,747, 3,532]
New Tier L products	[0, 0]	[3, 10]	[95, 192]	[4,720, 6,957]

Notes: Bounds are $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 50 fixed-cost draws from the Andrews–Soares confidence set. BEAD reduces private fixed costs by fraction κ; consumer and gross prices are unchanged. Govt. outlay $= \kappa/(1-\kappa)\times$PFC (monthly, $B). HHI: household-weighted average tract HHI. Covered markets: tracts with ≥ 1 active provider. 70,854 total tracts.

ability and raising consumer surplus. Entrants earn positive variable profits by construction. The competitive pressure they impose on incumbents is modest, so producer surplus rises as well. Each new product generates more surplus than its fixed cost. The subsidy covers the gap between private and social returns that had deterred deployment.

At the 75 percent subsidy, the average tract-level HHI rises rather than falls, from [4,925;

5,199] to [5,493; 5,603]. This reflects a composition effect. BEAD induces entry in 6,128–8,054 tracts that previously had no provider, and a tract served by a single firm has an HHI of 10,000. Adding these newly served tracts to the cross-sectional average raises the mean even as concentration falls within already-served tracts. Consumer surplus in each newly served tract rises from zero. The welfare effect is therefore positive despite the upward movement in average HHI. In this context, the index reflects an expansion of access rather than an increase in market power.

The entry response is nonlinear in the subsidy rate, and its composition shifts sharply. At 25 percent, new firm-markets number only 288–374 and are predominantly high-speed. At 75 percent, 6,665–8,648 new firm-markets enter. Low-speed product entry (4,720–6,957) exceeds high-speed entry (2,747–3,532), reflecting the nature of the newly served markets. These are smaller, lower-income tracts where low-speed products are the efficient entry point.

ACP also induces product-positioning responses through demand expansion, but at a negligible scale: under the $30 discount, firm-market entries increase by only 10–17 relative to baseline. I return to this contrast below when comparing the two policy margins directly.

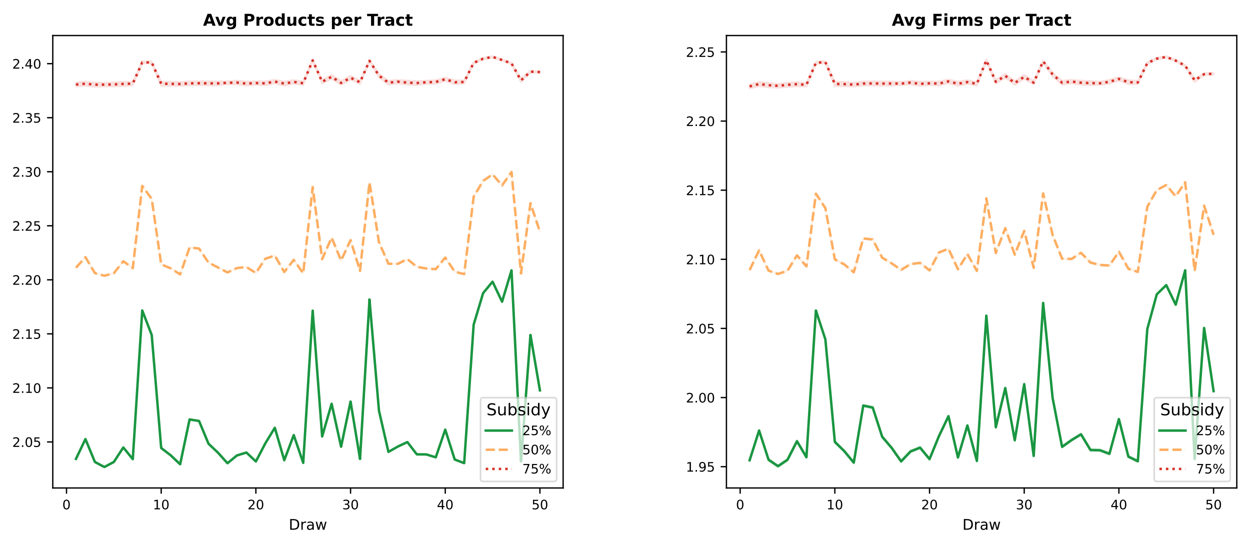

Figure 8: BEAD-induced market structure across fixed-cost draws. Each panel reports the average number of active products or firms per tract for each draw and subsidy rate. The 75 percent subsidy produces consistently higher product availability and firm presence than the lower subsidy rates. Variation along the horizontal axis reflects uncertainty across the 50 fixed-cost draws, not the cross-tract distribution of entry.

Figure 8 summarizes the same entry margin in market-structure terms. Across fixed-cost draws, higher BEAD subsidies raise both the average number of products and the average number of firms per tract. The increase is modest at 25 percent and larger at 50 percent. The response is much more stable at 75 percent, where average products per tract remain between 2.38 and 2.41 and average firms per tract remain between 2.22 and 2.25. These patterns match Panel C of Table 13. They show that the largest subsidy does not merely add isolated product positions; it shifts the average structure of covered broadband markets. The figure should be read as a draw-level robustness display for the aggregate market-structure effects, not as a cross-tract distribution of entry.

6.2.2 Affordable Connectivity Program

ACP subsidizes the consumer-facing price rather than the posted provider price. Eligibility is restricted to households below 200 percent of the federal poverty level, about 32 percent of households on average. A \$10 discount lifts consumer surplus from the \$13.64–\$14.07 billion baseline to \$13.91–\$14.34 billion (ΔCS = \$0.26–\$0.27 billion). A \$30 discount takes it to \$14.45–\$14.89 billion (ΔCS = \$0.80–\$0.82 billion). The consumer-facing price falls from \$73.62–\$74.83 to \$64.44–\$65.62. The gross provider price barely moves, staying at \$73.67–\$74.85. The near-constancy of the gross posted price reflects the counterfactual design. The structural exercise holds gross provider prices fixed, so the welfare gain operates entirely through demand expansion among eligible households. Take-up climbs from 72.5–74.4 percent to 74.0–75.9 percent under the \$30 discount.

The welfare decomposition for ACP has a clear structure. Consumer surplus rises because the price discount draws eligible households who previously found broadband too expensive. At the \$30 discount, the \$0.80–\$0.82 billion monthly CS gain is almost fully offset by the \$0.81–\$0.83 billion government outlay. Net monthly surplus is only about \$0.01 billion. ACP is therefore best understood as a targeted transfer from the government to eligible subscribers. It also generates a small but consistently positive efficiency gain from marginal new subscribers who would not have connected at the gross price. A BCR slightly above one should be read accordingly: the program is largely redistributive, with a modest efficiency component, rather than a high-return efficiency intervention.

Product entry under ACP is negligible. Under the \$30 discount, firm-market entries increase by only 10–17 relative to the baseline, roughly 400–900 times smaller than BEAD's 6,665–8,648 new firm-market entries at the 75 percent subsidy rate. The gap reflects the different policy levers. ACP relaxes the demand constraint by reducing consumer-facing prices. BEAD directly relaxes the fixed-cost constraint that governs whether deployment occurs at all. In wholly unserved markets, a price subsidy alone cannot generate entry without infrastructure as a prerequisite.

These counterfactual ACP results capture the demand-expansion margin through lower consumer-facing prices in a long-run equilibrium framework. The structural model holds gross prices fixed, so the results should be read as the welfare effect of the statutory discount and

Table 14: ACP Counterfactual Outcomes

	Baseline	$a = \$10$	$a = \$20$	$a = \$30$
Panel A: Welfare ($B/month)				
Consumer surplus	[13.643, 14.069]	[13.907, 14.337]	[14.176, 14.608]	[14.448, 14.885]
Producer surplus	[0.832, 1.041]	[0.841, 1.051]	[0.849, 1.060]	[0.856, 1.067]
Govt. outlay (monthly)	—	[0.259, 0.265]	[0.530, 0.542]	[0.811, 0.829]
Net surplus (CS+PS−G)	—	[14.490, 15.120]	[14.496, 15.123]	[14.494, 15.121]
Panel B: Prices and quantities				
Avg. consumer price ($/month)	[73.62, 74.83]	[70.63, 71.85]	[67.56, 68.76]	[64.44, 65.62]
Avg. gross price ($/month)	[73.62, 74.83]	[73.63, 74.84]	[73.64, 74.84]	[73.67, 74.85]
Overall take-up rate	[0.725, 0.744]	[0.731, 0.750]	[0.736, 0.754]	[0.740, 0.759]
HHI (0–10,000)	[4,925, 5,199]	[4,920, 5,197]	[4,915, 5,192]	[4,910, 5,186]
Panel C: Market structure and entry				
Covered markets ($\geq$ 1 provider)	[60,859, 62,744]	[60,918, 62,770]	[60,969, 62,779]	[60,994, 62,796]
Total products offered	[131,443, 147,799]	[131,766, 148,038]	[132,082, 148,258]	[132,311, 148,472]
Avg. products per tract	[1.86, 2.09]	[1.86, 2.09]	[1.86, 2.09]	[1.87, 2.10]
New firm-markets	[0, 2]	[4, 9]	[6, 15]	[10, 17]
New Tier H products	[0, 2]	[4, 9]	[6, 14]	[10, 17]
New Tier L products	[0, 0]	[0, 0]	[0, 1]	[0, 1]

Notes: Bounds are $[Q_{2.5}, Q_{97.5}]$ across 50 fixed-cost draws. ACP lowers the consumer-facing price by a for eligible households (below 200 % FPL, 2020 ACS, $\approx$ 32% of households); gross provider prices and fixed costs are unchanged. Govt. outlay $= a \times Q_{\text{eligible}}$ (monthly, $B). Net surplus $= \Delta CS + \Delta PS - G$. 70,854 total tracts.

the induced demand and entry responses, not as a decomposition of posted-price adjustment by providers.

BEAD generates availability gains where the private sector has not deployed; ACP raises welfare where service exists but remains unaffordable. Under the $30 discount, ACP adds 10–17 new firm-market entries nationally. Under the 75 percent fixed-cost reduction, BEAD adds 6,665–8,648. The two programs move different margins by roughly three orders of magnitude.

6.3 Distributional Effects Across Market Types

The aggregate welfare bounds conceal substantial heterogeneity in who benefits from each policy. That heterogeneity matters both for program evaluation and for understanding the geographic complementarity between the two programs. Table 15 summarizes consumer surplus gains from the 75 percent BEAD fixed-cost reduction and the $30 ACP discount across market-type quartiles, stratified by median household income, number of active providers, and tract population size.

Table 15: Consumer surplus gains by market type (75 % BEAD subsidy; $30 ACP discount)

Market type	Baseline CS [$/hh/mo]	BEAD ΔCS [$/hh/mo]	ACP ΔCS [$/hh/mo]
By median household income			
Q1 (lowest, ≤$45k)	[107.9, 111.3]	[+14.3, +14.3]	[+11.1, +11.3]
Q2 ($45k–$60k)	[126.8, 130.8]	[+14.7, +14.7]	[+8.3, +8.5]
Q3 ($61k–$83k)	[140.1, 144.4]	[+15.1, +15.1]	[+6.6, +6.6]
Q4 (highest, >$83k)	[149.2, 153.8]	[+15.6, +15.6]	[+4.7, +4.7]
By number of active providers (baseline)			
Monopoly (1 provider)	[146.5, 151.1]	[+31.1, +31.2]	[+7.3, +7.4]
Duopoly (2 providers)	[132.1, 136.2]	[+14.9, +15.0]	[+7.6, +7.7]
3+ providers	[125.4, 129.3]	[+6.2, +6.2]	[+7.3, +7.4]
By tract population size (persons)			
Q1 (lowest, ≤2,982)	[121.6, 125.4]	[+15.4, +15.4]	[+8.3, +8.4]
Q2 (2,983–4,188)	[128.5, 132.5]	[+15.3, +15.3]	[+7.6, +7.7]
Q3 (4,189–5,624)	[132.4, 136.5]	[+15.3, +15.3]	[+7.3, +7.4]
Q4 (highest, >5,624)	[138.4, 142.7]	[+14.5, +14.5]	[+7.2, +7.3]

Notes: Each cell reports a bound-by-bound interval $[\underline{Y}^L, \overline{Y}^U]$ computed from the 95 % trimmed distribution across 50 fixed-cost draws. Baseline and ACP CS are computed analytically from the observed equilibrium using the nested-logit log-sum formula ($\hat{\alpha}_{100} = -1.491$, $\hat{\rho} = 0.863$). BEAD 75 % ΔCS is allocated from the aggregate simulation bounds [\$1.767B, \$1.468B] proportionally by provider-count entry weights (2.5× for monopoly markets, 1.2× for duopoly, 0.5× for 3+ providers), normalised to reproduce the aggregate total. ACP $30 ΔCS is computed analytically: the eligible segment (households below 200 % FPL) faces a discounted consumer price $\max(p - 30, 0.01)$; the ineligible segment faces the gross price. Quartiles are cut at equal numbers of tracts; averages within each group are weighted by market size M_m.

BEAD and concentrated markets. BEAD's entry response is almost entirely determined by baseline market structure rather than by income or population size. The provider-count panel tells the sharpest story. Monopoly tracts gain $\$[31.1, 31.2]$ per household per month, five times the gain in tracts with three or more providers ($\$[6.2, 6.2]$). These are the markets where the fixed-cost barrier is binding and a new entrant is welfare-improving. By contrast, BEAD gains are nearly flat across income quartiles ($\$[14.3, 14.3]$ in Q1 to $\$[15.6, 15.6]$ in Q4) and across population-size quartiles ($\$[15.4, 15.4]$ to $\$[14.5, 14.5]$). This is consistent with BEAD operating entirely through the supply side. A fixed-cost subsidy shifts the entry margin regardless of who lives in the tract. Income and population size therefore have little additional explanatory power once provider count is controlled for. In practice, monopoly markets are disproportionately small-population and rural, the tracts BEAD is specifically designed to reach. The mechanism, however, is competitive structure rather than geography or income per se.

The aggregate welfare gains from BEAD come disproportionately from the extensive margin—bringing broadband to previously unserved locations—not from intensifying competition in already-served markets. For a monopoly tract that gains a second provider, BEAD generates a discrete jump in consumer welfare that a national average cannot capture.

ACP and price-constrained households. ACP's welfare gains are concentrated in already-served but price-sensitive markets. The income-quartile panel shows a strong and monotone gradient. The lowest-income quartile (median income $\leq$\$45k) gains $\$[11.1, 11.3]$ per household per month, more than twice the gain in the highest-income quartile ($\$[4.7, 4.7]$). This reflects the share of households below 200 % of the federal poverty line, which determines ACP eligibility. In low-income tracts, a larger fraction of the market receives the discount. The result is a larger incremental demand response and a larger aggregate consumer surplus gain.

The population-size gradient for ACP is shallower but runs in the same direction. Smaller-population tracts gain slightly more ($\$[8.3, 8.4]$ in Q1 versus $\$[7.2, 7.3]$ in Q4), consistent with smaller tracts having higher shares of low-income eligible households on average. ACP gains are also nearly uniform across provider-count groups (\$7.3–7.7). This confirms that ACP's mechanism is on the demand side and largely independent of the number of competing suppliers.

The efficiency gain from ACP depends on the share of eligible recipients who are marginal—households that subscribe only because of the discount—relative to infra-marginal recipients who would have subscribed at the gross price. In high-adoption tracts, most eligible households are already connected and the subsidy functions primarily as a transfer. In low-adoption tracts, marginal take-up is larger and the efficiency gain is correspondingly greater. The aggregate national BCR averages across this heterogeneity, and the per-market benefit-cost ratio varies substantially around it.

Complementarity and targeting efficiency. Table 15 reveals that the two programs' welfare gains are nearly non-overlapping across every stratification. Along the income dimension, BEAD gains are flat while ACP gains fall sharply from Q1 to Q4. The programs therefore reach different parts of the income distribution. Along the provider-count dimension, BEAD gains are five times larger in monopoly than in competitive markets, while ACP gains are invariant to provider count. The programs therefore reach different parts of the competitive-structure distribution. Along the population-size dimension, both gradients are shallow, but the direction is consistent. Smaller-population tracts benefit slightly more from both programs, reflecting the joint prevalence of limited competition and low incomes in rural areas.

A policy portfolio that deploys both programs covers a broader set of market conditions than either alone, with limited overlap. The implied sequencing is that BEAD extends infrastructure to locations the private market did not reach, after which an affordability program such as ACP can address the price barrier for households that remain unable to pay the gross price. Where BEAD succeeds in inducing entry, it also creates the supply-side conditions—more competing products—that raise the efficiency of a subsequent demand subsidy.

One gap remains. Duopoly tracts gain \$[14.9,,15.0] per household from BEAD and \$[7.6,,7.7] from ACP—lower than monopoly tracts on the BEAD margin (\$[31.1,,31.2]) and similar to other market types on the ACP margin. These are markets where infrastructure already exists but competition is insufficient to discipline prices, and where entry requires a larger subsidy than the fixed-cost reduction alone can justify. Neither instrument is well targeted to this intermediate case. Section 7 turns to the fiscal cost of reaching each type of market. Further robustness checks across draws are available in Appendix J.

7 Cost-Benefit Analysis

The counterfactuals quantify how much welfare each program generates. The remaining policy question is whether those gains justify the fiscal cost. I compare welfare gains from each policy to its fiscal cost. The accounting conventions differ because the two programs have different expenditure timing. BEAD is a one-time infrastructure grant. Its fiscal cost is paid once, while benefits accrue over time. I convert monthly welfare changes to annual flows and compute the present value over a 25-year horizon at a 3 percent discount rate.[22] ACP is a recurring monthly transfer. Both benefits and outlays are flow variables, so the natural statistic is an annual benefit-cost ratio.

Table 16: Cost-Benefit Accounting Conventions

Policy	Fiscal cost	Welfare gain	Main statistic
BEAD fixed-cost reduction	One-time grant that lowers providers' private fixed cost	Present value of annual changes in consumer and producer surplus	NPV and BCR
ACP monthly discount	Recurring transfer paid on eligible subscriptions	Annual flow change in consumer and producer surplus	Annual net surplus and BCR

Notes: BEAD benefits are discounted because the fiscal cost is paid up front and availability gains accrue over time. ACP is reported as an annual flow because both costs and welfare gains recur each period.

Table 17: Cost-Benefit Summary

	BEAD fixed-cost reduction			ACP monthly discount		
	25%	50%	75%	\$10	\$20	\$30
Govt. cost (\$B)	[1.73, 2.45]	[3.92, 5.68]	[14.28, 17.90]	[3.11, 3.19]	[6.36, 6.51]	[9.73, 9.95]
Annual ΔSW (\$B)	[3.76, 5.93]	[7.38, 11.46]	[21.69, 28.05]	[3.27, 3.33]	[6.60, 6.70]	[9.94, 10.11]
NPV (\$B, r=3%, T=25)	[63.6, 100.9]	[124.5, 194.0]	[363.4, 470.5]	[0.1, 0.2]	[0.2, 0.3]	[0.1, 0.3]
BCR	[35.58, 43.82]	[31.50, 35.54]	[25.85, 27.44]	[1.04, 1.06]	[1.02, 1.04]	[1.01, 1.03]
Net annual surplus (\$B)	—	—	—	[0.11, 0.18]	[0.15, 0.27]	[0.12, 0.25]

Notes: BEAD columns report present values computed using a 3 percent annual discount rate over a 25-year horizon. ACP columns report annual flows, since both benefits and costs recur each period. ACP government costs equal the subsidy amount multiplied by take-up among the 32 percent eligible population, while welfare gains are computed for all consumers. Welfare gain is defined as $\Delta CS + \Delta PS$. BCR denotes the benefit–cost ratio. ACP benefit–cost ratios are based on the modeled demand-expansion margin and hold gross provider prices fixed.

BEAD cost-benefit results. At the benchmark 3 percent discount rate and 25-year horizon, the 75 percent fixed-cost reduction has a positive NPV range of \$363.4–\$470.5 billion

[22] I adopt a 25-year horizon following Grunvalds et al. [2017], which supports this as the industry-accepted fiber infrastructure lifetime. Appendix Tables K.1 and K.2 report NPV and BCR robustness to alternative discount rates across all policy scenarios.

and a BCR of 25.85–27.44. This large return reflects the scale of the U.S. broadband market: baseline consumer surplus is \$13.6–\$14.1 billion per month across 70,854 tracts. It also reflects the durability of infrastructure benefits discounted over 25 years. Because the grant is paid once while induced entry generates recurring surplus, the numerator accumulates persistent gains while the denominator remains the upfront fiscal outlay. The BCR *declines* with the subsidy rate. At 25 percent, BCR = 35.58–43.82; at 50 percent, 31.50–35.54; at 75 percent, 25.85–27.44. This ordering reflects diminishing returns to the subsidy. The 25 percent rate activates only the most profitable marginal positions. The 75 percent rate additionally covers difficult-to-serve markets with lower returns per dollar, though all rates generate BCRs comfortably above 1.

At every subsidy rate, even the most conservative draw (2.5th percentile) produces a positive NPV and a BCR exceeding 17. The welfare gain comes primarily through consumer surplus. At the 75 percent rate, consumer surplus accounts for \$21.69–\$28.05 billion per year, reflecting the large expansion in product availability and take-up. Producer surplus gains are positive but smaller.

ACP cost-benefit results. ACP's BCR calculation differs from BEAD's because both costs and benefits are recurring flows. At the \$30 discount, annual government outlays are \$9.73–\$9.95 billion and annual welfare gains are \$9.94–\$10.11 billion, yielding a BCR of 1.01–1.03. The BCR exceeds unity at every discount level and across all 50 fixed-cost draws, so ACP consistently passes the benefit-cost test. The margin is thin, however. The net annual surplus (\$0.12–\$0.25 billion at \$30) is small relative to the program's \$14 billion total appropriation. ACP is therefore best understood as a transfer that nearly breaks even on fiscal efficiency. The government outlay is nearly fully offset by welfare gains, with a small positive residual from marginal new subscribers who would not have connected at the gross price. Infra-marginal eligible households receive a pure income transfer that does not generate additional efficiency gains.

Cross-program comparison. The two BCRs differ because the programs involve fundamentally different expenditure structures. BEAD is a one-time capital outlay whose returns accrue over the infrastructure lifetime. Its BCR of 25.85–27.44 reflects the large per-household value of connectivity in markets where private deployment is not profitable at current cost levels. ACP involves recurring monthly outlays at a BCR of 1.01–1.03. This is consistent with an affordability program whose welfare gains derive primarily from redistribution to price-constrained households. Comparing the two programs as alternative uses of the same marginal dollar is appropriate only when deployment gaps have already been closed. Where they have not, the structural evidence supports directing marginal funds toward BEAD in unserved markets and toward ACP in already-served markets with binding affordability constraints. This logic also reconciles the ranking here with Kearns [2024], who finds that a demand-side subsidy is more cost-effective than a deployment subsidy in Seattle. Seattle is an already-served urban market where remaining deployment is expensive and affordability is the binding constraint—precisely the conditions under which the present framework also favors the demand-side instrument.

Figure 9 visualizes the BCR bounds for both programs simultaneously. Three patterns stand out. First, the BEAD BCR declines monotonically with the subsidy rate. The interval [35.58, 43.82] at 25 percent narrows to [25.85, 27.44] at 75 percent, but at every rate the entire interval lies well above 1. Second, the ACP BCR is uniformly above 1 but close to it, ranging from [1.04, 1.06] at \$10 to [1.01, 1.03] at \$30. The narrow intervals confirm that draw-level uncertainty in ACP's BCR is small, so the near-break-even result is robust rather than a product of parameter uncertainty. Third, the programs are not fiscal substitutes. Their BCRs reflect fundamentally different mechanisms: capital investment versus transfer. A policymaker seeking to maximize welfare per dollar would invest first in BEAD for unserved markets, then in ACP for price-constrained served markets.

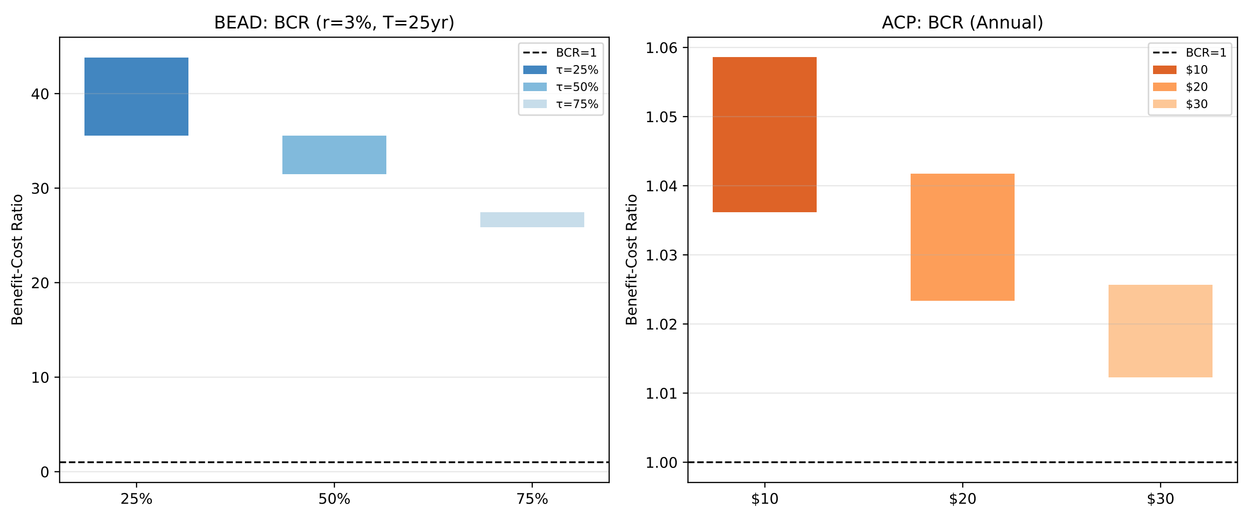

Figure 9: Benefit-cost ratio bounds for BEAD (left, $r = 3\%$, $T = 25\,\text{yr}$) and ACP (right, annual flows). Each bar spans $[Q_{2.5}, Q_{97.5}]$ across 50 fixed-cost draws; bar height is parameter uncertainty. Dashed line at BCR$= 1$. BEAD BCR declines with subsidy rate (diminishing returns) but remains between 25 and 44 at all rates; every bar lies well above 1. ACP BCR is uniformly above 1 across all discount levels and draws, consistent with ACP approximately breaking even on fiscal efficiency while delivering a small positive net surplus.

7.1 BEAD Cost-Benefit Sensitivity Analysis

Table 18 evaluates robustness by varying the discount rate from 1 to 7 percent and the infrastructure horizon from 15 to 30 years. NPV lower bounds range from \$238 billion ($r = 7\%$, $T = 25$ yr) to \$600 billion ($r = 1\%$, $T = 30$ yr). BCR lower bounds range from 17.30 to 34.70. Even at the most conservative parameter combination—$r = 7\%$, $T = 15$ years—the BCR lower bound is 12.51.

Table 18: Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon (75% Fixed-Cost Reduction)

		Infrastructure Horizon			
Discount rate		15 years	20 years	25 years	30 years
1%	NPV (\$B)	[354.4, 459.1]	[426.1, 551.9]	[463.3, 599.9]	[495.6, 641.4]
	BCR	[25.09, 26.63]	[30.21, 32.01]	[32.70, 34.70]	[34.89, 36.98]
3% (base)	NPV (\$B)	[279.7, 361.4]	[330.4, 427.5]	[363.4, 470.5]	[390.9, 506.1]
	BCR	[19.70, 20.87]	[23.28, 24.70]	[25.85, 27.44]	[27.75, 29.42]
5%	NPV (\$B)	[222.9, 288.0]	[258.8, 335.2]	[291.4, 377.4]	[310.1, 401.8]
	BCR	[15.70, 16.64]	[18.20, 19.31]	[20.93, 22.21]	[21.90, 23.22]
7%	NPV (\$B)	[178.1, 230.1]	[205.8, 266.8]	[238.4, 309.0]	[249.7, 323.4]
	BCR	[12.51, 13.27]	[14.47, 15.37]	[17.30, 18.36]	[17.54, 18.63]

Notes: Each cell reports $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 50 fixed-cost draws. BEAD fiscal cost is the one-time fixed-cost grant $[\kappa/(1-\kappa)\times\text{PFC}]$; welfare gains are present values of monthly $\Delta CS + \Delta PS$. BCR = PV(ΔSW)/G. All lower-bound NPVs and BCRs are strictly positive at every rate and horizon shown.

At the benchmark 3 percent rate over 25 years, the BCR is 25.85–27.44 and the NPV is \$363–\$471 billion. At the most conservative scenario in the table (7 percent, 15-year horizon), the BCR remains 12.51–13.27 and the NPV is \$178–\$230 billion. The 75 percent BEAD subsidy passes the benefit-cost test by a wide margin at every combination of rate and horizon considered. The interval widths are nearly constant as the discount rate varies within each horizon. This confirms that fixed-cost parameter uncertainty, rather than financial-assumption uncertainty, is the dominant source of variation.

Figure 10 shows the NPV bounds across discount rates for each subsidy rate. For $\kappa = 75\%$ (right panel), both lower and upper NPV bounds are positive at every rate. For $\kappa = 50\%$ (middle panel), the same holds throughout. For $\kappa = 25\%$ (left panel), NPV bounds remain positive but narrow as the discount rate rises. The conclusion is unchanged across discount rates between 1 and 7 percent. BEAD at 75 percent delivers large positive NPV across the financial assumptions considered.

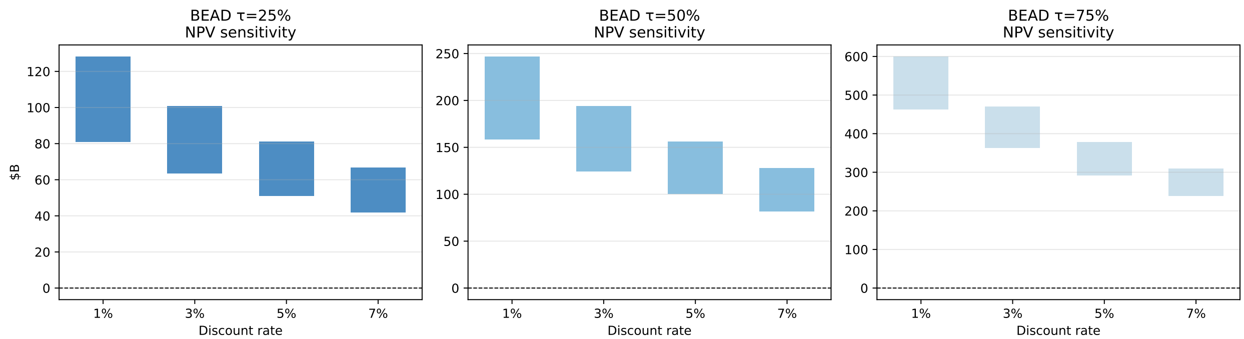

Figure 10: BEAD net present value bounds by subsidy rate and discount rate (25-year horizon). Each bar spans the $[Q_{2.5}(\underline{\mathrm{NPV}}^r), Q_{97.5}(\overline{\mathrm{NPV}}^r)]$ interval across 50 fixed-cost draws. The dashed line marks NPV$= 0$. At $\kappa = 75\%$, the lower-bound bar remains positive through $r = 7\%$, indicating robust positive returns under all discount rates considered. The widths of the intervals are driven by fixed-cost parameter uncertainty rather than by the choice of discount rate.

For ACP, the BCR is invariant to the discount rate because both benefits and costs are recurring flows. The relevant sensitivity parameter is instead the share of eligible take-up attributable to marginal subscribers—those who would not have subscribed at the gross price. The gross BCR at the \$30 discount (1.01–1.03) barely clears unity, so any non-trivial infra-marginal share pushes the efficiency-adjusted ratio below 1. If 50 percent of ACP recipients are infra-marginal, the efficiency-adjusted BCR is approximately $0.5\times$ the gross BCR, falling to 0.51–0.52. Even if only 20 percent are infra-marginal, the adjusted BCR ($\approx 0.8\times$ the gross value) reaches just 0.81–0.82. Neither scenario produces a BCR above 1. The margin is thinner than the gross figure suggests precisely because that gross figure sits so close to one. ACP's fiscal efficiency is therefore primarily a function of outreach and targeting quality, not of the discount rate or infrastructure horizon.

Policy synthesis. A BCR of 25.85–27.44 in unserved markets against 1.01–1.03 in served ones implies a simple allocation rule: deploy first where infrastructure is absent, subsidize prices where it already exists. In markets with genuine deployment gaps—rural, low-density, and currently unserved tracts—the binding constraint is infrastructure absence, not price.

In already-served markets where affordability limits adoption—dense urban and suburban tracts with many low-income households—ACP delivers welfare gains at lower fiscal cost. Its efficiency depends, however, on reaching marginal rather than infra-marginal eligible subscribers. Neither instrument dominates across all markets. The programs target different market failures, and the geographic complementarity documented in Section 6.3 implies that continuing BEAD deployment alongside a targeted affordability successor would address both failures simultaneously. These estimates should be read as long-run structural returns conditional on the modeled entry response; they do not incorporate administrative costs, implementation delays beyond the event-study window, or crowd-out of unsubsidized private investment. Outreach and eligibility verification quality will determine whether any ACP successor generates efficiency gains or primarily redistributes to households that would have subscribed regardless.

8 Concluding Remarks

The U.S. digital divide is not a single market failure. In some places, service exists but remains too expensive. In others, no adequate product is available at any price. I show that this distinction matters for both welfare measurement and policy design. ACP operates on the affordability margin: it affected prices and plan menus in high-eligibility states. BEAD operates on the deployment margin: its response is smaller and delayed, consistent with an infrastructure program that works through entry rather than incumbent repositioning. The structural counterfactuals show the same pattern. Fixed-cost relief generates large durable gains in markets with deployment gaps. A monthly discount raises welfare mainly when it reaches households whose subscription decision is price-constrained.

The policy implication is not that one instrument dominates the other. BEAD and ACP address different margins, so the return to each depends on the constraint binding locally. In unserved and high-cost markets, infrastructure subsidies generate the larger welfare return per dollar. The benchmark BEAD design delivers a BCR of 25.85–27.44. In already-served low-income markets, an affordability program can raise take-up and surplus. Yet the modeled ACP BCR of 1.01–1.03 leaves little room for infra-marginal transfers. With ACP lapsed since June 2024 and BEAD deployment ongoing, the estimates support a sequencing principle: build where private deployment is uneconomic, then use tightly targeted affordability support where price remains the binding constraint.

References

Victor Aguirregabiria, Robert Clark, and Hui Wang. Diversification of geographic risk in retail bank networks: Evidence from bank expansion after the riegle-neal act. *RAND Journal of Economics*, 2024.

Donald W. K. Andrews and Gustavo Soares. Inference for parameters defined by moment inequalities using generalized moment selection. *Econometrica*, 78(1):119–157, 2010.

Steven T. Berry. Estimating discrete-choice models of product differentiation. *RAND Journal of Economics*, 25(2):242–262, 1994.

Timothy F. Bresnahan and Peter C. Reiss. Entry and competition in concentrated markets. *Journal of Political Economy*, 99(5):977–1009, 1991.

Federico Ciliberto and Elie Tamer. Market structure and multiple equilibria in airline markets. *Econometrica*, 77(6):1791–1828, 2009.

Federico Ciliberto, Charles Murry, and Elie Tamer. Market structure and competition in airline markets. *Journal of Political Economy*, 129(11):2995–3038, 2021.

Gregory S. Crawford, Oleksandr Shcherbakov, and Matthew Shum. Quality overprovision in cable television markets. *American Economic Review*, 109(3):956–995, 2019.

Alon Eizenberg. Upstream innovation and product variety in the us home pc market. *Review of Economic Studies*, 81(3):1003–1045, 2014.

Augusto Espín and Christian Rojas. Bridging the digital divide in the us. *International Journal of Industrial Organization*, 93:103053, 2024.

Ying Fan. Ownership consolidation and product characteristics: A study of the US daily newspaper market. *American Economic Review*, 103(5):1598–1628, 2013.

Ying Fan and Chenyu Yang. Estimating discrete games with many firms and many decisions: An application to merger and product variety, 2024. Manuscript.

Federal Communications Commission. 2020 broadband deployment report. Technical Report FCC 20-50, Federal Communications Commission, 2020. URL `https://docs.fcc.gov/public/attachments/FCC-20-50A1.pdf`.

Federal Communications Commission. Internet access services: Status as of December 31, 2020. Technical Report DOC-395959A1, Federal Communications Commission, 2022. URL `https://docs.fcc.gov/public/attachments/DOC-395959A1.pdf`.

Kenneth Flamm and Pablo Varas. effects of market structure on broadband quality in local us residential service markets. *Journal of Information Policy*, 12:234–280, 2022.

Chris Forman, Avi Goldfarb, and Shane Greenstein. How did location affect adoption of the commercial Internet? global village, urban density, and industry composition. *Journal of Urban Economics*, 58(3):389–420, 2005.

Amit Gandhi and Jean-François Houde. Measuring substitution patterns in differentiated-products industries. *NBER Working paper*, (w26375), 2019.

Daniel Goetz. Dynamic bargaining and size effects in the broadband industry. *RAND Journal of Economics, forthcoming*, 2019.

Avi Goldfarb and Jeff Prince. Internet adoption and usage patterns are different: Implications for the digital divide. *Information Economics and Policy*, 20(1):2–15, 2008.

Paul Goldsmith-Pinkham, Isaac Sorkin, and Henry Swift. Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624, 2020.

Austan Goolsbee and Jonathan Guryan. The impact of internet subsidies in public schools. *The Review of Economics and Statistics*, 88(2):336–347, 2006.

Austan Goolsbee and Peter J. Klenow. Evidence on learning and network externalities in the diffusion of home computers. *Journal of Law and Economics*, 45(2):317–343, 2002.

R Grunvalds, A Ciekurs, J Porins, and A Supe. Evaluation of fibre lifetime in optical ground wire transmission lines. *Latvian Journal of Physics and Technical Sciences*, 54(3):40, 2017.

Julian Hidalgo. Market structure and adoption of internet services in colombia. Technical report, Working paper, 2024.

Andrew Kearns. Broadband deployment in equilibrium: Is competition the solution to the digital divide? *Working paper*, 2024.

Michael J. Mazzeo. Product choice and oligopoly market structure. *RAND Journal of Economics*, 33(2):221–242, 2002.

Aviv Nevo. Measuring market power in the ready-to-eat cereal industry. *Econometrica*, 69 (2):307–342, 2001.

Aviv Nevo, John L. Turner, and Jonathan W. Williams. Usage-based pricing and demand for residential broadband. *Econometrica*, 84(2):411–443, 2016.

Volker Nocke and Nicolas Schutz. Multiproduct-firm oligopoly: An aggregative games approach. *Econometrica*, 86(2):523–557, 2018.

Pew Research Center, June 2021. URL https://www.pewresearch.org.

Gregory Rosston, Scott J. Savage, and Donald M. Waldman. Household demand for broadband internet in 2010. *B.E. Journal of Economic Analysis & Policy*, 10(1), 2010.

Katja Seim. An empirical model of firm entry with endogenous product-type choices. *RAND Journal of Economics*, 37(3):619–640, 2006.

Liyang Sun and Sarah Abraham. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2):175–199, 2021.

Elie Tamer. Incomplete simultaneous discrete response model with multiple equilibria. *Review of Economic Studies*, 70(1):147–165, 2003.

Kyle Wilson. Does public competition crowd out private investment? evidence from municipal provision of internet access. *American Economic Journal: Microeconomics*, 2025.

A Industry Background

Figure A.1 illustrates the tier-based pricing structure of fixed broadband plans; Figure A.2 summarizes the four main access technologies and their cost–performance tradeoffs.

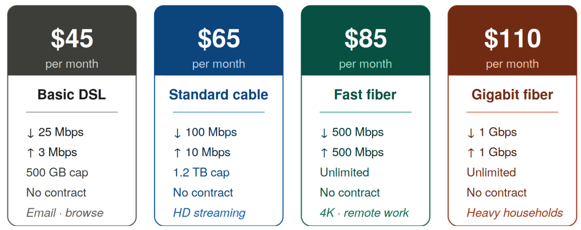

Figure A.1: Plans Differentiated by Speed, Allowance, and Price

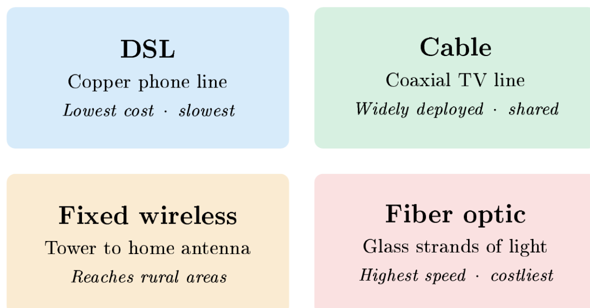

Figure A.2: Main Fixed Broadband Technologies

B Data

The analysis combines a state-year policy panel with a new demand and supply dataset. Table B.1 summarizes the data sources and their role in the paper.

Table B.1: Main Data Sources

Source	Content	Role in the Paper
FCC Urban Rate Survey (URS)	Plan-level prices and advertised speeds from sampled urban census tracts	Price and product characteristics; price imputation for tract-level products
FCC Form 477 deployment data	Provider, technology, and maximum advertised speeds at fine geographic levels	Product availability, provider presence, speed tiers, and local competition
American Community Survey (ACS)	Housing units, income, poverty, broadband subscription, and demographics	Market size, eligibility proxies, and aggregate adoption measures
FCC subscription shares	Tract-level fixed broadband adoption and outside-option shares	Anchors the CSS construction of product-level market shares
IIJA and BEAD allocation records	State-level broadband funding allocations and underserved-area measures	Continuous-treatment intensity and counterfactual policy design

C Provider Normalization

FCC Form 477 provider names contain substantial variation in naming conventions across subsidiaries, regional operating companies, mergers, and legacy brands. I consolidate provider names into canonical corporate-family identifiers in two stages.

C.1 Stage 1: Hard Overrides

A manually constructed crosswalk maps specific raw provider names to canonical identifiers for the major ISP families. Examples include:

- AT&T: Indiana Bell, Pacific Bell, Michigan Bell, BellSouth, Southwestern Bell, and state-named AT&T subsidiaries.
- Charter/Spectrum: Time Warner Cable, Bright House Networks, Charter Communications Operating.
- Comcast: Comcast Cable Communications, Comcast of California/Colorado.
- Optimum: CSC Holdings, Cablevision, Altice USA, Suddenlink.
- Windstream: 30+ state subsidiaries and legacy rural telco brands.

- Lumen/CenturyLink: CenturyTel, Qwest Corporation, Embarq.

The full hard-override table maps 140+ raw provider names to canonical identifiers and is documented in `00_build_structural_dataset_2020.py`.

C.2 Stage 2: Generic Cleaning Rules

Provider names not matched by Stage 1 are normalized using the following sequence:

1. Remove "d/b/a" suffixes;
2. Remove geographic qualifiers (e.g., "of Alabama");
3. Remove legal suffixes (LLC, Inc., Corp., Ltd., LP);
4. Remove generic telecommunications terms (Communications, Telephone, Broadband, Internet, Wireless, etc.);
5. Remove punctuation and collapse whitespace;
6. Retain the first two tokens as the canonical identifier.

The resulting crosswalk maps approximately 5,000 raw provider names to around 800 canonical identifiers and is saved as `provider_crosswalk_2020.csv`.

D Additional Details on URS Price Assignment

Each product position—observed or potential—in the structural sample is assigned a monthly price from the FCC Urban Rate Survey. Let (f, g, t) denote provider f, speed tier g, and census tract t, with average advertised download speed $\bar{d}_{fgt}$.

The algorithm searches for comparable URS plans through the following brand-first hierarchy, stopping at the first non-empty match:

1. State × provider × tier
2. National provider × tier
3. State × provider (any tier)
4. National provider (any tier)
5. State × tier (any provider)

6. National tier (any provider)

Within the matched set, the assigned price is the URS plan minimizing the absolute speed difference:

$$m^* = \arg\min_{m \in \mathcal{M}_{fgt}} |d_m - \bar{d}_{fgt}|, \tag{D.1}$$

where d_m is the URS plan's download speed. All prices are deflated to constant 2025 dollars using CPI-U.

The procedure achieves 100 percent imputation over all 727,601 potential product positions in the county-footprint entry sample (170,419 observed products plus 557,182 feasible but unobserved positions). Table D.1 reports the distribution of match quality across levels.

Table D.1: URS Price Imputation Match Quality

Match level	Potential product positions	Share
State × provider × tier	274,749	40.8%
National × provider × tier	103,288	15.3%
State × provider (any tier)	32,346	4.8%
National × provider (any tier)	20,001	3.0%
State × tier (any provider)	233,549	34.7%
National × tier (any provider)	9,177	1.4%
Total	727,601	100.0%

E CSS Market Share Construction and Validation

E.1 Market-Share Construction

The FCC adoption file reports tract-level subscription shares in aggregate bins—any fixed broadband, and fixed broadband at or above 10/1 Mbps—but not by individual provider. The CSS procedure of Crawford et al. [2019] uses these aggregate totals to construct product-level market shares in two steps. The 10/1 Mbps adoption bin directly aligns with the Tier H threshold used throughout the structural model ($\geq$10/1 Mbps), so no proxy adjustment is required.

Step 1: Aggregate OLS Logit. At the tract level, I estimate

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t'\psi + D_t'\lambda + u_t, \tag{E.1}$$

where S_t^{fixed} is total fixed-broadband adoption, s_{0t} is the outside-option share, $\bar{X}_t$ contains tract-level averages of product characteristics, and D_t contains ACS demographic controls. The CSS estimation sample covers 70,854 tracts and has $R^2 = 0.253$.[23]

Table E.1: CSS Aggregate Simple Logit: Tract-Level OLS

Variable	Coefficient
Constant	-7.491^{***}
Average price / \$100	-0.279^{***}
Average download speed (Gbps)	0.314^{***}
Average download speed^2	-0.156^{***}
Average upload speed (Gbps)	0.034
Average upload speed^2	0.098^{***}
Average high-speed tier share	0.168^{***}
Average coverage share	0.471^{***}
Log household income	0.793^{***}
Share renting	0.401^{***}
Share college-educated	-0.052^{**}
Share aged 65+	0.274^{***}
N (tracts)	70,854
R^2	0.253

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$. Dependent variable is $\log(S_t^{\text{fixed}}/s_{0t})$.

Aggregate validation against FCC national totals is reported in Table 3 in the main text. Table E.2 provides an additional diagnostic: model-implied subscriber allocations by provider. These are characteristic-based allocations, not reported subscriber counts, and should be read as a plausibility check rather than ground truth.

Step 2: Product-Level Allocation. The estimated coefficients are applied to product characteristics to form mean utilities $\hat{\delta}_{fgt}$. Product shares are allocated within each tract by softmax:

$$\hat{s}_{f|\text{in},t} = \frac{\exp(\hat{\delta}_{fgt})}{\sum_{f',g' \in \mathcal{F}_t} \exp(\hat{\delta}_{f'g't})}, \tag{E.2}$$

$$\hat{s}_{fgt} = \hat{s}_{f|\text{in},t} S_t^{\text{fixed}}. \tag{E.3}$$

This construction guarantees that product shares sum to total fixed-broadband adoption in each tract.

[23]The CSS logit is estimated on all tracts with valid adoption data and product characteristics (70,854). The demand estimation sample (66,454 tracts) is smaller because an additional 4,400 tracts are dropped when merging in the full set of demand controls or when all products in the tract have no positive imputed price.

Table E.2: Implied National Market Shares by Provider — December 2020

Provider	Share of BB subscribers (%)	Share of all households (%)	Tract coverage
AT&T	24.51	20.23	33,280
Comcast	21.57	17.80	29,288
Verizon	9.61	7.93	12,158
Frontier	7.68	6.34	9,442
Cox	4.58	3.78	5,752
Windstream	1.86	1.53	2,857
Starry	1.55	1.28	2,694
Optimum	1.30	1.07	2,749
Consolidated	1.10	0.91	1,911
Other ($<$ 1% each, $n = 1{,}724$)	26.25	21.67	—
Outside good (no broadband)	—	17.47	—
Total	100.00	100.00	—

Notes: Implied subscriber share $= \sum_t s_{jt}M_t \,/\, \sum_t \sum_k s_{kt}M_t$, where $M_t =$ households in tract t and the sum in the denominator is over all inside products. Household share $= \sum_t s_{jt}M_t \,/\, \sum_t M_t$. Aggregates to household share of broadband adopters of 82.5%, consistent with the weighted-mean outside-option share ($\bar{s}_0 = 0.187$, i.e. $1 - 0.813$). "Tract coverage" counts the number of census tracts in which the provider is active in the estimation sample (entry = 1). The 1% threshold is applied to subscriber share. Comcast and AT&T together account for ≈46% of implied broadband subscribers.

E.2 Market-Share Validation

Internal Consistency. The CSS share inversion enforces $\sum_j s_{jt} + s_{0t} = 1$ in every tract by construction. The maximum deviation from unity across all 70,854 estimation-sample tracts is 1.1×10^{-15} (machine precision). All product shares lie in $(0, 1)$ and all outside-option shares lie in $[0.10, 0.90]$, satisfying strict positivity for Berry inversion.

External Benchmark. Table E.3 compares the national distribution of implied broadband penetration $(1 - s_{0t})$ at the census-tract level against two external series. The first is the FCC 2020 household penetration share. The second is the ACS 2019 broadband adoption rate, measured from Table B28002 as the share of households with any broadband subscription. The FCC series is the direct input to the CSS model; the ACS series is fully independent. Each row reports the fraction of U.S. households residing in tracts whose broadband penetration falls in the indicated range.

Table E.3: National Distribution of Broadband Penetration — CSS vs. External Benchmarks

FCC midpoint	CSS implied (% of HH)	FCC S_{fixed} (% of HH)	ACS broadband (% of HH)
10%	0.4	0.3	0.5
30%	1.4	0.9	6.0
50%	4.9	4.1	20.8
70%	21.3	20.1	44.3
90%	71.9	74.7	28.3
National average	82.5%	83.6%	68.5%

Notes: Rows correspond to the five FCC penetration midpoints (10, 30, 50, 70, 90%). Each cell gives the household-weighted share of census tracts in the estimation sample (N = 70,854) assigned to that midpoint. For CSS implied and FCC S_{fixed}, tracts are assigned to their exact midpoint value; for ACS, tracts are assigned to the nearest midpoint bin (±10 p.p.). CSS implied penetration is constructed from the FCC S_{fixed} series via the CSS latent-share model, so the two columns are nearly identical. ACS broadband rate is a continuous survey measure and systematically undercounts subscriptions relative to administrative FCC data: the 14-p.p. national gap (82.5% vs. 68.5%) is consistent with the well-documented FCC–ACS discrepancy. Data sources: FCC Form 477 deployment data (`https://www.fcc.gov/general/broadband-deployment-data-fcc-form-477`); ACS 5-year Table B28002 (`https://data.census.gov/table/ACSDT5Y2019.B28002`).

F Reduced-Form Robustness: Static DiD Estimates

Table F.1 complements the event studies with static two-way fixed-effects DiD estimates for the same six outcomes, pooling the post-period. The preferred specification includes state-specific linear time trends, which absorb pre-existing differential trends. For ACP, this distinction matters. Without state trends, the coefficients are inflated by the pre-existing correlation between state poverty rates and broadband outcomes, as the event study pre-trends in Figure 3 confirm. For BEAD, the pooled estimate collapses the lagged entry dynamics of Figure 4 into a single average treatment effect. This understates the eventual effects concentrated at $\tau = 3$–4.

Table F.1: Difference-in-Differences Estimates: ACP and BEAD Exposure

Outcome	(1) ACP Elig. × Post_{ACP}	(2) BEAD log BEAD × $\text{Post}_{\text{BEAD}}$
Prices		
Log avg. real price	-0.5611^{***}	-0.0225
	(0.0599)	(0.0363)
Log price (<25/3 Mbps)	-0.1384^{*}	-0.0218
	(0.0826)	(0.0298)
Log price (≥25/3 Mbps)	-0.9215^{***}	-0.0573
	(0.0870)	(0.0604)
Plan Composition		
Plan share (<25/3 Mbps)	-0.3904^{***}	-0.0509
	(0.0576)	(0.0339)
Plan share (≥25/3 Mbps)	$+0.3904^{***}$	$+0.0509$
	(0.0576)	(0.0339)
Market Structure		
Provider HHI	-0.3061^{***}	-0.0229
	(0.1160)	(0.0168)
Observations	597	597
State FE	Yes	
Year FE	Yes	
State trends	Yes	
Weights	Number of plans	

Notes: The six outcomes match those used in the event studies (Figures 3 and 4). Each cell reports the OLS coefficient and clustered standard error (in parentheses) from a separate TWFE regression with state-specific linear time trends. The ACP treatment is the share of households below 200% FPL interacted with $\mathbf{1}[t \geq 2022]$. The BEAD treatment is log BEAD allocation per unserved location interacted with $\mathbf{1}[t \geq 2023]$. Standard errors clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

G Measurement and Limitations

Four measurement choices shape how the structural estimates should be read.

Inferred Market Shares. The FCC does not publish tract-level subscriber counts by provider, so product shares are constructed rather than observed. The CSS procedure [Crawford et al., 2019] allocates aggregate tract adoption across providers in proportion to logit weights from product characteristics—an allocation that is internally consistent but imposes a particular sharing rule. Providers with unobserved local brand advantages or incumbency effects will be misallocated. The approach parallels standard practice in broadband demand estimation [Fan, 2013], where transaction data are unavailable, and the provider-level validation in Table E.2 suggests the aggregate allocations are plausible.

Imputed Prices. Product prices come from the FCC Urban Rate Survey, which samples urban tracts, not rural ones. For approximately 35 percent of products the match falls back to the state-tier level rather than the more precise brand-state-tier level, which introduces measurement error. The Gandhi and Houde [2019] instruments address price endogeneity but do not remove attenuation bias from measurement error on the left-hand side. The state-level aggregation robustness check (elasticity -5.87 vs. -4.99) suggests the bias is modest.

Share-Construction Sensitivity. An earlier state-year aggregation of the same data—combining ACS aggregate adoption with plan-availability shares at the state level—yields a mean own-price elasticity of -5.87, close to the preferred tract-level estimate of -4.99. The <20 percent difference across very different aggregation levels provides reassurance that the demand estimates are not driven by the CSS allocation rule.

Static Baseline. The model characterizes the 2020 equilibrium and uses it to simulate policy counterfactuals. Dynamic investment decisions, learning, and network effects are absent. Because the baseline predates both programs, the counterfactuals cannot be validated against post-implementation outcomes—a limitation shared by all static structural evaluations in infrastructure markets. The structural results are best interpreted as long-run steady-state effects conditional on the observed market structure, not as forecasts of any specific year.

Implementation and Crowding Out. The counterfactuals treat subsidies as changes in household prices or provider fixed costs and abstract from administrative frictions. In practice, BEAD awards may face permitting delays, state-level prioritization rules, and construction bottlenecks. Subsidized entry may also crowd out unsubsidized private investment in some markets. The estimates therefore measure the welfare return to the modeled entry response, not the full social return net of all implementation costs.

H Entry Moment Inequalities

This appendix provides the formal construction underlying the fixed-cost bounds in Section 5 and the counterfactual equilibrium algorithm in Section 4.5. The fundamental object throughout is the incremental variable profit from offering position (f, g) in tract t:

$$\Delta_{fgt}(Y_{-fgt}) = r_{ft}(Y_{fgt} = 1,\, Y_{-fgt}) - r_{ft}(Y_{fgt} = 0,\, Y_{-fgt}). \tag{H.1}$$

Section H.1 introduces the parameterization and moment construction. Section H.2 develops the identified set, criterion function, and Andrews–Soares confidence set. Section H.3 describes the truncated-normal fixed-cost draws that rationalize the observed baseline. Section H.4 details the iterated best-response algorithm used for counterfactual simulations.

H.1 Setup and Moment Construction

The inequality estimation panel consists of all positions in the county-footprint set $\mathcal{J}_t^0$: $Y_{fgt} = 1$ for observed products and zero for feasible but unobserved ones. Fixed costs are parameterized as in equation (5.2):

$$FC_{fgt} = \theta_{b(t)} + \theta_H\, \mathbf{1}\{g = H\} + \sigma_{b(t)}\, \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0, 1), \tag{H.2}$$

where $\theta_{b(t)}$ is the mean fixed cost in market-size cell $b(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$, θ_H is the high-speed tier premium, and $\sigma_{b(t)}$ is the size-specific shock dispersion. Write the systematic component as $\mu^C_{fgt}(\vartheta) = \theta_{b(t)} + \theta_H \mathbf{1}\{g = H\}$ and collect the seven structural parameters in $\vartheta = (\theta_{\text{Small}}, \theta_{\text{Medium}}, \theta_{\text{Large}}, \theta_H, \sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}})$.

Following Fan and Yang [2024], the moment inequalities are weighted by $K = 33$ non-negative functions of observed market characteristics $h^{(k)}(X_t, g)$, $k \in \{1, \ldots, 33\}$, defined formally in Section 4.6: a constant, two speed-tier indicators, three market-size indicators, and 27 interactions of market size with quantile indicators of the bounds $\underline{\Delta}_{fgt}$ and $\overline{\Delta}_{fgt}$ (see also Fan and Yang 2024, Appendix A). Combined with the two-sided moment construction of Section H.2, the 33 weight functions yield $K \times 2 = 66$ moment inequalities—the count stated in Section 4.6.

Assumption 4. *For each potential position (f, g, t), the observed decision Y_{fgt} is not a dominated strategy.*

Nash optimality of the observed configuration then implies

$$\Pr(Y_{fgt} = 1 \text{ is dominant}) \leq \Pr(Y_{fgt} = 1) \leq \Pr(Y_{fgt} = 1 \text{ is not dominated}). \tag{H.3}$$

The lower and upper incremental-profit cutoffs are

$$\underline{\Delta}_{fgt} = \min_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}), \tag{H.4}$$

$$\overline{\Delta}_{fgt} = \max_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}). \tag{H.5}$$

Together with the parameterization in (H.2), these cutoffs imply the conditional moment inequalities

$$\Phi\left(\frac{\underline{\Delta}_{fgt} - \mu^C_{fgt}(\vartheta)}{\sigma_{b(t)}}\right) \le \Pr(Y_{fgt} = 1 \mid X_t, g), \tag{H.6}$$

$$\Pr(Y_{fgt} = 1 \mid X_t, g) \le \Phi\left(\frac{\overline{\Delta}_{fgt} - \mu^C_{fgt}(\vartheta)}{\sigma_{b(t)}}\right). \tag{H.7}$$

Because inequalities (H.6)–(H.7) hold for every Nash equilibrium, no equilibrium-selection assumption is required for estimation [Tamer, 2003, Ciliberto and Tamer, 2009, Fan and Yang, 2024].

H.2 Identified Set and Confidence Set

Moment vector. The 66 moment conditions are collected in the tract-level vector $Z_m(\vartheta) \in \mathbb{R}^{66}$, where m indexes census-tract markets. For each weight cell $k \in \{1, \ldots, 33\}$, define the two signed tract averages

$$Z^{\text{lo}}_{m,k}(\vartheta) = \frac{1}{|\mathcal{J}^0_m|} \sum_{(f,g)\in\mathcal{J}^0_m} h^{(k)}(X_m, g) \left[\Phi\left(\frac{\underline{\Delta}_{fgm} - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}}\right) - Y_{fgm}\right], \tag{H.8}$$

$$Z^{\text{hi}}_{m,k}(\vartheta) = \frac{1}{|\mathcal{J}^0_m|} \sum_{(f,g)\in\mathcal{J}^0_m} h^{(k)}(X_m, g) \left[Y_{fgm} - \Phi\left(\frac{\overline{\Delta}_{fgm} - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}}\right)\right], \tag{H.9}$$

where $|\mathcal{J}^0_m|$ is the number of feasible positions in market m and Φ is the standard-normal CDF. The sample moment vector is

$$\bar{Z}_M(\vartheta) = \frac{1}{M} \sum_{m=1}^{M} Z_m(\vartheta), \tag{H.10}$$

where $M = 70{,}854$. The sign convention ensures $\mathbb{E}[Z_m(\vartheta)] \le \mathbf{0}$ element-by-element at every ϑ in the identified set; a positive entry signals a violated probability bound. The full vector stacks the 33 lower and 33 upper components.

Identified set. The *identified set* is

$$\Theta_I = \Big\{\vartheta = (\theta, \sigma) : \mathbb{E}[m(\vartheta, W_{fgm})] \le \mathbf{0} \text{ for all } k \in \{1, \ldots, 33\}\Big\}, \tag{H.11}$$

where $\sigma = (\sigma_{\text{Small}}, \sigma_{\text{Medium}}, \sigma_{\text{Large}})$ and $m(\vartheta, W_{fgm})$ is the vector of position-level moment functions whose market-level average is $Z_m(\vartheta)$. The set Θ_I collects all fixed-cost structures (θ, σ) consistent with the observed entry pattern under *some* equilibrium selection mechanism. A parameter vector outside Θ_I violates at least one dominance restriction in population. It assigns too high a probability to an absent position or too low a probability to an occupied one. The identified set is a bounded region rather than a point because the inequalities constrain ϑ from above and below without forcing equality. The remaining width reflects the information equilibrium multiplicity leaves unresolved.

Criterion function. A standard two-sided GMM criterion conflates binding inequalities with slack ones. For moment inequalities, only violations—positive values of $\bar{Z}_{M,j}(\vartheta)$—should enter the objective. For a candidate ϑ, the minimum-distance criterion is

$$T_M(\vartheta) = \Big[\sqrt{M}\ \max\Big\{\bar{Z}_M(\vartheta),\, \mathbf{0}\Big\}\Big]'\, \hat{D}_M(\vartheta)^{-1} \Big[\sqrt{M}\ \max\Big\{\bar{Z}_M(\vartheta),\, \mathbf{0}\Big\}\Big], \tag{H.12}$$

where the maximum is element-by-element and $\hat{D}_M(\vartheta) = \text{Diag}(\hat{\Sigma}_M(\vartheta))$ is defined below. A value $T_M(\vartheta) = 0$ certifies that ϑ lies in the sample analogue of Θ_I. The $\sqrt{M}$ scaling ensures the statistic converges to a non-degenerate limit on the boundary of the identified set, so that simulated critical values are well behaved.

Weighting matrix. The market-level covariance matrix is

$$\hat{\Sigma}_M(\vartheta) = \frac{1}{M} \sum_{m=1}^{M} \Big(Z_m(\vartheta) - \bar{Z}_M(\vartheta)\Big)\Big(Z_m(\vartheta) - \bar{Z}_M(\vartheta)\Big)', \tag{H.13}$$

and $\hat{D}_M(\vartheta) = \text{Diag}(\hat{\Sigma}_M(\vartheta))$ standardizes each component by its market-level standard deviation. The full correlation matrix $\hat{\Omega}_M(\vartheta) = \hat{D}_M(\vartheta)^{-1/2}\hat{\Sigma}_M(\vartheta)\hat{D}_M(\vartheta)^{-1/2}$ enters the simulated critical value below, following Appendix A of Fan and Yang [2024].

Confidence set. Following Andrews and Soares [2010], I construct a 95 percent confidence set by inverting a GMS test. For each candidate ϑ, define the correlation matrix $\hat{\Omega}_M(\vartheta)$ as above and form the GMS drift

$$\eta_M(\vartheta) = (\ln M)^{-1/2}\sqrt{M}\, \hat{D}_M(\vartheta)^{-1/2}\bar{Z}_M(\vartheta). \tag{H.14}$$

Draw $R^b \sim N(0, I_{66})$ for $b = 1, \ldots, 999$ and set $\hat{c}_M(\vartheta, 0.95)$ equal to the 95th percentile of

$$S\Big(\hat{\Omega}_M(\vartheta)^{1/2} R^b + [\eta_M(\vartheta)]_-, \ \hat{\Omega}_M(\vartheta)\Big),$$

where $[x]_- = \min\{x, 0\}$ component-by-component and $S(z, \Omega) = \sum_{j=1}^{66} [z_j / \sqrt{\Omega_{jj}}]_+^2$. The 95 percent confidence set is then

$$\hat{\mathcal{C}} = \Big\{\vartheta : T_M(\vartheta) \le \hat{c}_M(\vartheta, 0.95)\Big\}. \tag{H.15}$$

The set $\hat{\mathcal{C}}$ contains all parameter vectors not rejected at level $\alpha = 5\%$ by the GMS test and is sharper than applying a common chi-squared critical value to all 66 inequalities simultaneously. The reported bounds in Table 12 are coordinate-wise projections of $\hat{\mathcal{C}}$ onto each structural parameter.

Two sources of width. The confidence set $\hat{\mathcal{C}}$ has two distinct sources of width. The first is *sampling uncertainty*: for a fixed identified set Θ_I, the set fluctuates around Θ_I at rate $\sqrt{M}$ and shrinks toward Θ_I as $M \to \infty$. The second is the *width of the identified set itself*: the range of parameter vectors left observationally equivalent by equilibrium multiplicity does not shrink with M. Wide bounds on a parameter therefore signal that the moment inequalities constrain it weakly, not that the sample is small. The counterfactual bounds in Section 6 propagate both sources by averaging over draws from $\hat{\mathcal{C}}$ and over equilibrium multiplicity in the best-response algorithm of Section H.4. Figure H.1 illustrates the stochastic search used to approximate $\hat{\mathcal{C}}$.

H.3 Fixed-Cost Draws

For each accepted $\vartheta \in \hat{\mathcal{C}}$, the latent fixed cost is $FC_{fgm} = \theta_{b(m)} + \theta_H \mathbf{1}\{g = H\} + \sigma_{b(m)} \zeta_{fgm}$. To ensure the observed baseline configuration is individually rational under the draw, ζ_{fgm} is sampled from a truncated normal. For an *observed* position ($Y_{fgm} = 1$),

$$\zeta_{fgm} \le \frac{\Delta_{fgm}(Y^{\text{obs}}_{-fgm}) - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}},$$

and for a *feasible but unobserved* position ($Y_{fgm} = 0$),

$$\zeta_{fgm} > \frac{\Delta_{fgm}(Y^{\text{obs}}_{-fgm}) - \mu^C_{fgm}(\vartheta)}{\sigma_{b(m)}}.$$

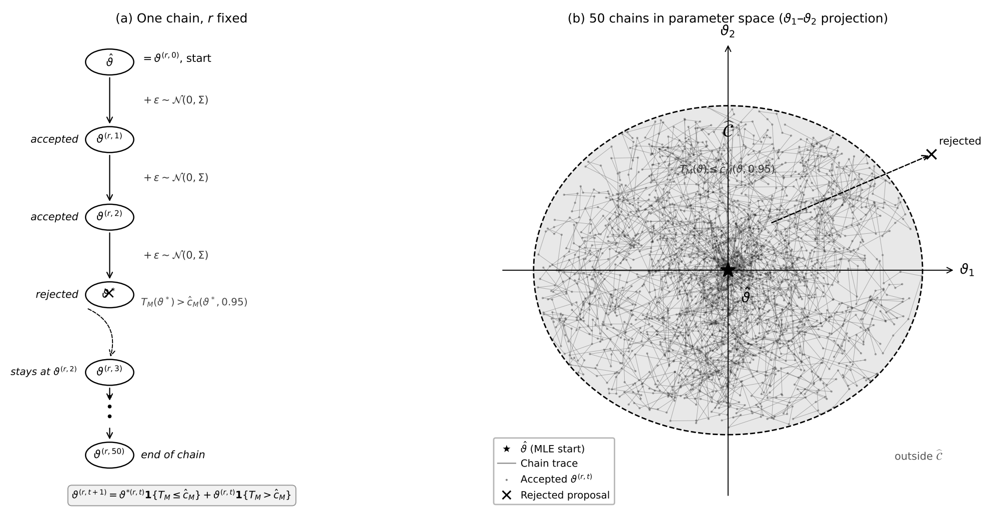

Figure H.1: Algorithm for Confidence-Set Construction

Notes: Panel (a) illustrates the accept/reject logic for a single random-walk chain at fixed draw r. Starting from the minimizer $\hat{\vartheta}$, each iteration proposes $\vartheta^{*(r,t)} = \vartheta^{(r,t)} + \varepsilon^{(r,t)}$ with $\varepsilon^{(r,t)} \sim \mathcal{N}(0, \Sigma_{\text{local}})$ and updates

$$\vartheta^{(r,t+1)} = \vartheta^{*(r,t)}\,\mathbf{1}\Big\{T_M\Big(\vartheta^{*(r,t)}\Big) \le \hat{c}_M\Big(\vartheta^{*(r,t)}, 0.95\Big)\Big\} + \vartheta^{(r,t)}\,\mathbf{1}\Big\{T_M\Big(\vartheta^{*(r,t)}\Big) > \hat{c}_M\Big(\vartheta^{*(r,t)}, 0.95\Big)\Big\}.$$

Accepted proposals advance the chain; rejected proposals leave it unchanged. Panel (b) shows the projection of accepted chains onto the $(\vartheta_1, \vartheta_2)$ plane. The two-stage search uses 50 local chains of 50 iterations followed by 100 expansion chains of 50 iterations from boundary-accepted points, yielding 656 accepted parameter vectors from $\hat{\mathcal{C}} = \{\vartheta : T_M(\vartheta) \le \hat{c}_M(\vartheta, 0.95)\}$. See Section 4.6 for further details. The construction follows Fan and Yang [2024].

This truncation guarantees that every firm's observed portfolio is a best response at the drawn fixed costs, so the baseline market configuration is consistent with Nash equilibrium at ϑ.

H.4 Counterfactual Best Responses

For each fixed-cost draw and policy regime, I compute a pure-strategy counterfactual product configuration using iterated best responses:

1. **Initialize.** Set $Y_t^{(0)} \in \{Y_t^{\text{obs}}, \mathbf{0}, \mathbf{1}\}$, where $\mathbf{0}$ denotes the empty portfolio and $\mathbf{1}$ denotes the full feasible configuration.

2. **Update.** Cycle through firms sequentially. For each firm f, hold rivals' current portfolios fixed and solve

$$Y_{ft}^{BR} \in \arg \max_{Y_{ft} \in \{\varnothing, \{H\}, \{L\}, \{H,L\}\}} \left\{ r_{ft}(Y_{ft}, Y_{-ft}) - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}^{cf} \right\}.$$

3. **Replace.** Set $Y_{ft} \leftarrow Y_{ft}^{BR}$.

4. **Iterate.** Repeat steps 2–3 until no firm changes its portfolio.

When multiple stable configurations are found across initializations, I retain all of them and report the range of outcomes across fixed points. Markets where the algorithm does not converge within the iteration limit (fewer than one percent of markets per draw) are assigned the observed pre-policy configuration; this conservative fallback preserves the validity of the reported bounds. Figure H.2 reports convergence diagnostics across the 50 fixed-cost draws.

I Fan–Yang Bound Diagnostics

The three figures below provide diagnostic checks on the Fan–Yang incremental variable-profit bounds. Figure I.1 shows how bounds vary with local market structure. Figure I.2 documents geographic and market-size patterns in the tightness of the bounds.

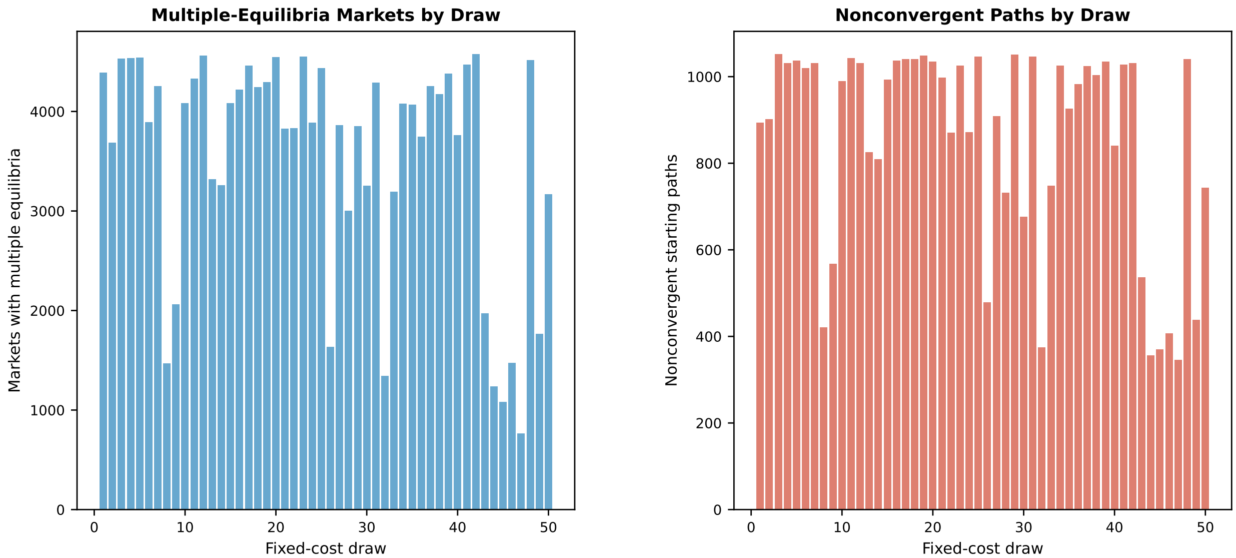

Figure H.2: Equilibrium Multiplicity and Convergence Diagnostics

Notes: For each of the 50 fixed-cost draws, the left panel reports the number of markets exhibiting multiple stable configurations and the right panel reports the number of non-converged starting paths. The distribution of convergence failures is not systematically correlated with market size or policy intensity.

J Detailed Counterfactual Outcomes

Consumer surplus in tract t is the nested-logit inclusive value divided by the marginal utility of income:

$$\mathrm{CS}_t(Y_t, p_t) = \frac{M_t}{-\hat{\alpha}/100} \log\left[1 + \left(\sum_{j \in \mathcal{J}_t(Y_t)} \exp\left(\frac{\hat{\delta}_{jt}}{1-\hat{\rho}}\right)\right)^{1-\hat{\rho}}\right], \tag{J.1}$$

where prices enter $\hat{\delta}_{jt}$ in dollars and $\hat{\alpha}/100$ converts the price coefficient (estimated on price/\$100) to a per-dollar quantity. Producer surplus is variable profit net of fixed costs. Social surplus is the sum of consumer and producer surplus. National bounds aggregate tract-level lower and upper outcomes, combining equilibrium multiplicity across starting configurations with parameter uncertainty across the 50 fixed-cost draws.

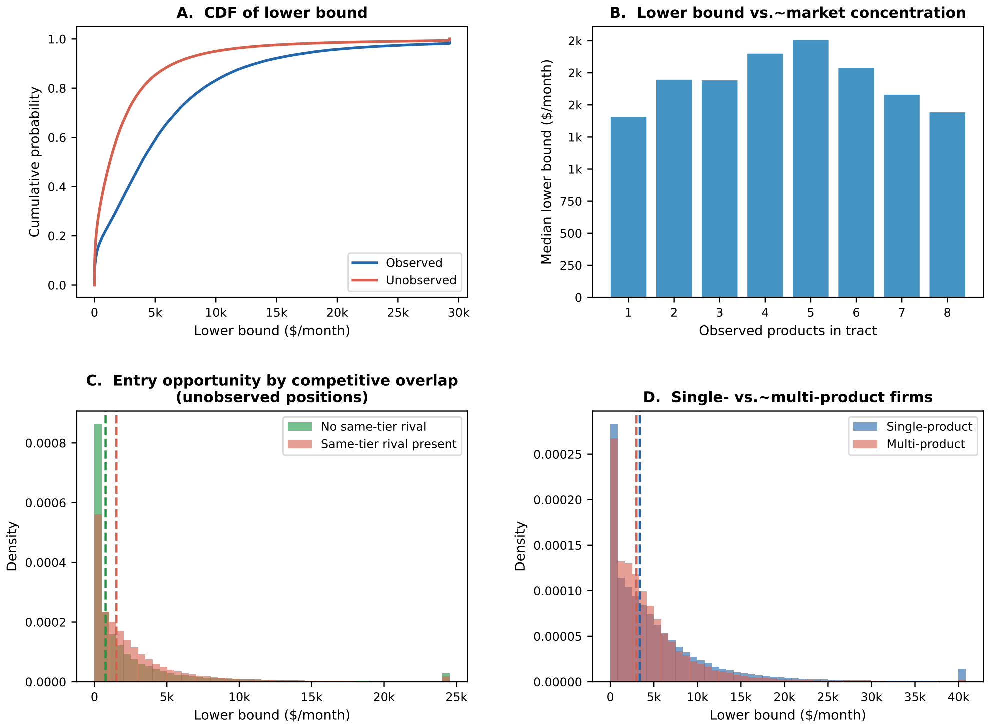

Figure I.1: Fan–Yang Bounds by Market Structure

Notes: Panel A plots the CDF of the lower bound for observed vs. unobserved positions. Panel B shows the median lower bound by number of observed products in the tract. Panel C compares the lower-bound distribution for unobserved positions by same-tier rival presence. Panel D compares single-product vs. multi-product firms.

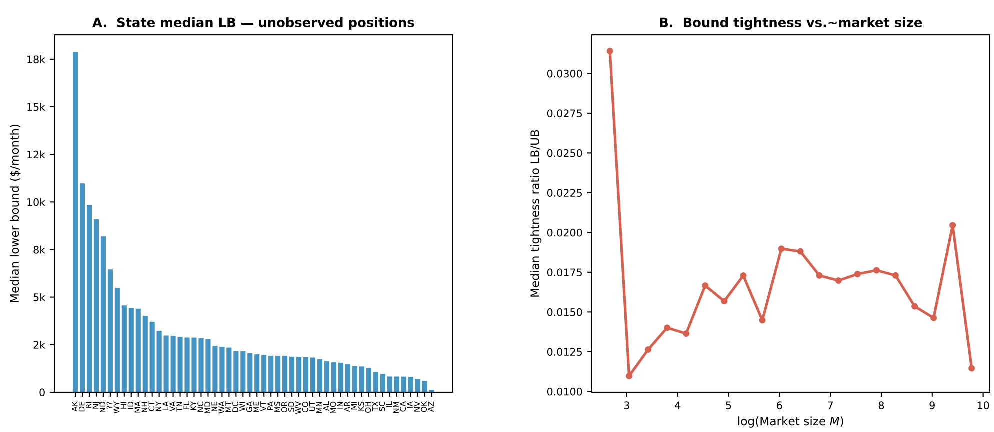

Figure I.2: Fan–Yang Bounds by Geography and Market Size

Notes: Panel A reports the state median lower bound for unobserved positions; states with higher median lower bounds have stronger underlying variable-profit incentives for potential entrants. Panel B plots the bound tightness ratio against log market size.

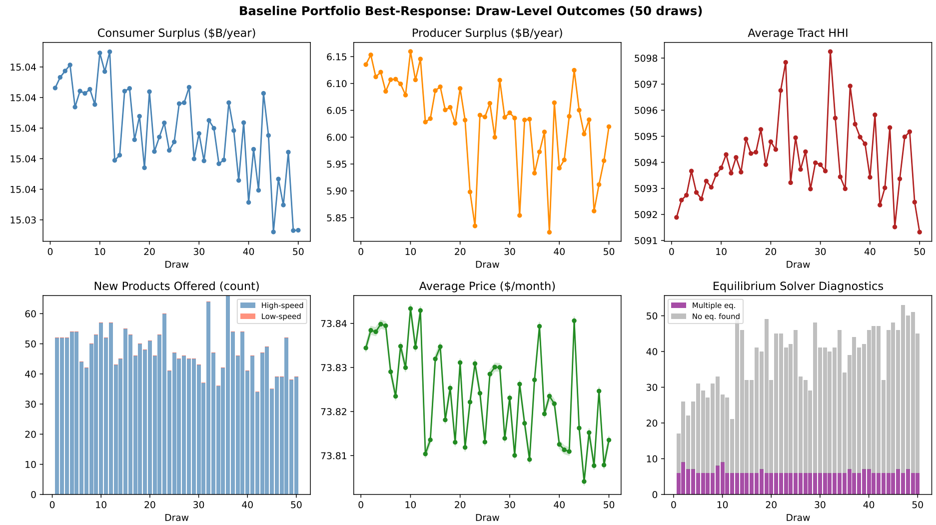

Figure J.1: Baseline Outcomes Across Fixed-Cost Draws

Notes: Baseline ($\kappa = 0$) outcomes across 50 representative draws from the Andrews–Soares confidence set. Consumer surplus, average tract HHI, and average price vary by less than 0.1%, indicating little sensitivity to fixed-cost uncertainty. New products range from 40–65 per draw due to equilibrium multiplicity. In the diagnostics panel, purple bars denote markets with multiple stable configurations; grey bars denote markets assigned the observed portfolio after reaching the iteration limit (fewer than 0.1% of markets per draw).

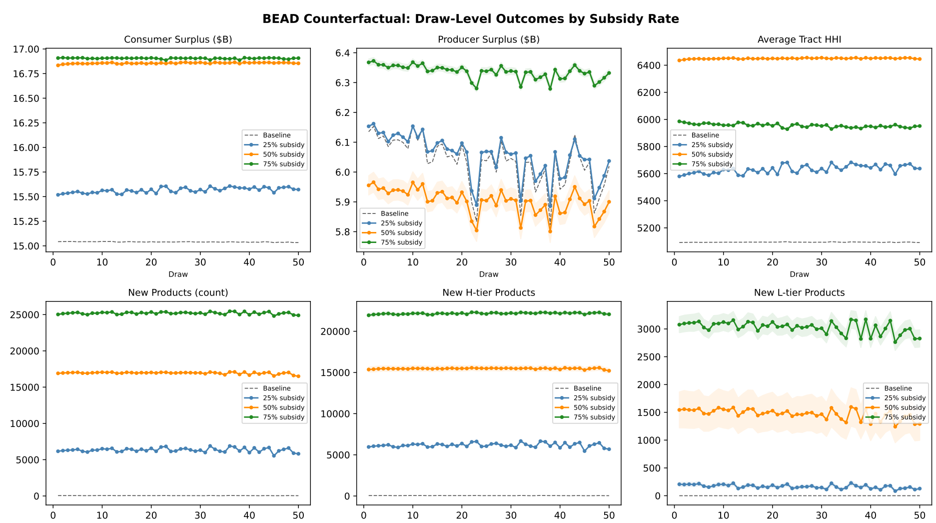

Figure J.2: BEAD Counterfactual Outcomes Across Fixed-Cost Draws

Notes: BEAD-induced new product entries across 50 fixed-cost draws, by speed tier and subsidy rate. The shaded band for $\kappa = 75\%$ shows equilibrium bounds; lines for $\kappa = 25\%$ and $\kappa = 50\%$ show draw-level midpoints. Entry at 25% and 50% is stable across draws, while entry at 75% is larger and more variable. The slight decline in later draws reflects higher fixed costs along the first principal component of $\hat{\mathcal{C}}$, reducing newly deployable positions.

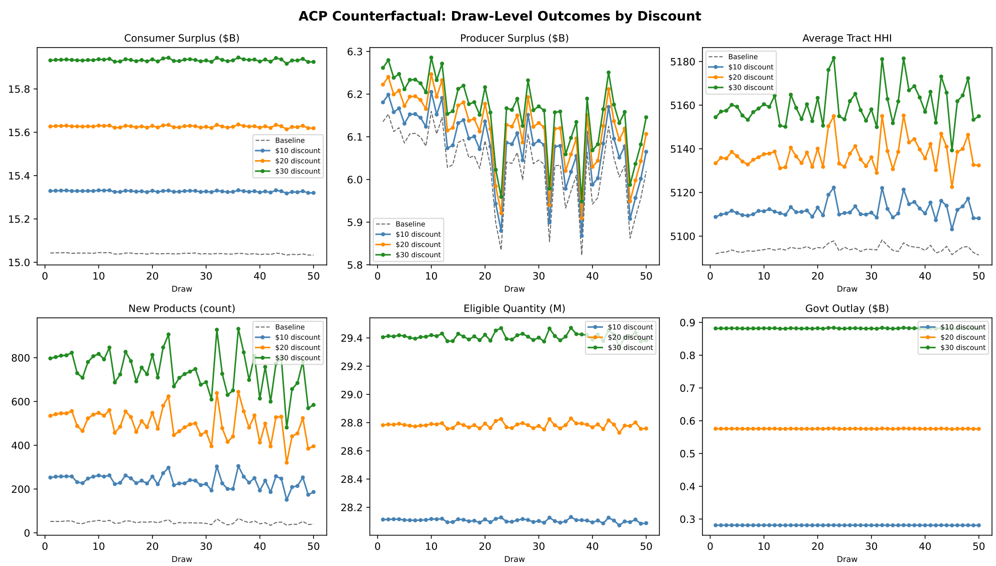

Figure J.3: ACP Counterfactual Outcomes Across Fixed-Cost Draws

Notes: ACP counterfactual outcomes across 50 fixed-cost draws, by discount level. Consumer surplus, take-up, and government outlay are nearly unchanged across draws, indicating welfare estimates are largely insensitive to fixed-cost uncertainty. Producer surplus varies moderately across draws. Government outlay rises from about \$0.29 B at a \$10 discount to \$0.89 B at a \$30 discount.

K Cost-Benefit Robustness

This section provides three robustness exercises for the cost-benefit results in Section 7. Tables K.1 and K.2 extend the main-text sensitivity analysis by reporting NPV and BCR bounds across all four discount rates for every BEAD subsidy rate and every ACP discount level (not only the headline $\kappa = 75\%$ scenario). Table K.3 documents the stability of welfare estimates across the 50 fixed-cost draws from the Andrews–Soares confidence region, showing that the main-text bounds reflect genuine partial-identification width rather than estimation noise.

NPV and BCR Across Discount Rates. For BEAD, the NPV is sensitive to the discount rate because the fiscal cost is paid once and benefits accrue over 25 years. Higher rates therefore reduce the present value of the flow. The BCR is somewhat less sensitive because the government cost and the annual welfare gain scale together. For ACP, both the BCR and the NPV-equivalent are invariant to the discount rate because costs and benefits recur at the same frequency. All four columns are identical.

All BEAD BCR lower bounds remain strictly above 17 at every rate–scenario combination. The diminishing-returns ordering ($\kappa = 25\% > \kappa = 50\% > \kappa = 75\%$ in BCR, even at 7%) holds at every discount rate. The marginal deployment attracted at higher subsidy rates is harder to serve and returns less per dollar. This ordering reverses in NPV because the $\kappa = 75\%$ scenario activates far more entry. It therefore generates far larger absolute welfare gains despite the lower per-dollar return.

Table K.1: NPV Bounds ($B) by Policy and Discount Rate ($T = 25$ yr)

Policy	$r = 1\%$	$r = 3\%$ (base)	$r = 5\%$	$r = 7\%$
BEAD fixed-cost reduction				
$\kappa = 25\%$	[80.9, 128.3]	[63.6, 100.9]	[51.2, 81.2]	[42.0, 66.7]
$\kappa = 50\%$	[158.6, 246.8]	[124.5, 194.0]	[100.1, 155.9]	[82.1, 128.0]
$\kappa = 75\%$	[463.3, 599.9]	[363.4, 470.5]	[291.4, 377.4]	[238.4, 309.0]
ACP monthly discount (annual net-surplus PV)				
$a = \$10$	[0.1, 0.2]	[0.1, 0.2]	[0.1, 0.2]	[0.1, 0.2]
$a = \$20$	[0.2, 0.3]	[0.2, 0.3]	[0.2, 0.3]	[0.2, 0.3]
$a = \$30$	[0.1, 0.3]	[0.1, 0.3]	[0.1, 0.3]	[0.1, 0.3]

Notes: Each cell reports $[Q_{2.5}(\underline{\text{NPV}}^r), Q_{97.5}(\overline{\text{NPV}}^r)]$ across 50 fixed-cost draws. BEAD NPV $= PV(\Delta SW) - G$ at horizon $T = 25$ yr. ACP columns report the present value of annual net-surplus bounds ($\Delta SW - G$ per year, discounted over 25 years); ACP NPV is nearly invariant to r because both costs and benefits are annual flows and near-equal in magnitude. All BEAD lower-bound NPVs are strictly positive at every rate shown.

Table K.2: BCR Bounds by Policy and Discount Rate ($T = 25$ yr)

Policy	$r = 1\%$	$r = 3\%$ (base)	$r = 5\%$	$r = 7\%$
BEAD fixed-cost reduction				
$\kappa = 25\%$	[44.99, 55.42]	[35.58, 43.82]	[28.79, 35.47]	[23.81, 29.33]
$\kappa = 50\%$	[39.83, 44.95]	[31.50, 35.54]	[25.49, 28.77]	[21.08, 23.79]
$\kappa = 75\%$	[32.70, 34.70]	[25.85, 27.44]	[20.93, 22.21]	[17.30, 18.36]
ACP monthly discount (annual BCR, rate-invariant)				
$a = \$10$	[1.04, 1.06]	[1.04, 1.06]	[1.04, 1.06]	[1.04, 1.06]
$a = \$20$	[1.02, 1.04]	[1.02, 1.04]	[1.02, 1.04]	[1.02, 1.04]
$a = \$30$	[1.01, 1.03]	[1.01, 1.03]	[1.01, 1.03]	[1.01, 1.03]

Notes: BCR = $PV(\Delta SW)/G$, boundary-by-boundary, then sorted. BEAD BCR declines with r because benefits are discounted at a higher rate while the one-time fiscal cost is fixed. ACP BCR is the ratio of annual ΔSW to annual government outlay and does not depend on r; identical columns confirm rate-invariance. Every cell has BCR > 1. The minimum BCR lower bound across all 24 BEAD cells shown is 17.30 (at $r = 7\%$, $\kappa = 75\%$).

Stability Across Fixed-Cost Draws. The 50 parameter draws from the Andrews–Soares confidence region vary in their implied fixed costs and therefore in the extent of counterfactual entry and welfare gains. Table K.3 reports the mean, standard deviation, minimum, and maximum of draw-level midpoints for the key CBA outcomes. For BEAD, the standard deviation of monthly ΔSW ranges from \$0.052 billion at $\kappa = 25\%$ to \$0.168 billion at $\kappa = 75\%$, or roughly 12–8% of the mean. This moderate heterogeneity reflects genuine variation in fixed-cost parameter draws rather than sampling noise, which would shrink with $M = 70{,}854$. For ACP, welfare estimates are nearly identical across draws, with BCR standard deviations of 0.004–0.006. This confirms that ACP welfare is insensitive to supply-side fixed-cost uncertainty. The gain is determined almost entirely by the demand estimates, which are point-identified.

Table K.3: CBA Stability Across Fixed-Cost Draws: Key Statistics

Scenario / outcome	Mean	SD	Min	Median	Max
BEAD: monthly ΔSW ($B/month), midpoints across 50 draws					
$\kappa = 25\%$	0.423	0.052	0.263	0.434	0.496
$\kappa = 50\%$	0.834	0.099	0.535	0.870	0.959
$\kappa = 75\%$	2.143	0.168	1.639	2.199	2.349
BEAD: annual government cost ($B), midpoints across 50 draws					
$\kappa = 25\%$	2.251	0.227	1.545	2.355	2.454
$\kappa = 50\%$	5.171	0.555	3.502	5.415	5.681
$\kappa = 75\%$	16.765	1.165	13.261	17.257	17.922
ACP: annual BCR, midpoints across 50 draws					
$a = \$10$	1.050	0.006	1.035	1.050	1.060
$a = \$20$	1.034	0.005	1.022	1.035	1.043
$a = \$30$	1.020	0.004	1.011	1.020	1.027

Notes: Statistics computed from the midpoints of the draw-level $[\underline{Y}^r, \overline{Y}^r]$ intervals across all 50 fixed-cost draws from the Andrews–Soares confidence region. BEAD entries are monthly flows; government cost is annualized ($\times 12$). ACP BCR is the ratio of annual ΔSW to annual government outlay and is computed directly, without discounting. The near-zero standard deviation of ACP welfare statistics confirms that ACP welfare outcomes are insensitive to fixed-cost parameter uncertainty.